PHU 系列关节模组用户手册

PHU Series Joint Module User Manual

V1.24

版本变更说明

Version Change Description

PHU 关节用户手册版本变更说明 Version Change Description for PHU Joint User Manual				
序号	版本号	变更日期	修订内容	变更人
1	v1.0	20250901	初版发布 First Edition Released	--
2	v1.1	20250928	重新排版，细化机械与电气安装，更新软件参数 Reformatting, Refining Mechanical and Electrical Installation, Updating Software Parameters	--
3	v1.11	20251114	更新关节规格表 Updated Joint Specifications Table	--
4	v1.12	20251125	更新电源插头母头型号 Updated Power Plug Female Model	--
5	v1.13	20251128	新增英文翻译 Add English translation	--
6	v1.14	20251209	新增 STO 功能说明、产品外观由银色变黑色 Added STO feature description, product appearance changed from silver to black	--
7	v1.2	20260121	新增力控功能，新增 PHU11L/H、PHU25、PHU-F Added force control function, new PHU11L/H and PHU25、PHU-F models	--
8	v1.21	20260206	新增规格参数名词解释，优化部分细节 Added explanations for specification terms and optimized some details	
9	1.22	20260224	更正描述错误	

			Correct some description errors	
10	1.23	20260312	新增 PHU32/40 型号，新增 FOE 固件升级方法，补充机械安装螺丝要求，复核母线电流要求及线束要求，更正其它错误描述。 Added PHU32/40 models, added FOE firmware upgrade method, supplemented mechanical installation screw requirements, corrected bus current requirements and wiring harness requirements, and corrected other description errors.	
11	1.24	20260409	新增 PHU08L，删除防护等级，修正部分描述 Add PHU08L, remove the protection rating, and revise part of the description.	

安全警告与重要须知

Safety Warnings and Important Notes

在进行本产品的安装、操作、维护或检查之前，请务必仔细阅读本用户手册及所有附录。错误操作关节模组可能导致人员受伤或财产损失。因此，操作人员必须确保已充分阅读并理解本手册全部内容，严格遵守其中所列的所有安全注意事项。

Before installing, operating, maintaining, or inspecting this product, please carefully read this user manual and all appendices. Incorrect operation of the joint module may lead to personal injury or property damage. Therefore, operators must ensure that they have fully read and understood all content in this manual and strictly adhere to all safety precautions listed.

下文所示标志用于定义本手册中所涉及的危险等级及相关说明，请在使用前予以关注。

The following symbols are used to define the hazard levels and related descriptions involved in this manual. Please pay attention to them before use.

■ 警告标志Warning Symbols

	**警告:** 这指的是即将引发危险的用电情况，如果不避免，可导致人员伤害或设备严重损坏。 This refers to an imminent electrical hazard that, if not avoided, could lead to personal injury or serious equipment damage.
	**警告:** 这指的是即将引发危险的情况，如果不避免，可导致人员伤害或设备严重损坏。 This refers to a potentially dangerous situation that, if not avoided, could lead to personal injury or serious equipment damage.
	**警告:** 这指的是可能引发危险的热表面，如果接触了，可造成人员伤害。 This refers to potentially dangerous hot surfaces that, if touched, could cause personal injury.

■ **请务必遵守以下事项，以避免造成模组损坏**

Please strictly follow the following precautions to avoid damage to the module:

警告标志 Warning Sign	警告说明 Warning Explanation	详细解释 Detailed Explanation
	1、请勿在带电时插拔电源线 Do not plug or unplug the power cord when powered on.	请确认在电源指示灯熄灭后，再进行接线和检查作业。 Ensure that the power indicator light is off before performing wiring and inspection.
	2、请勿在带电时插拔 UART 通讯线 Do not plug or unplug the UART communication line when powered on.	由于主站控制器与关节两端的GND在没接线时电平可能不一样，在接通的瞬间，两端GND之间的压差会导致UART接口损坏。 Since the GND between the main controller and the joint may have different voltage levels when not connected, the voltage difference between the GNDs at the moment of connection can damage the UART interface.
	3、请正确、牢固地连接电源端子 Ensure proper and secure connection of the power terminals.	确保电源端子牢固连接，避免因接触不良而造成电源问题。 Ensure that the power terminals are securely connected to avoid power issues caused by poor contact.

	4、注意外壳接地防护 Pay attention to grounding protection of the shell.	关节外壳如不进行合理的接地处理，可能导致外壳积累电荷而带电，应对外壳进行合理的接地处理。 If the shell is not properly grounded, it may accumulate charge and become energized. Proper grounding of the shell is required.
	5、请勿超过允许的电压和电流值 Do not exceed the allowed voltage and current values.	请根据所选产品规格确认连接条件，如果超过允许的电压值或电流值，则会损坏电机，比如电机出现过热的情况。 Verify the connection conditions according to the selected product specifications. Exceeding the allowed voltage or current can damage the motor, such as causing overheating.
	1、设置适当的保护限制 Set appropriate protection limits.	在操作本产品的力矩模式时，若上位控制器下发目标指令不正确，可能导致意外飞车情况。请注意在上位控制器和本产品驱动器设定适当的保护限制，包括位置限制、速度限制和电流限制，以确保设备和人身安全。 When operating in torque mode, if the upper controller sends incorrect target instructions, it may cause an

		unexpected runaway situation. Please ensure that appropriate protection limits, such as position limits, speed limits, and current limits, are set on both the upper controller and the product driver to ensure device and personal safety.
	2、进行空载试运行 Perform a no-load test run.	在试运行时，为防止意外事故的发生，请对关节模组进行空载（不与传动轴连接的状态）试运行，否则可能会导致受伤。 During the test run, to avoid accidents, please perform the test with the joint module unloaded (not connected to the drive shaft). Otherwise, it may result in injury.
	3、负载请勿超出容许转矩/弯矩 Do not exceed the allowed torque/bending moment.	施加转矩请不要超出瞬间容许最大转矩。否则可能会出现拧紧部螺丝松动、产生晃动、破坏等，导致产品故障。 Do not apply torque that exceeds the instantaneous maximum allowed torque. Otherwise, screws may loosen, cause vibration, or damage, leading to product failure.
	4、请勿拆解/更换产品组合 Do not disassemble/replace product	本产品为高精密设备，必须由专业人员进行安装、调试，严禁对组合

	combinations.	产品实施拆解、重新组装。非正常使用导致的产品异常将失去保修权利。 This product is a high-precision device and must be installed and debugged by professionals. Disassembling or reassembling the combined product is strictly prohibited. Any abnormal product behavior due to improper use will void the warranty.
	5、请在规定环境下使用 Use within the specified environment.	①工作温度：-20~60℃，力控系列（-10~60℃）；②使用与保存湿度：10%~90%RH；③无粉尘、金属粉、腐蚀性气体、易燃气体、油雾等。 ① Operating temperature: -20~60℃，Force Control Sensor Series (-10~60℃); ② Storage and operating humidity: 10%~90%RH; ③No dust, metal powder, corrosive gases, flammable gases, or oil mist.
	6、请小心取用及运输产品 Handle and transport the product with care.	①请勿使用坚硬物品敲击；②请勿摔落本产品；③运输时保持原包装，避免挤压碰撞。 ① Do not strike with hard objects; ② Do not drop the product; ③ Keep the original packaging during

		transportation to avoid pressure and collisions.
	7、请按要求使用力控关节 Please use the force-controlled joint as required	力控关节在输出端集成力控传感器，若超过其所能承受的轴向与径向力，会造成力矩传感器损坏。 The force-controlled joint integrates a force control sensor at the output end. If the axial and radial forces it can withstand are exceeded, it may cause damage to the torque sensor.
	1、当心高温表面 Caution: High Temperature Surface	电机在运行过程中，根据它们的防护类别，表面可能会非常烫。表面温度会超过 85℃，当心轻度烧伤。测量温度并等待，直到电机冷却到40℃以下再去触碰。 The motor surfaces may become very hot during operation, depending on their protection category. The surface temperature can exceed 85°C, which could cause mild burns. Measure the temperature and wait until the motor cools down below 40°C before touching.
	2、悬空空载运行 No-load operation in suspension	关节模组出厂时会通过严格的功耗测试。空载运行也会因摩擦，搅油损失保持制动器功耗等原因产

		生热量，模组的紧凑设计和光滑表面使其在未安装(如放置桌面)环境下不足以散发自身产生的热量而使表面温度上升，可能会触发温度保护并请注意高温烫伤风险。 The joint module undergoes strict power consumption testing during production. Even in no-load operation, heat is generated due to friction, oil loss, and the power consumed by the brake. The compact design and smooth surface of the module make it difficult to dissipate the generated heat in an uninstalled environment (e.g., placed on a table), causing the surface temperature to rise and potentially triggering temperature protection. Please be aware of the risk of burns from high temperatures.

目录 Table of Contents

一、产品概述

Product Overview

1.1 产品简介与特点

Product Introduction and Features

PHU 系列关节是意优科技集多年服务经验与技术积累，推出的新一代谐波关节模组。本系列产品基于自主开发的伺服驱动器与绝对值编码器技术进行优化升级，致力于为客户提供更简单、高效、安全的机器人解决方案。

该系列涵盖 PHU08L、PHU11L/H、PHU14H、PHU17H、PHU20H、PHU25H、PHU32H、PHU40H 等多款型号，具有结构紧凑、设计精巧的特点。在有限空间内高度集成伺服驱动器、电机端与输出端双绝对值编码器、无框力矩电机、制动器及精密谐波减速机等关键组件，大幅减少客户在器件选型、结构设计、采购与组装等方面的人力与时间投入，显著提升开发效率。

力控传感器仅 PHU14H、PHU17H、PHU20H、PHU25H 型号支持。

The PHU series of joints are a new generation of harmonic joint modules launched by EYou Technology, integrating years of service experience and technical expertise. This product series has been optimized and upgraded based on independently developed servo drives and absolute encoder technologies, aiming to provide customers with simpler, more efficient, and safer robotic solutions.

This series includes multiple models such as PHU08L, PHU11L/H, PHU14, PHU17, PHU20, PHU25, PHU32, and PHU40, and features a compact structure and delicate design. Key components such as the servo driver, dual absolute encoders (motor end and output end), frameless torque motor, brake, and precision harmonic reducer are highly integrated in a limited space, significantly reducing customers' labor and time investments in component selection, structural design, procurement, and assembly, thereby improving development efficiency.

The force control sensor is only supported by the PHU14H, PHU17H, PHU20H, and PHU25H models.

图 1-1 关节外形示意图

Figure 1-1 Schematic Diagram of Joint Shape

1.2 型号说明与命名规则

Model Explanation and Naming Rules

图 1-2 关节型号说明

Figure 1-2 Joint Model Explanation

表 1-1 产品型号详细说明

Table 1-1 Detailed Product Model Explanation

参数 Parameter	基本说明 Basic Explanation
PHU	谐波关节升级款 Upgraded Harmonic Actuator
11H	谐波减速器型号。注*： L、H 分别代表低扭矩与高扭矩减速器。 Harmonic Reducer Model. Note*:L、H represent low-torque and high-torque reducers respectively.
60	关节外径，单位 mm。 Outer diameter of the joint, in mm.
B	B：带制动器，N：不带制动器。注*：两款的尺寸一致，重量有差异。 B: With brake, N: Without brake. Note*: Both versions have the same dimensions but differ in weight.
101	减速器减速比 101，即内部电机旋转 101 圈，关节端旋转一圈。 Reducer reduction ratio 101, meaning the internal motor rotates 101 revolutions for one revolution of the actuator output end.
E	通信协议 E：EtherCAT，C：CANopen Communication Protocol E: EtherCAT, C: CANopen
F	指力控传感器版本。 Force-sensing control sensor version.

1.3 技术规格与性能参数表

Technical Specifications and Performance Parameter Table

表 1-2 关节参数表 A

Table 1-2 Joint Parameter Table A

	PHU系列关节规格书A PHU Series Joint Specification A								
	序号 No.	1	2	3	4	5	6	7	8
	模组型号 Model	PHU-08L-40		PHU-11L-52		PHU-11H-52		PHU-11H-60	
结构 Structure	减速器方式 Reducer Type	谐波 Harmonic		谐波 Harmonic		谐波 Harmonic		谐波 Harmonic	
	内部电机额定输出功率（W） Internal Motor Rated Output Power (W)	17		46		46		46	
	内部电机额定扭矩（N.m） Internal Motor Rated Torque (N.m)	0.04		0.11		0.11		0.11	
	内部电机峰值扭矩（N.m） Internal Motor Peak Torque (N.m)	0.12		0.33		0.33		0.33	
	减速比 Reduction Ratio	51	101	51	101	51	101	51	101
	关节额定输出功率（W）-寿命约束 Joint Rated Output Power (W)- Life Constraint	5	3	10	6	14	10	14	10
	输入2000rpm时额定扭矩（N.m） Rated Torque at Input 2000 rpm (N.m)	1.25	1.65	2.4	3	3.5	5	3.5	5
	输出端平均负载扭矩（N.m） Average Load Torque at Output (N.m)	1.6	2.3	3.5	5.5	5.5	8.9	5.5	8.9
	输出端启停峰值扭矩（N.m） Stop Peak Torque at Output (N.m)	2.3	3.3	5.8	6.5	8.3	11	8.3	11
	输出端瞬间容许外部最大扭矩（N.m） Instantaneous Maximum Allowable External Torque at Output (N.m)	4.6	6.3	11.5	15	17	25	17	25
	背隙（Arcsec） Backlash (Arcsec)	≤40	≤40	≤30	≤30	≤30	≤30	≤30	≤30
	输出端额定转速（rpm）-寿命约束 Rated Speed at Output (rpm) - Life Constraint	39	20	39	20	39	20	39	20
	输出端峰值转速（rpm） Output Peak Speed (rpm)	100	50	100	50	100	50	100	50
	是否抱闸（B/N） With Brake? (B/N)	N		B/N		B/N		B/N	
	尺寸（外径x轴向）（mm） Dimensions - Ø Dia. x Axial (mm)	Φ40 x 56		Φ52 x 65		Φ52 x 70		Φ60 x 68.5	
	重量（Kg）（B/N） Weight (Kg) (B/N)	0.196		0.39 / 0.35		0.44 / 0.40		0.59 / 0.54	
	中空直径（mm） Hollow Diameter (mm)	4.4		8		8		8	
	壳体外观（工艺颜色） Housing Appearance (Finish/Color)	黑色阳极氧化（不喷沙） Black Anodized (Non-Sandblasting)							
	减速器外观（工艺颜色） Reducer Appearance (Finish/Color)	外圈发黑 Outer Ring Blackened							
	输出端径向（动/静）允许载荷（N） Output Radial Load (N) (Dynamic/Static)	100		130		130		130	
	输出端轴向（动/静）允许载荷（N） Output Axial Load (N) (Dynamic/Static)	100		130		130		130	
	输出端负载弯矩（N.m） Output Load Moment (N.m)	13.5		16		16		16	
	减速器扭转刚性（N.m/rad） Reducer Torsional Stiffness (N.m/rad)	560	800	2250	2430	2250	2430	2250	2430
	减速器输入端转动惯量（Kg.mm2） Reducer Input Moment of Inertia (Kg.mm²)	1.97	1.97	11.236	11.236	13.248	13.248	16.88	16.88
	折算关节输出端转动惯量（Kg.mm2） Converted Joint Output Moment of Inertia (Kg.mm²)	5124	20096	29225	114618	34458	135143	43905	172193
硬件 Hardware	额定工作电压（VDC） Rated Operating Voltage (VDC)	48							
	工作电压范围（VDC） Operating Voltage Range (VDC)	24~48							
	驱动器 Driver	内置 Built-in							
	编码器/位数（bit） Encoder / Resolution (bit)	双绝对 / 电机端17Bit&减速器端19Bit Dual Absolute / 17-Bit at Motor End & 19-Bit at Reducer End		双绝对 / 电机端19Bit&减速器端19Bit Dual Absolute / 19-Bit at Motor End & 19-Bit at Reducer End					
	驱动器接口（端子型号见说明） Driver Interface	Power (DC in DC out)、CAN FD (In, Out)、EtherCAT (In, Out)、STO（1、2）							
软件 Software	通讯协议（详见通信说明） Communication Protocol	EtherCAT / CANopen							
	控制模式 Control Modes	轮廓位置模式(PP)、轮廓速度模式(PV)、轮廓转矩模式(PT)、循环同步位置模式(CSP)、循环同步速度模式(CSV)等 Profile Position (PP), Profile Velocity (PV), Profile Torque (PT), Cyclic Synchronous Position (CSP), Cyclic Synchronous Velocity (CSV), etc.							
可靠性与寿命 Reliability & Lifetime	工作环境温度（°C） Operating Ambient Temperature (°C)	-20~60							
	外壳工作温升（°C） Housing Operating Temperature Rise (°C)	≤60							
	噪音（dB） Noise (dB)	≤60							
	寿命（h） Lifetime (h)	>8000							

表 1-3 关节参数表 B

Table 1-3 Joint Parameter Table B

PHU系列关节规格书B
PHU Series Joint Specification B

	序号 No.	9	10	11	12	13	14	15	16	17	18
	模组型号 Model	PHU-14H-70		PHU-17H-80		PHU-20H-90			PHU-25H-110		
结构 Structure	减速器方式 Reducer Type	谐波 Harmonic		谐波 Harmonic		谐波 Harmonic			谐波 Harmonic		
	内部电机额定输出功率（W） Internal Motor Rated Output Power (W)	105		251		267			586		
	内部电机额定扭矩（N.m） Internal Motor Rated Torque (N.m)	0.25		0.6		0.85			1.6		
	内部电机峰值扭矩（N.m） Internal Motor Peak Torque (N.m)	0.75		1.8		2.55			4.8		
	减速比 Reduction Ratio	51	101	51	101	81	101	121	81	101	121
	关节额定输出功率（W）-寿命约束 Joint Rated Output Power (W)- Life Constraint	27	20	81	62	109	104	87	202	174	145
	输入2000rpm时额定扭矩（N.m） Rated Torque at Input 2000 rpm (N.m)	6.6	9.6	19.8	30	42	50	50	78	84	84
	输出端平均负载扭矩（N.m） Average Load Torque at Output (N.m)	8.6	13.5	32	49	58	61	61	107	133	133
	输出端启停峰值扭矩（N.m） Stop Peak Torque at Output (N.m)	23	34	42	66	91	102	108	169	194	207
	输出端瞬间容许外部最大扭矩（N.m） Instantaneous Maximum Allowable External Torque at Output (N.m)	43	66	86	134	158	182	182	315	351	376
	背隙（Arcsec） Backlash (Arcsec)	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20
	输出端额定转速（rpm）-寿命约束 Rated Speed at Output (rpm) - Life Constraint	39	20	39	20	25	20	17	25	20	17
	输出端峰值转速（rpm） Output Peak Speed (rpm)	90	45	88	44	38	31	25	44	36	30
	是否抱闸（B/N） With Brake? (B/N)	B/N		B/N		B/N			B/N		
	尺寸（外径x轴向）（mm） Dimensions - Ø Dia. x Axial (mm)	Φ70 x 75.4		Φ80 x 78		Φ90 x 85.8			Φ110 x 94		
	重量（Kg）（B/N） Weight (Kg) (B/N)	0.89 / 0.81		1.25 / 1.14		1.74 / 1.58			2.54 / 2.35		
	中空直径（mm） Hollow Diameter (mm)	10.8		10.8		13			13		
	壳体外观（工艺颜色） Housing Appearance (Finish/Color)	黑色阳极氧化（不喷沙） Black Anodized (Non-Sandblasting)									
	减速器外观（工艺颜色） Reducer Appearance (Finish/Color)	外圈发黑 Outer Ring Blackened									
	输出端径向（动/静）允许载荷（N） Output Radial Load (N) (Dynamic/Static)	270		400		650			900		
	输出端轴向（动/静）允许载荷（N） Output Axial Load (N) (Dynamic/Static)	270		400		650			900		
	输出端负载弯矩（N.m） Output Load Moment (N.m)	41		72		140			243		
	减速器扭转刚性（N.m/rad） Reducer Torsional Stiffness (N.m/rad)	4600	6000	10700	13000	23000			46000		
	减速器输入端转动惯量（Kg.mm2） Reducer Input Moment of Inertia (Kg.mm²)	36.733	36.733	85.734	85.734	137.378	137.378	137.378	251.975	251.975	251.975
	折算关节输出端转动惯量（Kg.mm2） Converted Joint Output Moment of Inertia (Kg.mm²)	95543	374713	222994	874573	901337	1401393	2011351	1653208	2570397	3689166
硬件 Hardware	额定工作电压（VDC） Rated Operating Voltage (VDC)	48									
	工作电压范围（VDC） Operating Voltage Range (VDC)	24~48									
	驱动器 Driver	内置									
	编码器/位数（bit） Encoder / Resolution (bit)	双绝对 / 电机端19Bit&减速器端19Bit Dual Absolute / 19-Bit at Motor End & 19-Bit at Reducer End									
	驱动器接口（端子型号见说明） Driver Interface	Power (DC in DC out)、CAN FD (In, Out)、 EtherCAT (In, Out)、STO（1、2）									
软件 Software	通讯协议（详见通信说明） Communication Protocol	EtherCAT / CANopen									
	控制模式 Control Modes	轮廓位置模式(PP)、轮廓速度模式(PV)、轮廓转矩模式(PT)、循环同步位置模式(CSP)、循环同步速度模式(CSV)等 Profile Position (PP), Profile Velocity (PV), Profile Torque (PT), Cyclic Synchronous Position (CSP), Cyclic Synchronous Velocity (CSV), etc.									
可靠性与寿命 Reliability & Lifetime	工作环境温度（°C） Operating Ambient Temperature (°C)	-20~60									
	外壳工作温升（°C） Housing Operating Temperature Rise (°C)	≤60									
	噪音（dB） Noise (dB)	≤60									
	寿命（h） Lifetime (h)	>10000									

表 1-4 关节参数表 C

Table 1-4 Joint Parameter Table C

EYOU意优科技		PHU系列关节规格书C PHU Series Joint Specification C							
序号		19	20	21	22	23	24	25	26
模组型号		PHU-32H-142				PHU-40H-170			
结构	减速器方式	谐波 Harmonic				谐波 Harmonic			
	内部电机额定输出功率（W）	603				905			
	内部电机额定扭矩（N.m）	2.4				3.6			
	内部电机峰值扭矩（N.m）	9.6				14.4			
	减速比	81	101	121	161	81	101	121	161
	关节额定输出功率（W）-寿命约束	377	350	293	220	659	664	554	416
	输入2000rpm时额定扭矩（N.m）	146	169	169	169	255	320	320	320
	输出端平均负载扭矩（N.m）	206	267	267	267	351	460	557	557
	输出端启停峰值扭矩（N.m）	376	411	436	459	641	702	762	800
	输出端瞬间容许外部最大扭矩（N.m）	702	800	848	848	1210	1334	1458	1458
	背隙（Arcsec）	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20
	输出端额定转速（rpm）-寿命约束	25	20	17	12	25	20	17	12
	输出端峰值转速（rpm）	30	24	20	15	30	24	20	15
	是否抱闸（B/N）	B/N				B/N			
	尺寸（外径x轴向）（mm）	Φ142 x 144				Φ170 x 149			
	重量（Kg）（B/N）	6.31 / 5.83				9.34 / 8.70			
	中空直径（mm）	22				23			
	壳体外观（工艺颜色）	黑色阳极氧化（不喷沙） Black Anodized (Non-Sandblasting)							
	减速器外观（工艺颜色）	外圈发黑 Outer Ring Blackened							
	输出端径向（动/静）允许载荷（N）	1350				2000			
	输出端轴向（动/静）允许载荷（N）	1350				2000			
	输出端负载弯矩（N.m）	460				600			
	减速器扭转刚性（N.m/rad）	99000				186000			
	减速器输入端转动惯量（Kg.mm2）(B)	1171.326	1171.326	1171.326	1171.326	1876.452	1876.452	1876.452	1876.452
	折算关节输出端转动惯量（Kg.mm2）(B)	7685070	11948697	17149384	30361941	12311402	19141687	27473134	48639512
硬件	额定工作电压（VDC）	48							
	工作电压范围（VDC）	24~48							
	驱动器	内置 Built-in							
	编码器/位数（bit）	双绝对 / 电机端17Bit&减速器端19Bit Dual Absolute / 17-Bit at Motor End & 19-Bit at Reducer End							
	驱动器接口（端子型号见说明）	Power (DC in DC out)、CAN FD (In, Out)、 EtherCAT (In, Out)、STO（1、2）							
软件	通讯协议（详见通信说明）	EtherCAT / CANopen							
	控制模式	轮廓位置模式(PP)、轮廓速度模式(PV)、轮廓转矩模式(PT)、循环同步位置模式(CSP)、循环同步速度模式(CSV)等 Profile Position (PP), Profile Velocity (PV), Profile Torque (PT), Cyclic Synchronous Position (CSP), Cyclic Synchronous Velocity (CSV), etc.							
可靠性与寿命	工作环境温度（°C）	-20~60							
	外壳工作温升（°C）	≤60				≤80			
	噪音（dB）	≤60							
	寿命（h）	>10000							

表 1-5 关节参数表 D

Table 1-5 Joint Parameter Table D

PHU-F系列关节规格书
PHU-F Series Joint Specification

	序号 No.	1	2	3	4	5	6	7	8	9	10
	模组型号 Model	PHU-14H-70-F		PHU-17H-80-F		PHU-20H-90-F			PHU-25H-110-F		
结构 Structure	减速器方式 Reducer Type	谐波 Harmonic		谐波 Harmonic		谐波 Harmonic			谐波 Harmonic		
	内部电机额定输出功率（W） Internal Motor Rated Output Power (W)	105		251		267			586		
	内部电机额定扭矩（N.m） Internal Motor Rated Torque (N.m)	0.25		0.6		0.85			1.6		
	内部电机峰值扭矩（N.m） Internal Motor Peak Torque (N.m)	0.75		1.8		2.55			4.8		
	减速比 Reduction Ratio	50	100	50	100	80	100	120	80	100	120
	关节额定输出功率（W）-寿命约束 Joint Rated Output Power (W)- Life Constraint	28	20	83	63	110	105	87	204	176	147
	输入2000rpm时额定扭矩（N.m） Rated Torque at Input 2000 rpm (N.m)	6.6	9.6	19.8	30	42	50	50	78	84	84
	输出端平均负载扭矩（N.m） Average Load Torque at Output (N.m)	8.6	13.5	32	49	58	61	61	107	133	133
	输出端启停峰值扭矩（N.m） Stop Peak Torque at Output (N.m)	23	34	42	66	91	102	108	169	194	207
	输出端瞬间容许外部最大扭矩（N.m） Instantaneous Maximum Allowable External Torque at Output (N.m)	43	66	86	134	158	182	182	315	351	376
	背隙（Arcsec） Backlash (Arcsec)	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20	≤20
	输出端额定转速（rpm）-寿命约束 Rated Speed at Output (rpm) - Life Constraint	40	20	40	20	25	20	17	25	20	17
	输出端峰值转速（rpm） Output Peak Speed (rpm)	90	45	88	44	38	31	25	44	36	30
	是否抱闸（B/N） With Brake? (B/N)	B/N		B/N		B/N			B/N		
	尺寸（外径x轴向）（mm） Dimensions - Ø Dia. x Axial (mm)	Φ70 x 84		Φ80 x 86		Φ90 x 94			Φ110 x 106		
	重量（Kg）（B/N） Weight (Kg) (B/N)	0.98 / 0.9		1.35 / 1.24		1.9 / 1.74			2.97 / 2.78		
	中空直径（mm） Hollow Diameter (mm)	9		10		13			13		
	壳体外观（工艺颜色） Housing Appearance (Finish/Color)	黑色阳极氧化（不喷沙） Black Anodized (Non-Sandblasting)									
	减速器外观（工艺颜色） Reducer Appearance (Finish/Color)	外圈发黑 Outer Ring Blackened									
	输出端径向（动/静）允许载荷（N） Output Radial Load (N) (Dynamic/Static)	270		400		650			900		
	输出端轴向（动/静）允许载荷（N） Output Axial Load (N) (Dynamic/Static)	270		400		650			900		
	输出端负载弯矩（N.m） Output Load Moment (N.m)	41		72		140			243		
	减速器扭转刚性（N.m/rad） Reducer Torsional Stiffness (N.m/rad)	4600	6000	10700	13000	23000			46000		
	减速器输入端转动惯量（Kg.mm2） Reducer Input Moment of Inertia (Kg.mm²)	36.733	36.733	85.734	85.734	137.378	137.378	137.378	251.975	251.975	251.975
	折算关节输出端转动惯量（Kg.mm2） Converted Joint Output Moment of Inertia (Kg.mm²)	91833	367330	214335	857340	879219	1373780	1978243	1612640	2519750	3628440
力控 Force Control	力矩传感器量程（N.m） Torque Sensor Range (N.m)	35		70		110			220		
	力矩传感器扭矩精度（%F.S） Torque Sensor Accuracy (%F.S.)	≤±0.5									
	力矩传感器分辨率（%F.S） Torque Sensor Resolution (%F.S.)	0.1									
	力矩传感器极限过载（%F.S） Torque Sensor Overload Limit (%F.S.)	300									
	力矩传感器采样频率（KHZ）-数字量 Torque Sensor Sampling Frequency (kHz) – Digital	8									
	力控闭环带宽（HZ） Force Control Closed-loop Bandwidth (Hz)	30									
	阻抗控制频率（KHZ） Impedance Control Frequency (kHz)	1									
硬件 Hardware	额定工作电压（VDC） Rated Operating Voltage (VDC)	48									
	工作电压范围（VDC） Operating Voltage Range (VDC)	24~48									
	驱动器 Driver	内置 Built-in									
	编码器/位数（bit） Encoder / Resolution (bit)	双绝对 / 电机端19Bit&减速器端19Bit Dual Absolute / 19-Bit at Motor End & 19-Bit at Reducer End									
	驱动器接口（端子型号见说明） Driver Interface	Power (DC in DC out)、CAN FD (In, Out)、EtherCAT (In, Out)、STO（1、2）									
软件 Software	通讯协议（详见通信说明） Communication Protocol	EtherCAT / CANopen									
	控制模式 Control Modes	轮廓位置模式(PP)、轮廓速度模式(PV)、轮廓转矩模式(PT)、循环同步位置模式(CSP)、力控模式(CSF)等Profile Position Mode (PP), Profile Velocity Mode (PV), Profile Torque Mode (PT), Cyclic Synchronous Position Mode (CSP), Force Control Mode (CSF), etc.									
可靠性与寿命 Reliability & Lifetime	工作环境温度（°C） Operating Ambient Temperature (°C)	-10~60									
	外壳工作温升（°C） Housing Operating Temperature Rise (°C)	≤60									
	噪音（dB） Noise (dB)	≤60									
	寿命（h） Lifetime (h)	>10000									

1.4 规格参数名词解释

Specification Parameter Terminology Explanation

内部电机额定输功出率：在特定工况条件下，谐波减速器内部电机输出轴能够持续传递且不致使内部组件温升、疲劳寿命或运动精度超过允许限值的最大机械功率，该工况条件包括额定输出扭矩与转速、规定热环境以及充分润滑状态。

Rated power output rate of the internal motor: Under specific operating conditions, the maximum mechanical power that the output shaft of the harmonic reducer's internal motor can continuously transmit without causing the internal components' temperature rise, fatigue life, or motion accuracy to exceed permissible limits. These conditions include rated output torque and speed, specified thermal environment, and adequate lubrication.

内部电机额定扭矩：是谐波减速器内部电机在标准工况下可长期、连续输出的最大稳态扭矩。

The rated torque of the internal motor refers to the maximum steady-state torque that the harmonic reducer's internal motor can output continuously under standard operating conditions.

内部电机峰值扭矩：是谐波减速器内部电机在极短时间内可承受或输出的最大瞬时扭矩。

Internal motor peak torque: refers to the maximum instantaneous torque that the harmonic reducer's internal motor can withstand or output in an extremely short period.

减速比：指输入轴转速与输出轴转速之比，由弹性齿轮与刚性齿轮的齿轮数决定。本公司 PHU 系列非力控版本是帽型，PHU-F 系列力控版本是杯型。在下列公式中弹性齿轮的齿轮数为Zf，刚性齿轮的齿轮数为Zc，额定表的减速比数值用R_1，R_2表示。

Reduction ratio: The ratio of input shaft speed to output shaft speed, determined by the gear ratios of elastic and rigid gears. Our PHU series non-force-controlled versions are cap-shaped, while the PHU-F series force-controlled versions are cup-shaped.In the following formula, the number of gears for the elastic gear is Zf, the number of gears for the rigid gear is Zc, and the reduction ratios in the rated table are denoted as R_1 and R_2.

下面的公式为杯型的减速比公式：

The following formula is the cup-type reduction ratio formula:

$$\mathrm{i}_1 = \frac{1}{\mathrm{R}_1} = \frac{Z_f - Z_c}{Z_f}$$

下面的公式为帽型的减速比公式：

The following formula is the deceleration ratio formula for the cap type.

$$i_2 = \frac{1}{R_2} = \frac{Z_c - Z_f}{Z_c}$$

注：杯型与帽型输入均为波发生器，但是输出轮与固定轮是相反的。杯型输出轮为弹性齿轮，固定轮为刚性齿轮；帽型输出轮为刚性齿轮，固定轮为弹性齿轮。

Note: Both cup-type and cap-type inputs are wave generators, but the output wheels and fixed wheels are opposite. The cup-shaped output wheel is a flexible gear, and the fixed wheel is a rigid gear; the cap-shaped output wheel is a rigid gear, and the fixed wheel is a flexible gear.

关节额定输出功率-寿命约束：指在减速器输入 2000rpm 时且满足温升与寿命要求时的输出机械功率。

Joint rated output power-lifetime constraint: refers to the output mechanical power when the reducer input is 2000rpm and the temperature rise and lifetime requirements are met.

输入 2000rpm 时额定扭矩：在波发生器输入轴以 2000rpm 的恒定转速旋转，且谐波减速器处于标准工作温度、充分润滑及规定安装精度的条件下，其输出轴能够长期、连续、稳定输出的最大机械扭矩，且在此扭矩下运行时，减速器内部关键部件的应力、温升及磨损率均不超出设计允许的安全极限。

Rated torque at 2000 rpm: When the wave generator input shaft rotates at a constant speed of 2000 rpm, and the harmonic reducer operates under standard temperature, adequate lubrication, and specified installation precision, the output shaft can deliver a maximum mechanical torque for prolonged, continuous, and stable operation. Under this torque, the stress, temperature rise, and wear rates of the reducer's critical components remain within the design-specified safety limits.

输出端的平均负载扭矩(平均负载转矩的容许最大值)：当负载转矩或输入转速变化时，需要另行计算负载转矩的平均值。规格书中的数值，即为此平均负载转矩的容许值。若平均负载转矩超过规格书中的数值，将因发热导致润滑剂提前劣化，或加剧齿轮磨损。请务必注意。

The rated average load torque (maximum allowable value) at the output end: When load torque or input speed changes, the average load torque must be recalculated. The specification value represents the permissible average load torque. Exceeding this value may cause premature lubricant degradation due to overheating or accelerated gear wear. This must be strictly observed.

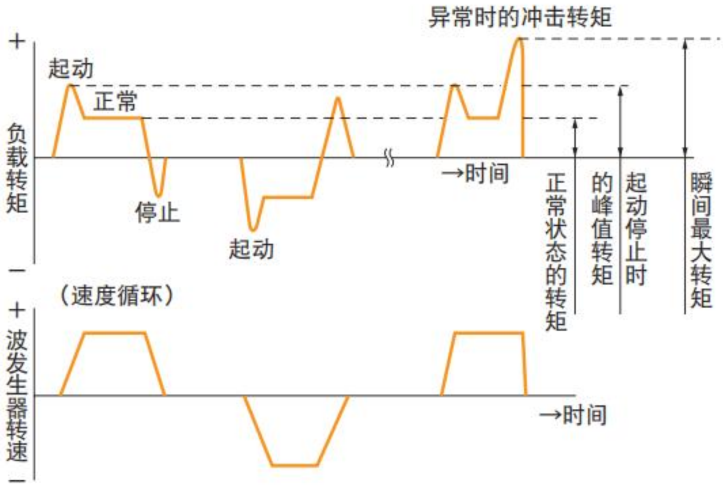


图 1-3 负载转矩模式示例

Figure 1-3 Load Torque Mode Example

输出端的启停峰值扭矩(起动、停止时的容许峰值转矩)： 在启动、停止时，因负载惯性力矩的影响，输出端将承受比恒定转矩更大的负载。规格书中的数值，为此时的峰值转矩容许值。（如图 1-3 所示）

Peak torque at output during start/stop (permissible peak torque during operation): During startup and shutdown, the output end will experience a load exceeding the constant torque due to load inertia. The value specified in the technical data sheet represents the permissible peak torque under these conditions.

输出端瞬时容许外部最大扭矩(瞬时容许最大转矩)： 除了通常负载转矩、启动或停止时的负载转矩外，也可能出现来自外部无法预期的冲击转矩。但冲击转矩的最大值不得超过规格书中的瞬时最大转矩。（如图 1-3 所示）

Maximum allowable external torque at the output end (transient maximum allowable torque): In addition to standard load torque and torque during startup or shutdown, unexpected external shock torque may occur. However, the maximum value of shock torque must not exceed the specified transient maximum torque. (As shown in the figure below)

背隙：迟滞损耗主要由内部摩擦产生，转矩极小时则几乎不存在，线图上仅显示微小的背隙。这一量值即表现为背隙量。关节处于制动条件下，输出端在±3%额定转矩作用下输出端的转角值之差。[来源:GB/T 35089—2018, 3.7, 有修改]

Back clearance: Hysteresis loss is primarily caused by internal friction and is virtually non-existent at minimal torque, with only a negligible back clearance displayed on the line graph. This quantity is referred to as the back clearance value. When the joint is under braking conditions, it represents the difference in output torque angle values at ±3% of the rated torque.

输出端额定转速-寿命约束：输入端 2000rpm 经减速比换算的转速，即满足输出额定扭矩时的温升与寿命要求的转速。

Output rated speed-lifetime constraint: The input speed of 2000 rpm converted through a reduction ratio, which satisfies the temperature rise and lifespan requirements for achieving the rated output torque.

输出端峰值转速：指谐波减速器输出轴在极短时间内可达到的最高允许转速，超过此转速可能导致零件过热、润滑失效或结构共振，进而损害传动精度与使用寿命。

Peak output speed: refers to the maximum allowable rotational speed that the output shaft of a harmonic reducer can reach in an extremely short time. Exceeding this speed may cause parts to overheat, lubrication failure, or structural resonance, thereby damaging transmission accuracy and service life.

抱闸：是产品内部一种通过机械方式强制锁止旋转运动的制动装置，产品分为携带抱闸（B）和不携带抱闸(N)。

Brake: A mechanical braking device that forcibly locks the rotational movement of a product. The product is categorized as either with brake (B) or without brake (N).

重量:产品净重，分为携带抱闸和不携带抱闸两种质量。（误差在±20g 以内）

Weight: Net weight of the product, divided into two categories based on whether the brake is carried or not. (Error margin within ±20g)

中空直径：实际产品中中空部分圆柱横截面的直径。

Hollow diameter: The diameter of the cylindrical cross-section of the hollow part in the actual product.

输出端径向（动/静）允许载荷：是指谐波减速器在静止或运动过程中在径向方向上所能承受的最大外力的允许值。

The radial (dynamic/static) allowable load at the output end refers to the maximum permissible external force that the harmonic reducer can withstand in the radial direction during both stationary and dynamic operation.

输出端轴向（动/静）允许载荷：是指谐波减速器在静止或运动过程中在其输出轴方向上所能承受的最大外力的允许值。

The allowable load at the output shaft (dynamic/static) refers to the maximum external force that the harmonic reducer can withstand along its output shaft direction during both stationary and dynamic operation.

输出端负载弯矩：是指在谐波减速器的输出法兰或输出轴上，由外部负载作用产生的弯曲力矩的允许值。

Output load bending moment: The permissible bending moment value induced external loads on the output flange or output shaft of a harmonic reducer.

最大负载静力矩(M_{max})的计算方法如下：

The calculation method for the maximum load static moment (M_{max}) is as follows:

$$M_{\max} = Fr_{max}(Lr + R) + Fa_{\max} \cdot La$$

请确认$M_{\max} \leq Mc$，Mc为容许静力距，这种情况下符合使用弯矩的要求。

注：Fr_{max}为最大径向负载，$Fa_{\max}$为最大轴向负载，R 为偏置量

Verify that $M_{max} \leq Mc$, where Mc is the permissible static moment, ensuring compliance with the bending moment requirements.

Note: $Fr_{max}$is the maximum radial load, $Fa_{\max}$ is the maximum axial load, and R is the offset.

减速器扭转刚性：将输入侧（波发生器）固定，向输出侧（柔轮）施加转矩后，输出轴会产生几乎与转矩成正比的扭转。

Torsional rigidity of the reducer: When the input side (wave generator) is fixed and torque is applied to the output side (flexible wheel), the output shaft will generate torsion almost proportional to the torque.

减速器输入端转动惯量：是指从减速器的输入轴（即与电机直接相连的轴）处观测，整个传动链上所有旋转部件质量的总惯性效应在该轴上的动力学等效值。

The input shaft of the reducer is directly connected to the motor. The input shaft of the reducer is the shaft where the dynamic equivalent value of the inertial effect of the mass of all rotating parts in the transmission chain is observed.

折算关节输出端转动惯量：经过减速比统一折算到关节的输出轴上的总等效转动惯量。

The total equivalent moment of inertia at the output shaft of the joint, converted uniformly to the joint output shaft through the reduction ratio.

额定工作电压：可确保产品长期安全可靠运行并实现全部额定性能的理想输入电压值，额定性能包括额定转速、峰值转速、额定转矩等。

Rated operating voltage: The ideal input voltage that ensures long-term safe and reliable operation of the product while achieving all rated performance parameters, including rated speed, peak speed, and rated torque.

工作电压范围：本产品在确保整体系统能实现基本性能、稳定控制、安全运行及符合电磁兼容标准的前提下，其主电源输入端可长期耐受并正常工作的直流电压区间。

Operating voltage range: The DC voltage range that the main power input of this product can withstand and operate normally under long-term conditions, while ensuring the system achieves basic performance, stable control, safe operation, and compliance with electromagnetic compatibility standards.

编码器/位数：本产品采用双绝对编码器，在一个闭环运动控制单元同时配置两个绝对式位置传感器，双绝对/电机端 19Bit&减速器端 19Bit 指输入与输出都是 19 位绝对式编码器，双绝对/电机端 17Bit&减速器端 19Bit 指输入 17 位与输出 19 位绝对式编码器。

Encoder/Bit Depth: This product uses dual absolute encoders, with two absolute position sensors configured in a single closed-loop motion control unit. Dual absolute/motor side 19Bit & reducer side 19Bit means both input and output are 19-bit absolute encoders, while dual absolute/motor side 17Bit & reducer side 19Bit means the input is a 17-bit and the output is a 19-bit absolute encoder.

寿命：关节模组的使用寿命取决于减速器的使用寿命,而减速器的使用寿命又取决于波发生器轴承的使用寿命。与普通滚珠轴承相同,可通过转速和负载转矩计算出来。

Service life: The service life of the joint module depends on the service life of the reducer, which in turn is determined by the service life of the wave generator bearings. Similar to conventional ball bearings, it can be calculated based on rotational speed and load torque.

实际运转条件下使用寿命(L_h)的计算公式：

The calculation formula for the service life (Lh) under actual operating conditions：

$$L_h = L_n \times \left(\frac{T_r}{T_{avg}}\right)^3 \times \left(\frac{N_r}{N_{avg}}\right)$$

注：L_h为使用寿命，L_n一般是L_{10}或L_{50}的使用寿命时间（L_{10}指产品 10%破损率的使用寿命，L_{50}指产品平均使用寿命）;T_r是减速机额定转矩;N_r是用于减速机额定扭矩测试的电机端额定测试转速，一般是 2000RPM;T_{avg}指减速器输出端的平均负载转矩;N_{avg}指减速机输入（或电机输出）平均转速。

Note: L_h denotes the service life, while $L_n$typically represents the service life at either L_{10} (10% failure rate) or L_{50} (50% failure rate); T_r is the rated torque of the reducer; N_r is the rated test speed at the motor end for torque measurement, usually 2000 RPM; T_{avg} indicates the average load torque at the reducer's output end; and N_{avg} denotes the average speed at the reducer's input (or motor output).

1.5 应用领域

Application Areas

PHU 系列凭借其高度集成、结构紧凑及高精度力控的特性，广泛应用于工业机器人、协作机器人、人型机器人、精密自动化设备（半导体封装设备、精密检测仪器）、医疗机器人（手术机器人关节、康复训练设备）及航空航天等领域，为高扭矩密度、高动态响应及精密力控要求的应用场景提供可靠的核心关节解决方案。

The PHU series, with its high integration, compact structure, and high-precision force control characteristics, is widely used in industrial robots, collaborative robots, humanoid robots, precision automation equipment (semiconductor packaging equipment, precision testing instruments), medical robots (surgical robot joints, rehabilitation training equipment), and aerospace fields. It provides reliable core joint solutions for applications requiring high torque density, high dynamic response, and precise force control.

二、结构介绍与工作原理

Structure Introduction and Working Principle

2.1 结构爆炸图

Structural Explosion Diagram

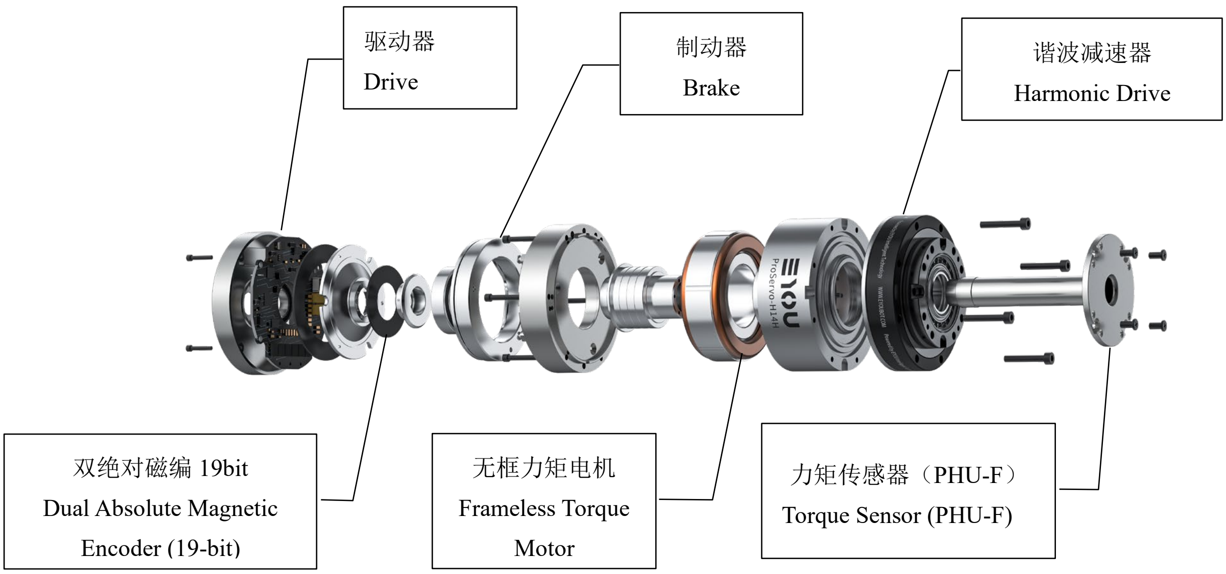


图 2-1 关节爆炸示意图

Figure 2-1 Joint Explosion Diagram

2.2 各部件功能简介

Component Function Introduction

2.2.1 驱动器

Driver

内置高性能伺服驱动器，是整套关节模组的控制核心与“智能大脑”。它高度集成于关节内部，负责接收指令、处理信息、驱动电机并确保系统安全、可靠地运行。

The built-in high-performance servo driver is the core control unit and "brain" of the entire joint module. It is highly integrated within the joint, responsible for receiving instructions, processing information, driving the motor, and ensuring the system operates safely, and reliably.

2.2.2 编码器

Encoder

关节集成双路 19 位高分辨率绝对值编码器，分别位于电机端与输出端，共同构成了实现超高运动精度的核心感知与反馈系统。该设计是本产品实现真正全闭环控制、超越传统单编码器方案的关键。

The joint integrates dual 19-bit high-resolution absolute encoders located at both the motor end and output end, forming the core sensing and feedback system that achieves ultra-high motion accuracy. This design is the key to realizing true closed-loop control, surpassing traditional single-encoder solutions.

编码器分辨率是指编码器能够检测并反馈的最小位置变化量。旋转一圈的脉冲是 2^{19} 即 524288，意味着 19 位的绝对值编码器能够将电机的每一转精确地划分为 524288 个独立的脉冲或步长。

The encoder resolution refers to the smallest position change the encoder can detect and feedback. The pulses per revolution are 2^19, or 524288, meaning that the 19-bit absolute encoder can divide each revolution of the motor into 524288 independent pulses or steps.

角度与脉冲换算：位置反馈脉冲数 / 524288 * 360°= 当前角度°

速度换算：速度反馈脉冲数 / 524288 * 60 = 转速 rpm

Angle and Pulse Conversion: Position feedback pulse count / 524288 * 360° = Current Angle°

Speed Conversion: Speed feedback pulse count / 524288 * 60 = RPM

2.2.3 制动器

Brake

内置的制动器是一个关键的安全与功能部件，其主要作用可归纳为以下几点：

The built-in brake is a key safety and functional component. Its main functions can be summarized as follows:

①静态位置保持：在机器人断电或伺服未使能（下使能）时，制动器会自动闭合（抱

闸），产生巨大的静摩擦力矩以锁住关节。

Static Position Holding: When the robot is powered off or the servo is disabled (disabled mode), the brake will automatically close (engage), generating significant static friction torque to lock the joint.

②安全制动：当系统触发急停、故障或意外断电时，制动器会立即动作，紧急刹停机构运动，作为最后一道安全屏障，防止事故发生。

Emergency Braking: When the system triggers an emergency stop, fault, or unexpected power failure, the brake will immediately engage, halting the mechanism's movement as the last line of safety defense to prevent accidents.

③供电与控制：制动器由驱动板电源供电，无需额外接线。其状态（松开/抱闸）由使能信号（上使能/下使能） 自动控制。

Power Supply and Control: The brake is powered by the driver board's power supply, requiring no additional wiring. Its state (release/engaged) is automatically controlled by the enable signal (enable/disable).

④时序要求：制动器物理动作有延迟（约 150ms）。下使能后，必须等待至少 300ms 才能重新上使能；上使能后，需等待至少 500ms（确保制动器完全释放）再发送运动指令。

Timing Requirements: The brake's physical action has a delay (approximately 150ms). After disabling, you must wait at least 300ms before re-enabling; after enabling, wait at least 500ms (to ensure the brake is fully released) before sending motion commands.

⑤使用限制：避免在高转速、重载荷工况下触发急停，以免制动器闭合造成剧烈冲击损坏设备。严禁将其作为频繁工作的动态制动器使用，它主要设计用于静态保持。避免在强磁场环境中使用，否则需做磁屏蔽。

Usage Restrictions: Avoid triggering emergency stops under high-speed or heavy-load conditions to prevent severe impact damage from the brake closing. It is strictly prohibited to use it as a dynamic brake for frequent work; it is primarily designed for static holding. Avoid using in strong magnetic field environments unless proper magnetic shielding is provided.

⑥使用寿命：拥有超过 10 万次的吸合寿命，并能承受偶发的强制拖动，可靠性高。

Service Life: The brake has over 100,000 engage cycles and can withstand occasional forced dragging, offering high reliability.

2.2.4 无框力矩电机

Frameless Torque Motor

无框力矩电机是一种高性能直驱电机，由独立的转子和定子组成，集成在关节内部。谐波减速器与无框力矩电机深度融合，构成紧凑、高精度的动力单元。其优势在于通过直驱方式提供零背隙的传动，实现了极高的定位精度、刚性及扭矩密度，同时，省去传统传动部件，大幅简化结构，使关节更轻量、紧凑、高效。

The frameless torque motor is a high-performance direct-drive motor composed of an independent rotor and stator, integrated within the joint. The harmonic reducer and frameless torque motor are deeply integrated to form a compact, high-precision power unit. Its advantages lie in providing zero-backlash transmission through direct drive, achieving extremely high positioning accuracy, rigidity, and torque density, while eliminating traditional transmission components, greatly simplifying the structure and making the joint lighter, more compact, and efficient.

2.2.5 谐波减速器

Harmonic Reducer

谐波减速器是一种利用柔性元件弹性变形来传递运动和动力的精密减速装置。其核心优势在于超高的单级传动比、极小的运动背隙、高精度定位、紧凑轻量的结构及高扭矩输出。它通过波发生器、柔轮和刚轮三者的相互作用实现运动传递，广泛用于机器人关节、航空航天等高精度需求领域，是实现精密传动和精简设计的理想解决方案。

The harmonic reducer is a precision reduction device that transmits motion and power using the elastic deformation of flexible components. Its core advantages include an extremely high single-stage transmission ratio, minimal motion backlash, high-precision positioning, a compact and lightweight structure, and high torque output. It achieves motion transmission through the interaction of the wave generator, flexible wheel, and rigid wheel, and is widely used in robotic joints, aerospace, and other high-precision fields. It is the ideal solution for achieving precise transmission and simplified design.

2.2.6 力矩传感器

Torque Sensor

力矩传感器的工作原理主要基于应变效应：当传感器弹性体受外力作用发生微小形变时，粘贴其上的应变片电阻值会发生变化，通过测量电路转换为电信号，从而计算出力/力矩值。

The working principle of a torque sensor is mainly based on the strain effect: when the elastic body of the sensor undergoes slight deformation under an external force, the resistance of the strain gauges attached to it changes. This change is converted into an electrical signal through a measurement circuit, enabling the calculation of force/torque values.

2.3 系统工作原理框图

System Working Principle Block Diagram

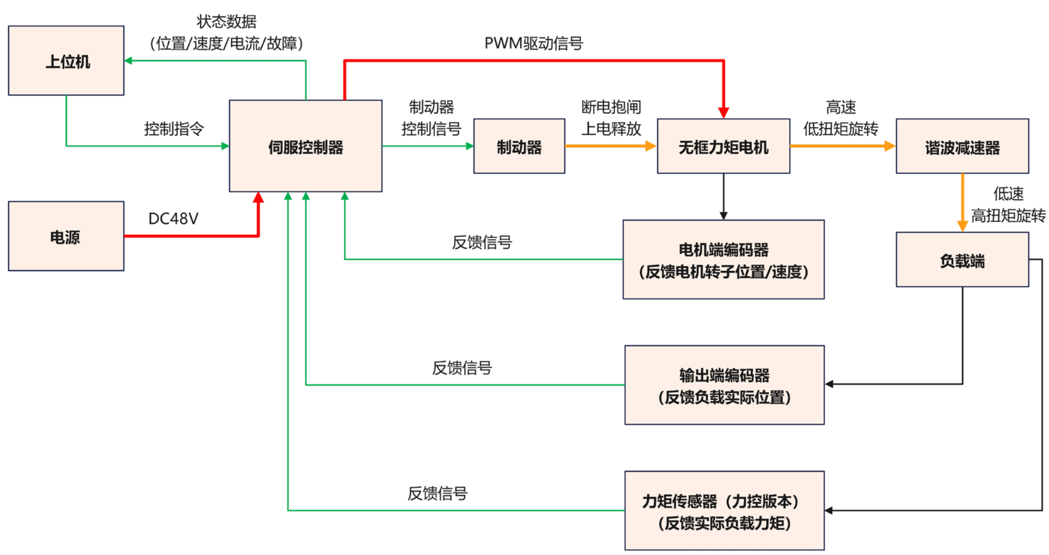


图 2-2 系统工作原理框图

Figure 2-2 System Operation Principle Block Diagram

1） 指令输入

Instruction Input

路径：上位机控制器（如 PLC、运动控制卡）通过通信总线（如 EtherCAT、CANopen）向关节内置的伺服驱动器发送目标位置、速度或扭矩指令。

Path: The upper controller (e.g., PLC, motion control card) sends target position, speed, or torque commands to the servo driver built into the joint via a communication bus (e.g., EtherCAT, CANopen).

2） 核心控制与驱动

Core Control and Drive

核心：伺服驱动器作为系统大脑，接收指令并与反馈信号进行实时比对，基于先进的控制算法计算出所需的驱动电流，通过功率电路输出 PWM（脉宽调制）信号来驱动无框力矩电机旋转。

Core: The servo driver acts as the system's brain, receiving instructions and comparing them in real-time with feedback signals. Using advanced control algorithms, it calculates the required driving current and outputs PWM (pulse-width modulation) signals through the power circuit to drive the frameless torque motor.

3） 执行与传动

Execution and Transmission

电机执行：无框力矩电机在驱动信号的作用下产生高速、低扭矩的旋转运动。

减速增矩：电机的旋转运动通过精密谐波减速机进行减速，并显著增大输出扭矩，最终驱动负载端（机械臂连杆等）运动。

Motor Execution: The frameless torque motor generates high-speed, low-torque rotational motion under the driving signal.

Speed Reduction and Torque Increase: The motor's rotational motion is slowed down by the precision harmonic reducer, significantly increasing the output torque, which eventually drives the load end (e.g., robotic arm linkage) movement.

4） 双闭环反馈-系统精度的关键

Dual Closed-loop Feedback - Key to System Accuracy

电机端反馈：电机端绝对值编码器实时检测电机的转子位置和速度，并反馈给驱动器，形成内环（电流/速度环） 闭环，确保电机本身的控制精度。

输出端反馈：输出端绝对值编码器直接检测负载端的实际位置，并反馈给驱动器，形成外环（位置环） 闭环。此举消除了谐波减速机本身的传动误差、背隙和弹性变形的影响，实现了全闭环控制，达到了极高的末端定位精度。

Motor-End Feedback: The motor-end absolute encoder detects the motor rotor's position and speed in real-time, feeding it back to the driver, forming an inner loop (current/speed loop) to ensure motor control accuracy.

Output-End Feedback: The output-end absolute encoder directly detects the actual position of the load end and feeds it back to the driver, forming an outer loop (position loop). This eliminates transmission errors, backlash, and elastic deformation in the harmonic reducer, achieving full closed-loop control and extremely high terminal positioning accuracy.

5） 安全与辅助功能

Safety and Auxiliary Functions

制动器：在断电或系统故障时，驱动器发送信号使制动器自动抱闸，防止负载下滑，保障设备与人员安全。

状态监控：驱动器持续监控系统状态（如电流、位置、温度、错误代码），并将这些实时数据上传至上位机，实现可视化监控和故障诊断。

Brake: In the event of a power failure or system malfunction, the drive sends a signal to automatically engage the brake, preventing the load from dropping and ensuring the safety of both equipment and personnel.

Status Monitoring: The drive continuously monitors system status parameters (such as current, position, temperature, and error codes). This real-time data is uploaded to the host system, enabling visual monitoring and facilitating fault diagnosis.

2.4 接口与指示灯定义

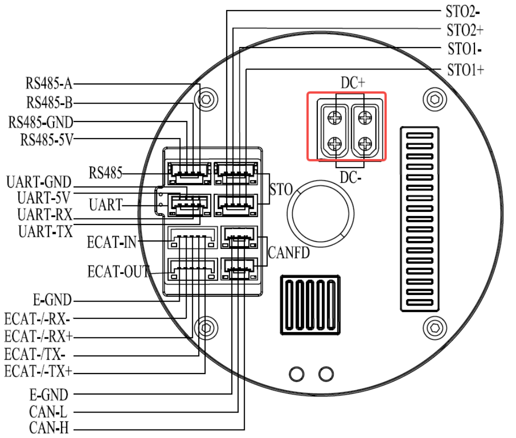
Interface and Indicator Light Definitions

2.4.1 48V 供电电源接口

48V Power Supply Interface

表 2-1 供电电源接口说明

Table 2-1 Power Supply Interface Specifications

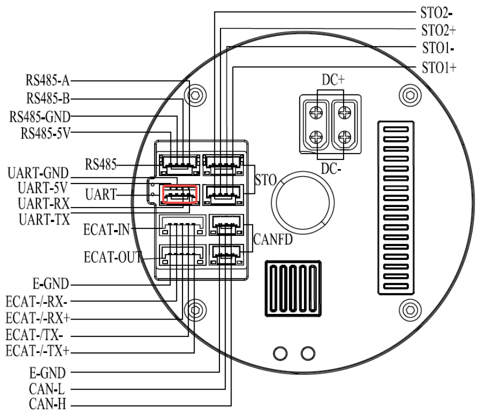
Pin	**电源端子引脚定义** **Terminal Pin Definition**	**功能** **Function**
1	DC+	电源正极 Power Positive
2	DC-	电源负极 Power Negative
电源引脚位置示意 **Power Pinout Diagram**		**接线端子** **Terminal Block**
		
接插件信息 **Connector Information**		**配线端子信息** **Terminal Block Information**
型号：XT30AW-M，AMASS Model：XT30AW-M，AMASS		型号：XT30UW-F，AMASS Model：XT30UW-F，AMASS
规格：2Pin，脚间距 5mm，脚长 2mm Specification：2Pin, 5mm pitch, 2mm pin length		规格：2Pin，脚间距 5mm Specification：2Pin, 5mm pitch

2.4.2 UART 通信接口

UART Communication Interface

表 2-2 UART 通信接口说明

Table 2-2 UART Communication Interface Specifications

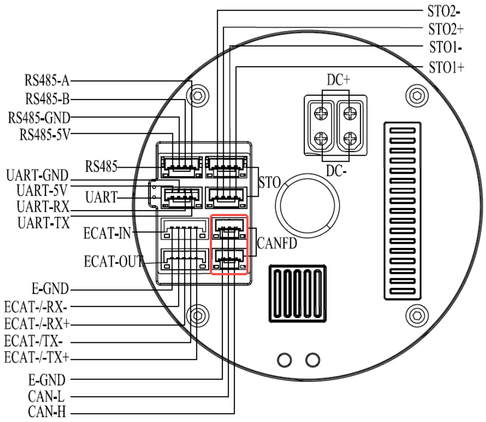
<table>
<tr><th>Pin</th><th>UART 端子引脚定义
UART Terminal Pin Definition</th><th>功能
Function</th></tr>
<tr><td>1</td><td>TX</td><td>串口输出
Serial Output</td></tr>
<tr><td>2</td><td>RX</td><td>串口输入
Serial Input</td></tr>
<tr><td>3</td><td>GND</td><td>GND</td></tr>
<tr><td>4</td><td>+5V</td><td>+5V</td></tr>
<tr><th colspan="2">UART 引脚位置示意
UART Pinout Diagram</th><th>接线端子
Terminal Block</th></tr>
<tr><td colspan="2">
</td><td></td></tr>
<tr><th colspan="2">接插件信息
Connector Information</th><th>配线端子信息
Terminal Block Information</th></tr>
<tr><td colspan="2">型号：BM04B-GHS-TBT，JST
Model：BM04B-GHS-TBT，JST</td><td>型号：GHR-04V-S，JST
Model：GHR-04V-S，JST</td></tr>
<tr><td colspan="2">规格：单排 4Pin，间距 1.25mm
Specification：Single Row 4Pin, 1.25mm pitch</td><td>规格：单排 4Pin，间距 1.25mm
Specification：Single Row 4Pin, 1.25mm pitch</td></tr>
</table>

2.4.3 CAN 通信接口

CAN Communication Interface

表 2-3 CAN 通信接口说明

Table 2-3 CAN Communication Interface Specifications

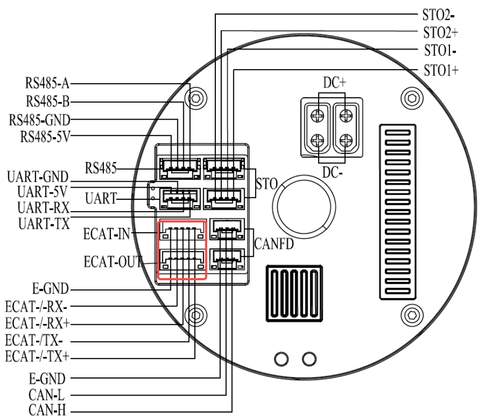
Pin	**CAN 端子引脚定义** **CAN Terminal Pin Definition**	**功能** **Function**
1	CAN-H	CAN 信号高 CAN High Signal
2	CAN-L	CAN 信号低 CAN Low Signal
3	E_GND	金属外壳地 Metal Shell Ground
CAN 引脚位置示意 **CAN Pinout Diagram**		**接线端子** **Terminal Block**
		
接插件信息 **Connector Information**		**配线端子信息** **Terminal Block Information**
型号：BM03B-GHS-TBT，JST Model: BM03B-GHS-TBT，JST		型号：GHR-03V-S，JST Model: GHR-03V-S，JST
规格：单排 3Pin，间距 1.25mm Specification: Single Row 3Pin, 1.25mm pitch		规格：单排 3Pin，间距 1.25mm Specification: Single Row 3Pin, 1.25mm pitch

2.4.4 EtherCAT 通信接口

EtherCAT Communication Interface

表 2-4 EtherCAT 通信接口说明

Table 2-4 EtherCAT Communication Interface Specifications

	ECAT IN		**ECAT OUT**	
Pin	**端子引脚定义 Terminal Pin Definition**	**功能 Function**	**端子引脚定义 Terminal Pin Definition**	**功能 Function**
1	ECAT IN_TX+	输入发送+ Input Send+	ECAT OUT_TX+	输出发送+ Output Send+
2	ECAT IN_TX-	输入发送- Input Send-	ECAT OUT_TX-	输出发送- Output Send-
3	ECAT IN_RX+	输入接收+ Input Receive+	ECAT OUT_RX+	输出接收+ Output Receive+
4	ECAT IN_RX-	输入接收- Input Receive-	ECAT OUT_RX-	输出接收- Output Receive-
5	E_GND	金属外壳地 Metal Shell Ground	E_GND	金属外壳地 Metal Shell Ground
引脚位置示意 Pinout Diagram			**接线端子 Terminal Block**	
				

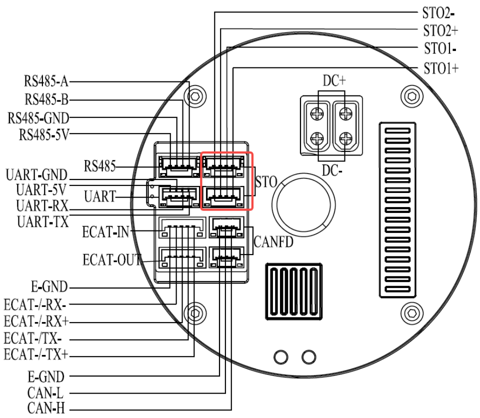
接插件信息 Connector Information	配线端子信息 Terminal Block Information
型号：BM05B-GHS-TBT，JST Model: BM05B-GHS-TBT，JST	型号：GHR-05V-S，JST Model: GHR-05V-S，JST
规格：单排 5Pin，间距 1.25mm Specification: Single Row 5Pin, 1.25mm pitch	规格：单排 5Pin，间距 1.25mm Specification: Single Row 5Pin, 1.25mm pitch

2.4.5 STO 接口

STO Communication Interface

表 2-5 STO 通信接口说明

Table 2-5 STO Communication Interface Specifications

Pin	STO 端子引脚定义 STO Terminal Pin Definition	功能 Function
1	STO1+	安全输入通道 1 正极
2	STO1-	安全输入通道 1 负极
3	STO2+	安全输入通道 2 正极
4	STO2-	安全输入通道 2 负极
STO 引脚位置示意 STO Pinout Diagram		**接线端子 Terminal Block**
		
接插件信息		**配线端子信息**

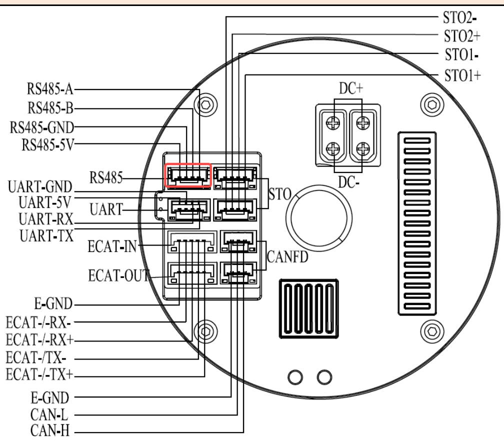
Connector Information	Terminal Block Information
型号：BM04B-GHS-TBT，JST Model：BM04B-GHS-TBT，JST	型号：GHR-04V-S，JST Model：GHR-04V-S，JST
规格：单排 4Pin，间距 1.25mm Specification：Single Row 4Pin, 1.25mm pitch	规格：单排 4Pin，间距 1.25mm Specification：Single Row 4Pin, 1.25mm pitch

2.4.6 RS485 接口（力矩传感器）

RS485 Interface (Torque Sensor)

表 2-6 RS485 通信接口说明

Table 2-6 RS485 Communication Interface Specifications

Pin	RS485 端子引脚定义 RS485 Terminal Pin Definition	功能 Function
1	A	RS485-A
2	B	RS485-B
3	GND	RS485-GND
4	+5V	RS485-5V
RS485 引脚位置示意 RS485 Pinout Diagram		**接线端子 Terminal Block**
		
接插件信息 Connector Information		**配线端子信息 Terminal Block Information**

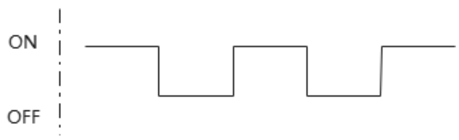
型号：BM04B-GHS-TBT，JST Model：BM04B-GHS-TBT，JST	型号：GHR-04V-S，JST Model：GHR-04V-S，JST
规格：单排 4Pin，间距 1.25mm Specification：Single Row 4Pin, 1.25mm pitch	规格：单排 4Pin，间距 1.25mm Specification：Single Row 4Pin, 1.25mm pitch

2.4.7 指示灯功能

Indicator Light Function

表 2-7 关节运行状态指示灯说明

Table 2-7 Joint Operational Status Indicator Specifications

LED 标识 **LED Marking**	**LED 说明** **LED Description**	**LED 状态** **LED Status**	**LED 状态图** **LED Status Image**
RUN	系统运行正常 System Running Normally	绿色常亮 Solid Green	
	系统运行异常 System Running Abnormally	红色闪烁 Solid Red	
关节运行状态指示灯位置示意 **Joint Operational Status Indicator Location Diagram**			

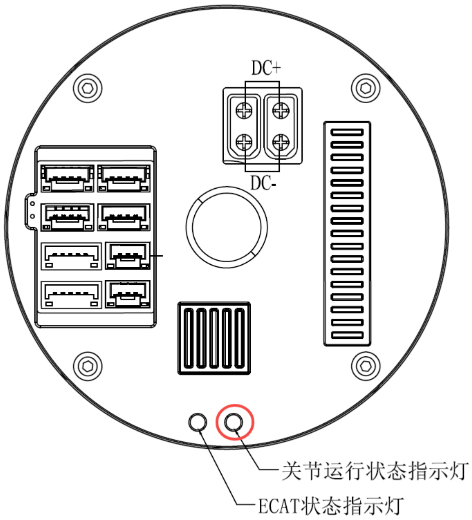
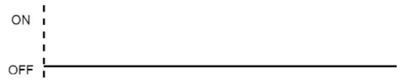
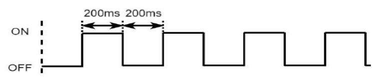
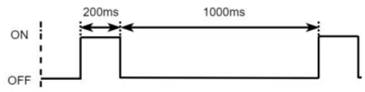
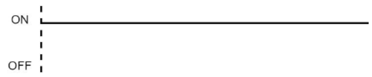
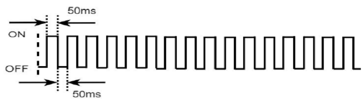


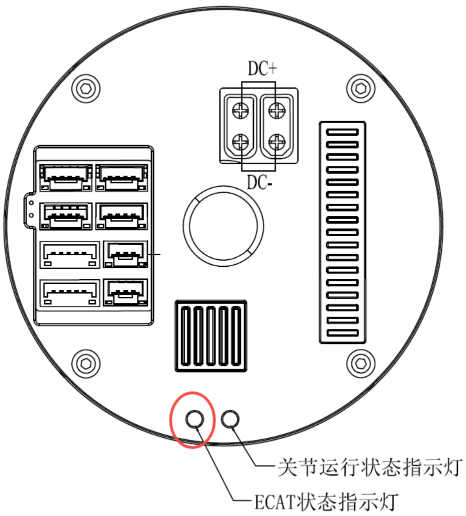
表 2-8 EtherCAT 状态指示灯说明

Table 2-8 EtherCAT Status Indicator Light Explanation

LED 标识 LED Label	**LED 说明 Description**	**ECAT 状态机 ECAT State Machine**	**LED 状态 LED State**	**LED 状态图 LED State Diagram**
ECAT STA	ECAT 状态机指示 ECAT State Machine Indicator	INIT	熄灭 Off	
		PREOP	绿色闪烁 Green Blinking	
		SAFEOP	绿色单闪 Green, Single Flash	
		OP	绿色常亮 Green Steady On	
		BOOT	绿色快速闪烁 Green Fast Blinking	

EtherCAT 状态指示灯位置示意

EtherCAT Status Indicator Location Diagram



三、安装与机械集成

Installation and Mechanical Integration

3.1 开箱检查与内容清单

Unboxing Inspection and Item Checklist

1） 内容清单

Contents Checklist

开箱后，请首先对照本清单清点所有物品。清单通常置于箱内最上层或粘贴于外箱内侧。

After unpacking, please first check and count all items against this checklist. The checklist is usually placed at the top of the box or pasted on the inside of the outer box.

表 3-1 开箱清单

Table 3-1 Unpacking Checklist

序号 No.	物品名称 Item Name	型号/规格 Model/Specification	数量 Quantity	备注/序列号 Remarks/Serial Number
1	关节模组 Joint Module	PHU*	1	--
2	电源转接线 Power Conversion Cable	XT30UW-F	1	--
3	调试信号线 Debugging Signal Cable	GHR-04V-S	1	--
4	CAN 或者 EtherCAT 信号线 CAN or EtherCAT Signal Cable	GHR-03V-S or GHR-05V-S	1	--

注意：

Notes:

① 以上清单为示例，请以实际收到的产品配置为准；

The above list is an example. Please refer to the actual product configuration you receive.

② 请核对产品型号和序列号是否与您的订单一致；

Please verify that the product model and serial number match your order.

③ 请妥善保管所有包装材料（内衬、泡沫等），以便在需要返厂维修或运输时使用。

Please keep all packaging materials (inner liners, foam, etc.) properly for use in case factory return, repair, or transportation is needed.

2） 开箱检查

Unpacking Inspection

清点完毕后，请立即对产品进行仔细的外观和初步功能检查。

After completing the inventory, please immediately conduct a careful visual and preliminary functional inspection of the product.

① 基本外观检查：仔细检查关节外壳、减速器外壳、输出法兰端是否有任何**磕碰、划伤、凹陷或裂纹**；

Basic Visual Inspection: Carefully check the joint housing, reducer housing, and output flange end for any bumps, scratches, dents, or cracks.

② 机械接口检查：检查输出法兰的安装孔和端面是否有**损伤或变形**；

Mechanical Interface Inspection: Check the mounting holes and end face of the output flange for damage or deformation.

③ 电气接口检查：检查连接器的针脚是否平直、完好，有无弯曲或锈蚀。

Electrical Interface Inspection: Check the pins of the connectors to ensure they are straight, intact, and free from bending or corrosion.

3） 问题处理流程

Problem Handling Procedure

如果在开箱检查中发现任何问题，请立即停止并按照以下流程处理：

If any issues are found during the unpacking inspection, please stop immediately and follow this procedure:

① 拍照取证：对问题部位（包装损坏、产品外观损伤、配件缺失等）进行多角度、清晰地拍摄；

Photograph for Evidence: Take clear, multi-angle photographs of the problematic areas (e.g., damaged packaging, product exterior damage, missing accessories).

② 联系我方：立即通过电话或邮件联系我们的客户服务部门（联系方式见附录 9.3 章节）；

Contact Us: Immediately contact our customer service department via phone or email (refer to Appendix 9.3 for contact details).

③ 切勿自行处理：请不要自行尝试拆卸、修理或安装存在问题的产品，以免造成进一步损坏或影响保修权益。等待我司技术支持人员给出处理方案。

Do Not Attempt Self-Repair: Do not attempt to disassemble, repair, or install the faulty product yourself, as this may cause further damage or void the warranty. Wait for instructions from our technical support personnel.

3.2 安装前准备

Pre-installation Preparation

警告：

Warning

- 不正确的安装和操作可能导致产品性能下降、损坏或人员伤害。

  Incorrect installation and operation may result in performance degradation, damage, or personal injury.
- 安装工作必须由经过培训的、了解机械和电气危险的专业技术人员进行。

  Installation must be carried out by trained professionals familiar with mechanical and electrical hazards.
- 在进行任何操作前，请务必阅读并理解本手册的全部相关内容。

  Before performing any operations, please ensure that you read and understand all relevant sections of this manual.

1） 工具与材料准备

Tools and Materials Preparation:

请预先准备好以下工具和材料，以确保安装过程高效、准确。

Please prepare the following tools and materials in advance to ensure the installation process is efficient and accurate.

表 3-2 安装工具材料

Table 3-2 Installation Tools and Materials

类别 Category	推荐工具/材料 Recommended Tools/Materials	说明与要求 Description and Requirements
机械工具 Mechanical Tools	扭矩扳手 Torque Wrench	【必备】量程需覆盖螺丝的紧固扭矩范围 **Essential.** The range must cover the fastening torque specification of the screws.
	内六角扳手组 Hex Wrench Set	必须为**高质量**产品，与螺丝头完美匹配，防止拧圆。 Must be high-quality and perfectly match the screw heads to prevent rounding.
清洁用品 Cleaning supplies	无绒布（如拭镜纸） Lint-free Cloth (e.g., Lens Paper)	【必备】用于清洁配合表面。 Essential. For cleaning mating surfaces.
	高纯度异丙醇或专用清洗剂 High-Purity Isopropyl Alcohol or Dedicated Cleaner	【必备】用于清洗螺丝孔和配合面，禁止使用汽油等刺激性溶剂。 **Essential.** For cleaning screw holes and mating surfaces. The use of irritating solvents such as gasoline is prohibited.
	不起毛的手套 Lint-free Gloves	防止手部油脂和汗液污染精密部件。 To prevent contamination of precision components by skin oils and sweat.
材料 Material	螺纹锁固剂（中等强度） Thread locker (Medium	【必备】用于防止螺丝松动。 **Essential.** Used to prevent screw

	Strength)	loosening.

2） 环境与人员准备

Environment and Personnel Preparation

环境要求：工作环境温度-20~60℃。

人员要求：操作人员需熟悉装配流程，理解扭矩控制的重要性，并能正确使用扭矩扳手。

Environmental Requirements: Operating environment temperature: -20~60°C.

Personnel Requirements: Operators must be familiar with the assembly process, understand the importance of torque control, and be able to correctly use the torque wrench.

3） 关键数据确认与查阅

Confirmation and Review of Key Data

在开始安装前，请务必在本手册中查阅以下关键数据，并记录下来：

Before starting installation, please ensure to review the following key data in this manual and record them:

①紧固扭矩值：查阅螺丝紧固扭矩表，确认输出法兰连接螺丝和输入轴锁紧螺钉的准确扭矩值。该值取决于螺丝的规格、等级和螺纹锁固剂的使用情况；

Screw Tightening Torque Values: Refer to the screw tightening torque table to confirm the correct torque values for output flange screws and input shaft locking screws.

②允许的负载：查阅"性能参数"章节，确认模组的最大（弯矩负载）、最大（径向负载） 和 最大（轴向负载）。确保您的设计负载在此范围内；

Allowable Loads: Refer to the "Performance Parameters" section to confirm the module's maximum (bending moment load), maximum (radial load), and maximum (axial load). Ensure that your design load falls within this range.

③同心度与垂直度要求：查阅"安装指导"章节，确认对相连机械部件的径向跳动和端面跳动的公差要求；

Concentricity and Verticality Requirements: Refer to the "Installation Guidelines" section to confirm the tolerance requirements for radial runout and end face runout of the connected mechanical components.

④电气参数：查阅“规格表”额定电压等参数。

Electrical Parameters: Refer to the "Specification Table" for rated voltage and other parameters.

4） 部件检查与安全准备

Component Check and Safety Preparation

①部件准备：确保配合部件正常，螺丝长度合适，涂抹适量螺纹紧固剂。

Component Preparation: Ensure that the mating parts are in good condition, the screw lengths are appropriate, and the correct amount of thread locking agent is applied.

②安全准备：设备系统断电，穿戴防护手套。

Safety Preparation: Ensure the system is powered off and wear protective gloves.

3.3 机械安装步骤与指导

Mechanical Installation Steps and Guidelines

警告：

Warning:

- 严禁使用任何锤击、冲击或碾压的方式将模组装入机座！ 这将导致精密齿轮的永久性损坏。

  Do not use any hammering, impact, or crushing methods to install the module into the housing! This will cause permanent damage to the precision gears.

- 严禁对输出法兰施加任何径向或轴向的冲击力！

  Do not apply any radial or axial impact force to the output flange!

- 严禁以输出法兰为杠杆来移动或抬起整个模组，应用手托住外壳。

  Do not use the output flange as a lever to move or lift the entire module. Always support the shell with your hands.

- 所有螺丝必须使用**扭矩扳手**按规定的扭矩和顺序拧紧。

  All screws must be tightened using a torque wrench according to the specified torque and order.

3.3.1 安装步骤

Installation Steps

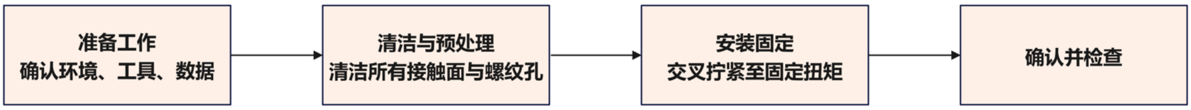


图 3-1 安装步骤

Figure 3-1 Installation Steps

1） 准备工作：确认环境、工具、相关扭力数据。

Preparations: Confirm the environment, tools, and relevant torque data.

2） 清洁与预处理：使用无绒布和高纯度清洗剂，彻底清洁模组输出法兰端面和负载连接板的配合面，确保无任何油污、灰尘和毛刺。

Cleaning and Pre-treatment: Use a lint-free cloth and high-purity cleaning agents to thoroughly clean the module's output flange face and the mating face of the load connection plate, ensuring no oil, dust, or burrs.

3） 螺丝安装：在螺丝螺纹上涂抹适量的中强度螺纹锁固剂，使用扭矩扳手，按对角交叉顺序分三步拧紧螺丝。第一步： 将所有螺丝拧至规定扭矩值的 30%，第二步：将所有螺丝拧至规定扭矩值的 60%，第三步： 再次按对角顺序，将所有螺丝拧至 100%的规定扭矩值。

Screw Installation: Apply an appropriate amount of medium-strength thread locking agent to the screw threads. Use a torque wrench to tighten the screws in a cross-diagonal order in three steps. Step 1: Tighten all screws to 30% of the specified torque value. Step 2: Tighten all screws to 60% of the specified torque value. Step 3: Tighten all screws to 100% of the specified torque value.

注意：务必查阅扭矩表，使用准确的扭矩值。

Note: It is imperative to consult the torque table and use the accurate torque values.

3.3.2 连接法兰要求

Flange Connection Requirements

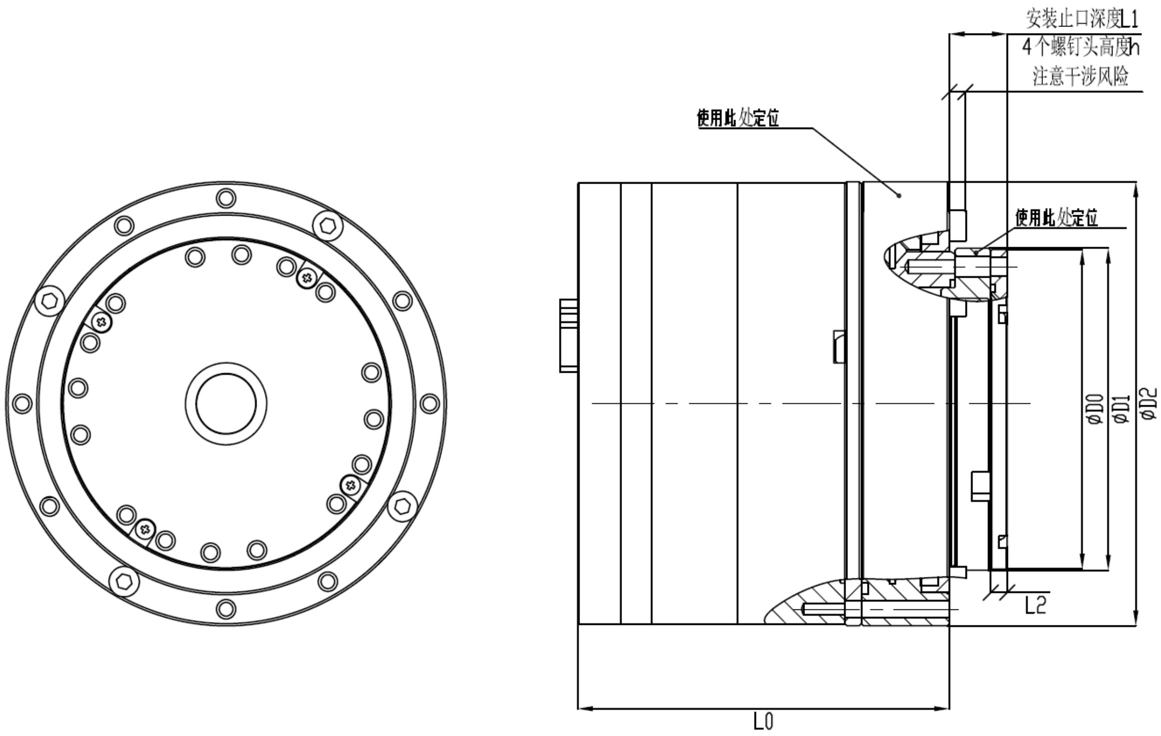


图 3-2 安装示意图

Figure 3-2 Installation Diagram

负载端止口的深度须严格按图 3-2 所示位置进行控制，具体模组型号对应的深度值见表 3-3。连接法兰端面不得存在披锋或其他凸起；钻孔时需注意避免孔口产生凸起或披锋。

The depth of the load end stop must be strictly controlled according to the position shown in Figure 3-2. The specific depth values corresponding to different module models are shown in Table 3-3. The flange end face must not have burrs or other protrusions; care should be taken to avoid burrs or protrusions at the hole opening when drilling.

注释：L0 关节长度，**L1 止口深度**，L2 输出轴长度，D0 输出轴直径，D1 输出钢轮直径，D2 减速器直径，h 指带有螺钉头的高度，固定螺丝即固定关节螺丝，法兰螺丝即输出旋转端固定螺丝。

Notes: L0: Joint Length, L1: Mounting Register Depth, L2: Output Shaft Length, D0: Output Shaft Diameter, D1: Output Steel Wheel Diameter, D2: Reducer Diameter, h: Height including the screw head, Fixing Screw (joint fixing screw),Flange Screw (output rotating end fixing screw).

表 3-3 安装尺寸(单位为 mm)

Table 3-3 Installation Dimensions (Unit: mm)

型号 Model	L0	L1	L2	D0	D1	D2	h	止口圆度 Register Roundness	止口平面度 Register Flatness
PHU08L	51.5	4.5	2.5	29.5	30	40	/	0.01	0.01
PHU11L	57.5	8.5	3	37.5	38	52	/	0.01	0.01
PHU11H-52	59	9	2.5	37.5	38	52	2.5	0.01	0.01
PHU11H-60	60	8.5	4	41.5	42	60	2.5	0.01	0.01
PHU14H	65.4	10	3	49.5	50	70	/	0.01	0.01
PHU17H	67.5	10.5	3	59.5	60	80	/	0.01	0.01
PHU20H	73	12.5	4	69.5	70	90	/	0.01	0.01
PHU25H	78	16	4	84.5	85	110	/	0.01	0.01
PHU32H	122	22	5	109.5	110	142	/	0.01	0.01
PHU40H	123	26	5	134.5	135	170	/	0.01	0.01

注：公差范围详见 2D 总成图

Note: For tolerance ranges, please refer to the 2D assembly drawing.

表 3-4 固定螺钉型号

Table 3-4 Fixing Screw Specifications

	固定关节螺丝 Joint Fixing Screw			法兰转接螺丝 Flange Adapter Screw		
型号 Model	**螺丝型号 Screw Type**	**螺丝孔深度 Screw Hole Depth**	**个数 Qty**	**螺丝型号 Screw Type**	**螺丝孔深度 Screw Hole Depth**	**个数 Qty**
PHU08L	M2	4	8	M1.6	5+3（有效）	8
PHU11L	M2.5	5	8	M2.5	3+5（有效）	8
PHU11H-52	M2.5	5	8	M2.5	2.5+5（有效）	8
PHU11H-60	M2.5	5	8	M2.5	4+5（有效）	8

PHU14H	M3	5.5	8	M3	9+6（有效）	8
PHU17H	M3	6	12	M3	9.5+6（有效）	16
PHU20H	M3	6	12	M3	11.5+6（有效）	16
PHU25H	M4	8	12	M4	14+8（有效）	16
PHU32H	M5	10	12	M5	21+8（有效）	16
PHU40H	M6	12.5	12	M6	25+10（有效）	16

注：深度数字，+号前面代表通孔长度，+后面代表螺纹孔长度

Note: Regarding the depth notation (e.g., X+Y), the number before the '+' sign represents the through-hole length, and the number after the '+' sign represents the threaded hole length.

注意：若未按关节输出端连接法兰的安装要求执行，可能导致输出轴变形，造成不可逆的损坏。

Important: Failure to adhere to the installation requirements for the joint output flange may cause deformation of the output shaft, resulting in irreversible damage.

L1 安装止口深度的设定旨在确保：①输出法兰的径向定位精度②负载力矩可通过输出法兰止口有效传递至直接支撑外部负载的精密交叉滚子轴承，建议安装深度为止口 3/5 以上。

The setting of the L1 installation stop depth is intended to ensure ① the radial positioning accuracy of the output flange, and ② that the load torque can be effectively transmitted via the output flange register to the precision cross roller bearing which directly supports the external load.

客户负载法兰止口的圆度与平面度应尽量达到或接近规定数值。如因圆度或平面度偏差过大导致模组输出端受力不均，可能引发模组异响或抖动。

The roundness and flatness of the customer's load flange register should ideally meet or closely approach the specified values. If the module output end experiences uneven force due to excessive roundness or flatness deviation, it may cause abnormal noise or vibration in the module.

3.3.3 螺丝扭矩标准

Screw Torque Standard

螺丝锁紧力参照下表 3-5

Screw fastening forces shall comply with Table 3-5 below.

表 3-5 螺丝锁紧扭矩表

Table 3-5 Screw Fastening Torque Table

Screw Size	M1.6	M2	M2.5	M3	M4	M5	M6
锁紧扭矩（N*m） Fastening Torque (N.m)	0.3	0.6	1.2	2	4	9	15

注：锁紧扭矩不仅与螺丝强度有关，而且与内螺纹强度有关，螺丝强度推荐 12.9 级。

Note: The fastening torque depends not only on the screw strength grade but also on the strength of the internal threads. A screw strength grade of 12.9 is recommended.

3.4 负载安装指南

Load Installation Guide

警告与原则：

Warnings and Principles:

①严禁锤击：绝对禁止使用锤子或其他冲击工具将负载敲击到位。冲击载荷会瞬间损坏柔轮和轴承。

No Hammering: It is strictly forbidden to use a hammer or other impact tools to drive the load into position. Impact loads can instantly damage the flex wheel and bearings.

②精确对中：负载轴孔与输出法兰必须完美对齐。任何强制性的偏心安装都会产生巨大的附加弯矩，显著降低模组寿命。

Precise Alignment: The load shaft hole and output flange must be perfectly aligned. Any forced eccentric installation will generate significant additional bending moments, drastically reducing module life.

③扭矩控制：必须使用扭矩扳手，并严格按照规定的扭矩值紧固螺栓。这是防止松动或过紧导致失效的唯一方法。

Torque Control: A torque wrench must be used, and bolts must be tightened strictly according to the specified torque values. This is the only method to prevent failure caused by loosening or over-tightening.

④清洁无尘：微小的颗粒物都会影响连接刚度和平行度，安装必须在清洁环境下进行。

Cleanliness: Minute particles can affect connection stiffness and parallelism. Installation must be performed in a clean environment.

1）预安装与对准

Pre-installation and Alignment

①在输出法兰螺栓上涂抹适量的螺纹锁固剂；

Apply an appropriate amount of thread locker to the output flange bolts.

②将螺栓先用手轻轻旋入负载的螺纹孔中 2-3 牙，不要拧紧；

Hand-start the bolts into the load's threaded holes for 2-3 threads; do not tighten.

③将负载小心地套在输出法兰上，利用螺栓进行初步的对准和悬挂。

Carefully place the load onto the output flange, using the bolts for preliminary alignment and temporary support.

2）分布交叉紧固

Cross-pattern Tightening

①使用扭矩扳手，按照“对角交叉”顺序 分步拧紧螺栓。第一步将所有螺栓拧至最终扭矩值的 30%，第二步再次按对角顺序，将所有螺栓拧至最终扭矩值的 60%，第三步按顺序将所有螺栓拧至 100%的规定扭矩值；

Using a torque wrench, tighten the bolts in a "diagonal crisscross pattern" in multiple steps. First, tighten all bolts to 30% of the final torque value. Second, following the same diagonal sequence, tighten all bolts to 60% of the final torque value. Finally, tighten all bolts to 100% of the specified torque in sequence.

②**注意：**此分步交叉方法能确保负载被均匀地拉紧，贴合平顺，避免法兰面产生变形。

Note: This incremental cross-pattern method ensures the load is pulled evenly and seats flatly, preventing distortion of the flange face.

3）常见错误与后果

Common Errors and Consequences

①错误：使用锤子强行安装。后果：立即导致柔轮产生塑性变形或齿面压溃，模组报废；

Error: Using a hammer to forcibly install. Consequence: Immediately causes plastic deformation of the flexible wheel or tooth surface crushing, resulting in module scrapping.

②错误：螺栓未按顺序或扭矩拧紧。后果：结合面受力不均，连接刚度下降，运行时产生振动和异响，螺栓易松动脱落；

Error: Bolts not tightened in sequence or to the proper torque. Consequence: Uneven stress on the joint surface, reduced connection stiffness, vibration, and abnormal noise during operation, with bolts prone to loosening and falling off.

③错误：安装面有污物或毛刺。后果：负载与法兰接触不良，导致实际跳动远大于测量跳动，影响精度，并产生附加弯矩；

Error: Contaminants or burrs on the installation surface. Consequence: Poor contact between the load and the flange, causing the actual runout to be much larger than the measured runout, affecting accuracy, and generating additional bending moments.

④错误：未使用螺纹锁固剂或未等待固化即运行。后果：在振动下螺栓极易松动，可能导致负载脱落，引发严重事故。

Error: Not using thread-locking agent or running before it has cured. Consequence: Under vibration, bolts are highly likely to loosen, potentially causing the load to fall off and triggering serious accidents.

四、电气连接

Electrical Connections

4.1 供电要求

Power Supply Requirements

4.1.1 额定电压电流

Rated Voltage and Current

本设备采用 DC48V 电源供电（电源允许波动范围±10%），母线电压允许范围为 24V 至 60V。当检测到母线电压超过 60V 时，驱动器将触发过压报警；当电压低于 24V 时，则会出现欠压报警。单关节额定电压电流如下表 4-1 所示。

This equipment uses a DC 48V power supply (with an allowable fluctuation range of ±10%), and the bus voltage range is 24V to 60V. If the bus voltage exceeds 60V, the driver will trigger an overvoltage alarm; if the voltage is below 24V, an undervoltage alarm will be triggered. The rated voltage and current for a single joint are shown in Table 4-1 below.

表 4-1 额定电压电流

Table 4-1 Rated Voltage and Current

Model	Voltage (V)	Current (A)
PHU08L	48±10%	0.4
PHU11L/H	48±10%	1.3
PHU14H	48±10%	1.7
PHU17H	48±10%	3.8
PHU20H	48±10%	5
PHU25H	48±10%	8
PHU32H	48±10%	10
PHU40H	48±10%	18

4.1.2 正常运行最高电压

Maximum Operating Voltage

关节正常运行的最高电压为 DC60V，输入超过 60V 将导致关节无法正常运行，报警过压。

使用电池（常见锂电池或其他种类蓄电池）供电时，再生能量可回灌至电池，因此通常不需要外接额外的泄放电阻来抑制母线电压升高，但需注意电池的充电接受能力，并且还需修改关节的最高容许与最低容许电压设定。

The maximum operating voltage for the joint is DC 60V. If the input voltage exceeds 60V, the joint will not operate normally, and an overvoltage alarm will be triggered.

When powered by a battery (commonly lithium batteries or other types of rechargeable batteries), regenerative energy can be returned to the battery. Therefore, it is generally not necessary to connect external discharge resistors to suppress the bus voltage rise. However, attention should be paid to the battery's charging capacity, and the joint's maximum and minimum allowable voltage settings should be modified.

4.1.3 正常运行最低电压

Minimum Operating Voltage

关节正常运行的最低电压为 DC24V，低于 24V 将导致制动器无法正常打开。当输入电压低于额定电压时将会降低实际最高转速，但不会降低转矩性能。

The minimum operating voltage for the joint is DC 24V. If the voltage drops below 24V, the brake will fail to release properly. When the input voltage is below the rated voltage, the actual maximum speed will decrease, but the torque performance will not be affected.

4.2 接线图与引脚定义

Wiring Diagram and Pin Definitions

4.2.1 电源接线说明

Power Wiring Instructions

本公司关节模组电源接线提供三种方式：单轴直连、树形拓扑连接与链型拓扑连接，示意图分别参见图 4-1 至图 4-3。基于多轴协作应用中采用 48V 开关电源供电时的实际压降测

试结果，推荐按以下顺序选择接线方式：

The company's joint module power wiring provides three options: single-axis direct connection, tree topology connection, and chain topology connection. The schematic diagrams are shown in Figures 4-1 to 4-3. Based on actual voltage drop test results when using a 48V switch power supply in multi-axis collaborative applications, the following wiring options are recommended:

注意：关节在减速运行时会发电，导致母线电压升高，接入电源泄放模块可以将多余能量通过电阻消耗，从而避免动能回馈导致的电源电压升高。

Note: When the joint is decelerating, it generates power, which leads to an increase in bus voltage. Connecting a power discharge module can consume the excess energy through resistors, preventing the bus voltage from rising due to energy recovery.

① **单轴直连：**为最优方式，接线电阻最小，线路损耗压降最低，适用于大功率模组；

Single-axis direct connection: This is the optimal method, with the least wiring resistance and minimal line loss, suitable for high-power modules.

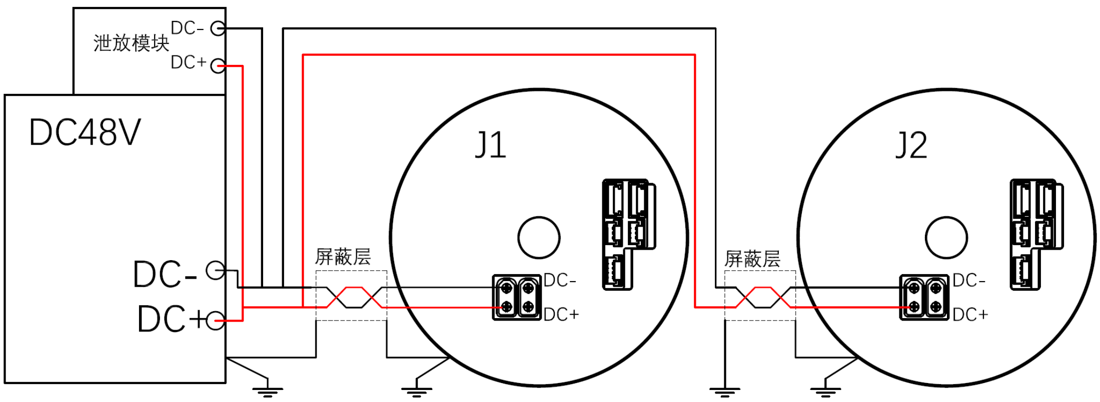


图 4-1 单轴电源串联

Figure 4-1 Single-Axis Power Series Connection

② **树形拓扑连接：**性能较优；

Tree topology connection: This method offers better performance.

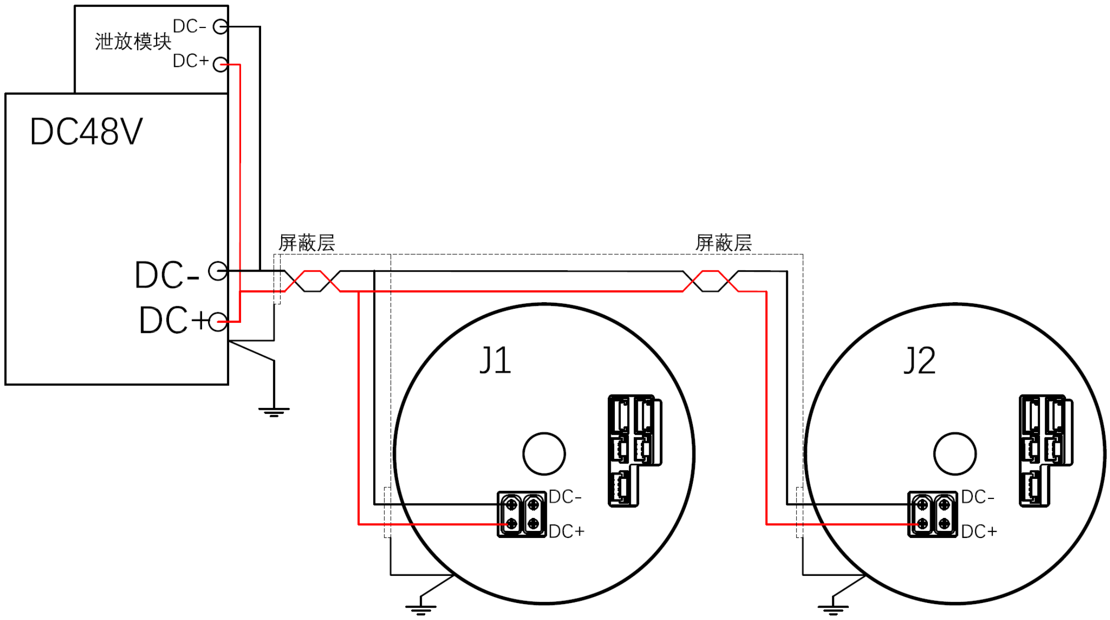


图 4-2 树形拓扑连接

Figure 4-2 Tree Topology Connection

③ **链型拓扑连接**：相较单轴直连与树形拓扑接线电阻较大，线路损耗压降明显，仅推荐用于小功率模组。

Chain topology connection: Compared to single-axis direct and tree topology connections, this method has greater wiring resistance and noticeable line loss. It is only recommended for small-power modules.

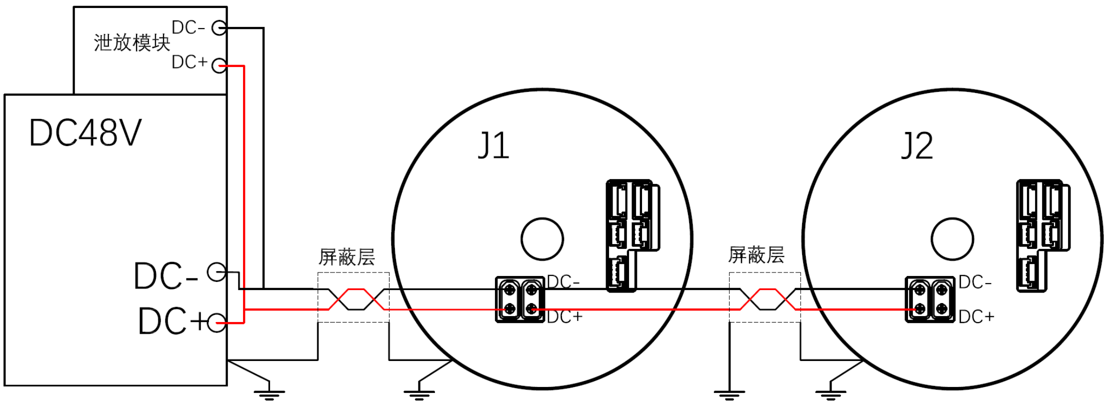


图 4-3 链型拓扑连接

Figure 4-3 Chain Topology Connection

警告：严禁在电源回路中串联其他用电设备，以免因不可预知的压降或电压波动导致模组工作异常。

Warning: Never connect other electrical equipment in series within the power circuit. Any unforeseen voltage drop or fluctuation may cause abnormal module operation.

4.2.2 CAN 通信接线说明

CAN Communication Wiring Instructions

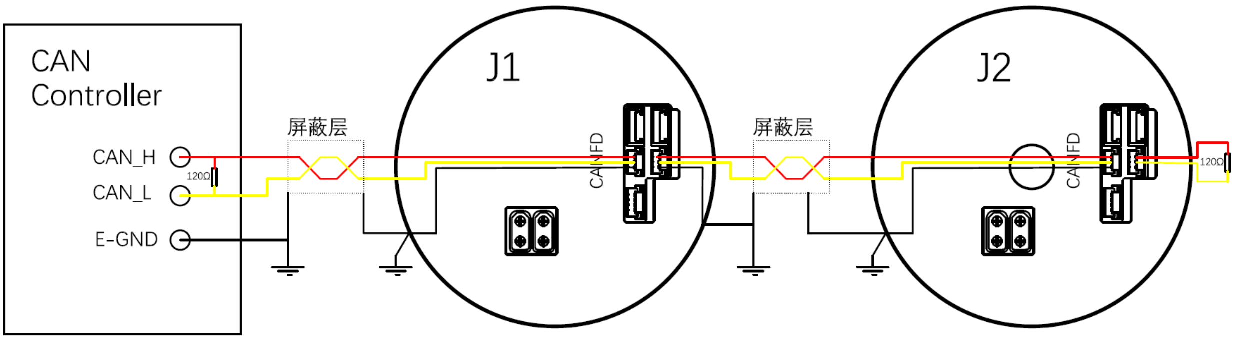


图 4-4 CAN 通信接线

Figure 4-4 CAN Communication Wiring

注意事项：

Notes:

①CAN 通信需采用双绞线，并应带独立屏蔽层。意优关节模组固定以 1Mbps 速率通信，CAN 总线最远通信距离为 25m（指距离最远两节点间）；

CAN communication must use a twisted-pair cable with an independent shield. The joint modules are set to communicate at a fixed rate of 1Mbps. The maximum communication distance for the CAN bus is 25m (refers to the distance between the two farthest nodes).

②请在控制器端与末端伺服的 CAN 接口处各并联一个 120Ω 终端电阻；

Please connect a 120Ω termination resistor at both the controller end and the CAN interface of the last servo drive.

③建立 CAN 通信前，请为每个关节模组设置唯一的 CAN ID；

Before establishing CAN communication, set a unique CAN ID for each joint module.

④E-GND 是机壳地，为电磁干扰和静电提供低阻抗泄放路径，提高设备稳定性。

E-GND is the chassis ground. It provides a low-impedance discharge path for electromagnetic interference (EMI) and electrostatic discharge (ESD), improving equipment stability.

4.2.3 EtherCAT 通信接线说明

EtherCAT Communication Wiring Instructions

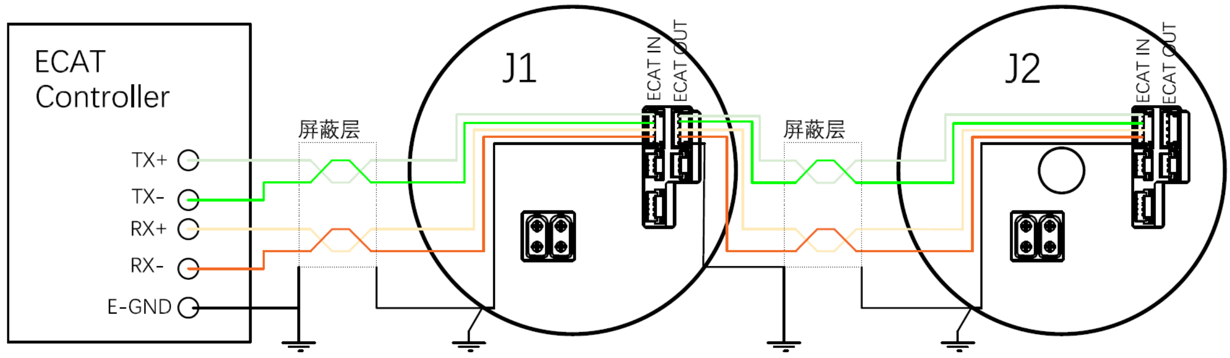


图 4-5 EtherCAT 通信接线

Figure 4-5 EtherCAT Communication Wiring

注意事项：

Notes:

EtherCAT 通信线应使用双绞屏蔽线，并独立屏蔽，以保证信号传输质量。关节之间的 ECAT In 与 ECAT Out 接口接反，仍可能建立通信连接，但将导致 EtherCAT 主站控制器识别到的伺服轴顺序与实际物理连接顺序不一致，可能影响多轴协同控制。

EtherCAT communication cables should use independently shielded twisted-pair cables to ensure signal transmission quality. If the ECAT In and ECAT Out interfaces between joints are connected in reverse, communication may still be established. However, this will cause the servo axis order recognized by the EtherCAT master controller to be inconsistent with the actual physical connection sequence, which may affect multi-axis coordinated control.

4.2.4 UART 通信接线说明

UART Communication Wiring Instructions

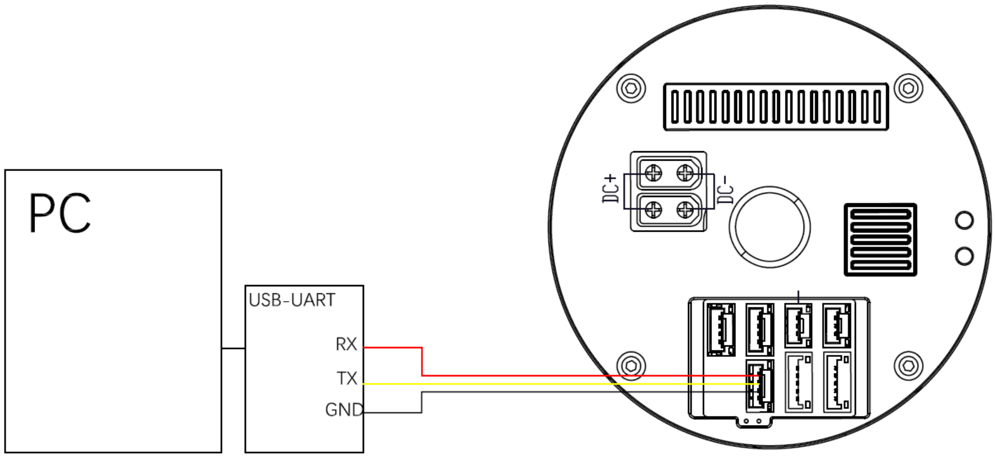


图 4-6 UART 通信接线

Figure 4-6 UART Communication Wiring

4.2.5 STO 接线说明

STO Wiring Instructions

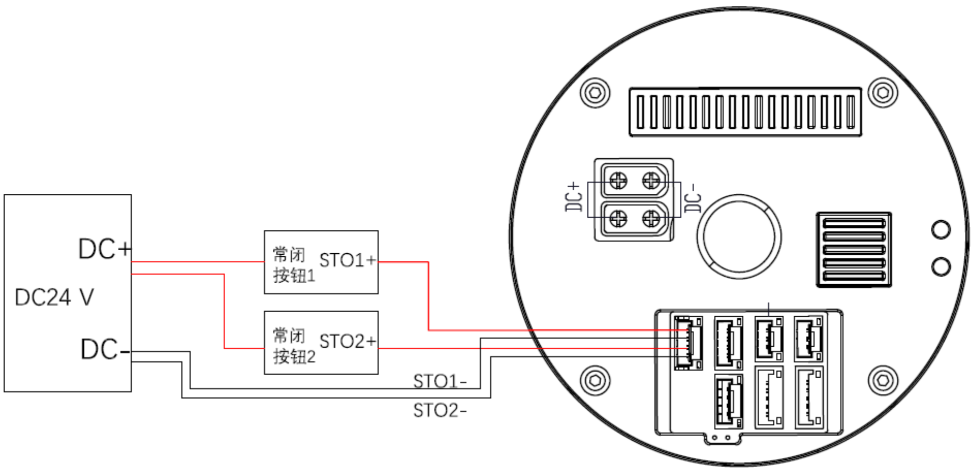


图 4-7 STO 接线

Figure 4-7 STO Wiring

注意事项：两路 STO 必须同时接通 24V，关节才能正常运行。（两路冗余，防止一路硬件失效）

Precautions: Both STO circuits must be connected to 24V at the same time for the joint to operate normally. (Two circuits provide redundancy to prevent hardware failure in one path)

4.3 线缆与连接器选型建议

Cable and Connector Selection Recommendations

关节电源线与信号线选型规格如下表 4-2。

The recommended specifications for joint power and signal cables are shown in Table 4-2 below.

表 4-2 关节线缆规格

Table 4-2 Joint Cable Specifications

模组型号 Module Series	信号接口 Signal Interface	所需连接器型号 Recommended Connector Model	持续额定推荐线缆规格 Continuous Duty Recommended Cable Specification		机械臂推荐线缆规格 Robotic Arm Application Recommended Cable Specification	
			截面积 Cross-sectional Area(mm^2)	AWG#	截面积 Cross-sectional Area (mm^2)	AWG#
全系列 Full series	CAN	GHR-03V-S	0.05~0.13	30~26	0.05~0.13	30~26
	EtherCAT	GHR-05V-S	0.05~0.13	30~26	0.05~0.13	30~26
	UART	GHR-04V-S	0.05~0.13	30~26	0.05~0.13	30~26
	STO	GHR-04V-S	0.05~0.13	30~26	0.05~0.13	30~26
PHU08L	48V 电源 DC IN	XT30UW-F	0.32	22	0.2	24
PHU11*		XT30UW-F	0.32	22	0.32	22
PHU14H		XT30UW-F	0.32	22	0.32	22
PHU17H		XT30UW-F	0.81	18	0.52	20
PHU20H		XT30UW-F	0.81	18	0.52	20
PHU25H		XT30UW-F	1.31	16	1.31	18
PHU32H		XT30UW-F	1.31	16	1.31	16
PHU40H		XT30UW-F	1.31	16	1.31	16

4.4 回收动能方法建议

Energy Recovery Methods Recommendations

在 48V 开关电源供电的模组系统中，电机在正常工作状态下由电源提供电能，驱动负载运转。但在减速或制动过程中，电机转变为发电机状态，将机械动能转化为电能，回馈至电源端。这一过程会导致电源母线电容电压上升，其回馈功率与负载扭矩及电机转速成正比。若回馈能量过大，母线电压超过驱动器所设定的最高允许值（如 60V），系统将触发过压保护并报错停机。表 4-3 提供了 3 种动能回收方式。

In a module system powered by a 48V switch power supply, the motor is powered by the supply and drives the load during normal operation. However, during deceleration or braking, the motor switches to generator mode, converting mechanical energy into electrical energy and feeding it back to the power supply. This process causes the bus capacitor voltage to rise, and the recovery power is proportional to the load torque and motor speed. If the recovered energy is too high and the bus voltage exceeds the maximum allowable value set by the driver (e.g., 60V), the system will trigger overvoltage protection and shut down. Table 4-3 provides three energy recovery methods.

推荐采用泄放电阻方式进行放电，这是一种成本低、结构简单且可靠性高的解决方案。建议使用本公司 PD50 专用放电模块，该模块内置 50W 标准放电能力，并支持并联外接扩展电阻，最大可将泄放功率提升至 550W，满足不同应用场景下的放电需求。

The use of a bleeder resistor (dump resistor) for discharge is recommended, as it is a low-cost, simple, and highly reliable solution. We recommend using our company's dedicated PD50 discharge module. This module has a built-in standard 50W discharge capability and supports connecting an external parallel expansion resistor, allowing the maximum bleed power to be increased up to 550W to meet the discharge requirements of different application scenarios.

表 4-3 动能回收方式

Table 4-3 Regenerative Braking Methods

方法 **Method**	能量处理方式 **Energy**	适用场景 **Applicable Scenarios**	关键特点 **Key Characteristics**

	Handling		
泄放电阻 Bleeder Resistor	消耗为热能 Dissipated as heat	小功率、间歇性制动 Low power, intermittent braking	成本低，简单可靠效率低 Low cost, simple, reliable, low efficiency
超级电容 Supercapacitor	暂存并复用 Temporarily stored and reused	频繁启停、短时大功率回馈 Frequent start-stop, short-term high-power regeneration	响应快，循环寿命长 Fast response, long cycle life
蓄电池 Battery	储存并复用 Stored and reused	需要能量回收的长期运行 系统 Long-term operating systems requiring energy recovery	能效高，系统复杂 High energy efficiency, complex system

使用 PD50 泄放模块，各型号关节推荐并联外接扩展电阻规格如表 4-4 所示。

Table 4-4 Recommended Parallel Expansion Resistors for the PD50 Module

表 4-4 PD50 并联扩展电阻推荐

Table 4-4 Recommended Parallel Expansion Resistors for the PD50 Module

关节型号 **Joint Model**	直流电源规格 **DC Power Supply Specification**	外接并联扩展电阻规格 **Recommended External Parallel Expansion Resistor**
PHU11L/H	100W、48VDC	无需外接扩展 Not required
PHU14	200W、48VDC	无需外接扩展 Not required
PHU17	400W、48VDC	50W、5Ω
PHU20	400W、48VDC	50W、5Ω
PHU25	800W、48VDC	250W、5Ω
PHU32	800W、48VDC	250W、5Ω
PHU40	1200W、48VDC	250W、5Ω

五、软件配置与调试

Software Configuration and Debugging

5.1 调试工具介绍与下载

Debugging Tool Introduction and Download

EYouServoStudio 是意优关节串口调试上位机，是一款专为关节设计的调试与控制软件。该软件通过串口通信协议与谐波关节硬件连接，为用户提供直观的设备状态监控、参数配置和运动控制功能。

EYouServoStudio is the upper computer serial debugging tool developed by EYou for joint modules. It is a debugging and control software specifically designed for the joints. This software connects with the harmonic joint hardware through the serial communication protocol, providing users with intuitive device status monitoring, parameter configuration, and motion control functions.

下载方式：

Download Methods:

① 云链接下载：访问意优科意优关节资料包 - 飞书云文档，根据您的产品型号搜索并下载对应的调试软件和所需驱动。

Cloud Link Download: Access the EYou joint module documentation on Feishu Cloud Documents, search for your product model, and download the corresponding debugging software and required drivers.

② 联系技术支持：通过技术支持获取最新版本软件资源。

Contact Technical Support: Obtain the latest version of the software resources through technical support.

5.2 通信协议说明

Communication Protocol Explanation

PHU 系列采用标准 CANopen 或者 EtherCAT 通信协议，详细说明见《EYou-PHU 系列关节 CANopen 与 EtherCAT 通信手册》，请前往意优帮助中心查找，或者联系技术支持。

The PHU series uses the standard CANopen or EtherCAT communication protocol. For detailed instructions, please refer to the "EYou-PHU Series Joint CANopen and EtherCAT Communication Manual," available at the EYou Help Center, or contact technical support.

5.3 首次上电配置向导

First Power-Up Configuration Wizard

5.3.1 硬件接线

Hardware Wiring

串口引脚定义详见 2.4.2 UART 通信接口，接线说明详见 4.2.4 UART 通信接线。电源引脚定义详见 2.4.1 48V 电源引脚，接线说明详见 4.2.1 电源接线。

The serial port pin definitions are detailed in section 2.4.2 UART Communication Interface, and the wiring instructions are described in section 4.2.4 UART Communication Wiring. The power pin definitions are detailed in section 2.4.1 48V Power Supply Pins, and the wiring instructions are described in section 4.2.1 Power Wiring.

5.3.2 软件通信

Software Communication

①通信端口连接，下拉菜单选择可用端口（COM 口），通信参数默认，点击连接。

Communication Port Connection: From the dropdown menu, select the available port (COM port), use default communication parameters, and click Connect.

②控制权选择：控制权决定使用哪种通信协议下发指令。PC 是指电脑串口上位机，EtherCAT 是指 EtherCAT 主站，CANopen 是指 CANopen 主站（固件版本 V0.0.0.37 之后支持）。

Control Selection: Control determines which communication protocol is used to issue commands. PC refers to the computer serial port host, EtherCAT refers to the EtherCAT master, and CANopen refers to the CANopen master (supported in firmware versions after V0.0.0.37).

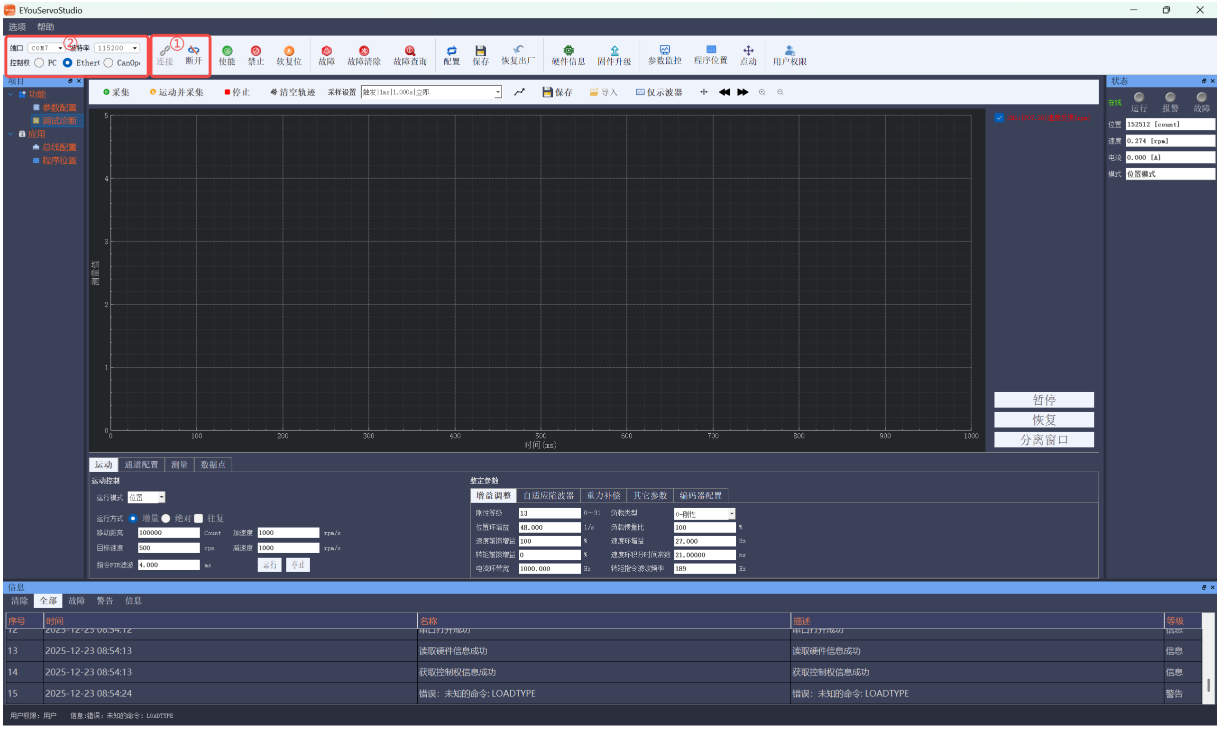


图 5-1 通信连接

Figure 5-1 Communication Connection

③关节型号与固件信息确认。

Confirm Joint Model and Firmware Information.

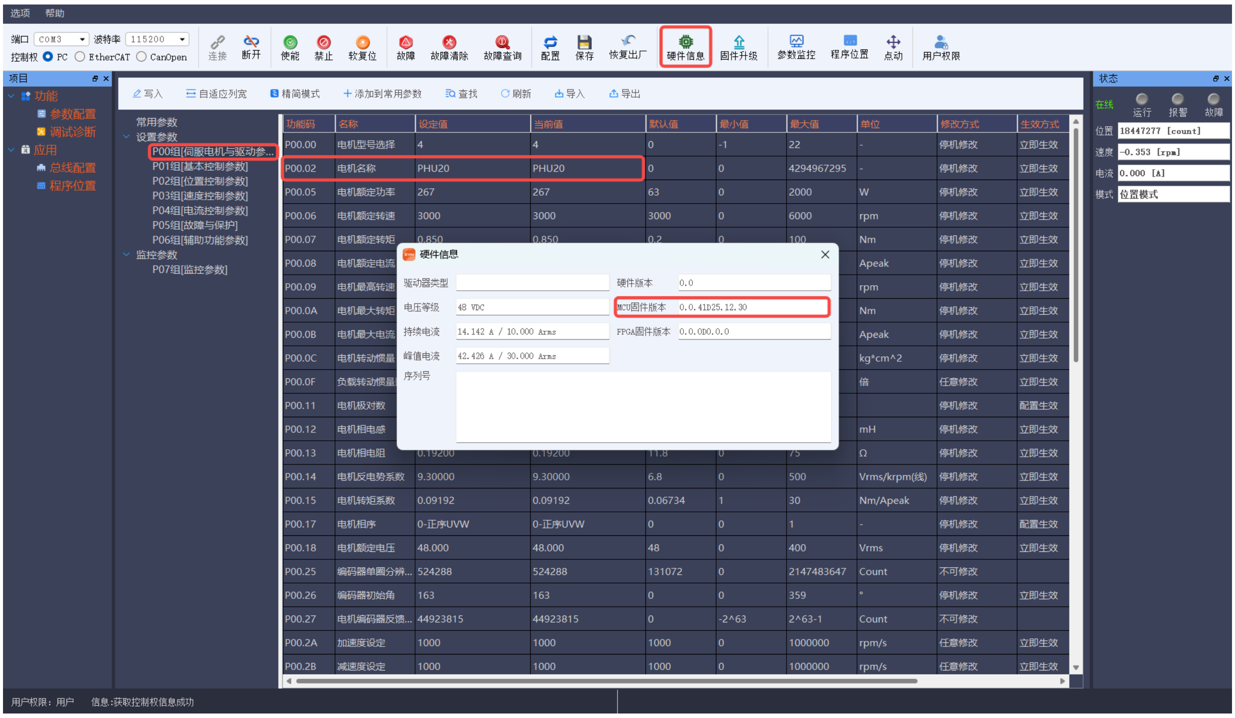


图 5-2 关节型号

Figure 5-2 Joint Model

5.4 参数设置与功能说明

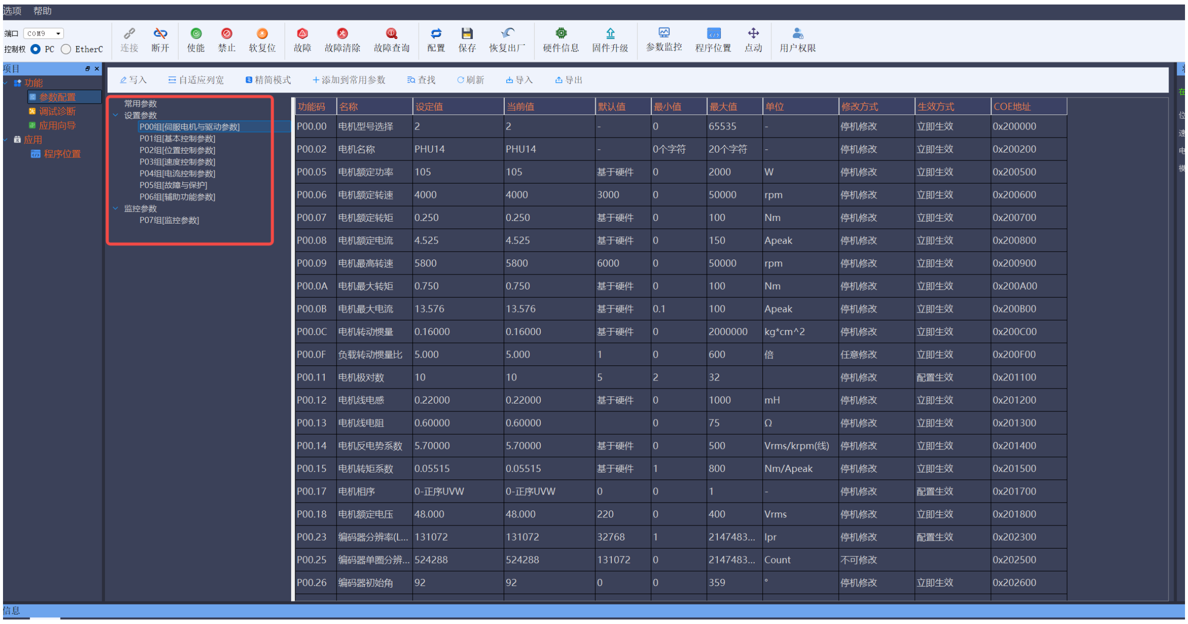
Parameter Settings and Function Descriptions

出厂默认参数：P00（伺服电机参数）、P01（基本控制参数）、P02（位置控制参数）、P03（速度控制参数）、P04（电流控制参数）、P05（故障与保护）、P06（辅助功能参数）、P07（监控参数）

注：默认参数无特殊情况无需修改。

Factory Default Parameters: P00 (Servo Motor Parameters), P01 (Basic Control Parameters), P02 (Position Control Parameters), P03 (Speed Control Parameters), P04 (Current Control Parameters), P05 (Faults and Protection), P06 (Auxiliary Functions), P07 (Monitoring Parameters).

Note: The default parameters do not need modification unless specified otherwise.



功能码	名称	设定值	当前值	默认值	最小值	最大值	单位	修改方式	生效方式	COE地址
P00.00	电机型号选择	2	2	-	0	65535	-	停机修改	立即生效	0x200000
P00.02	电机名称	PHU14	PHU14	-	0个字符	20个字符	-	停机修改	立即生效	0x200200
P00.05	电机额定功率	105	105	基于硬件	0	2000	W	停机修改	立即生效	0x200500
P00.06	电机额定转速	4000	4000	3000	0	50000	rpm	停机修改	立即生效	0x200600
P00.07	电机额定转矩	0.250	0.250	基于硬件	0	100	Nm	停机修改	立即生效	0x200700
P00.08	电机额定电流	4.525	4.525	基于硬件	0	150	Apeak	停机修改	立即生效	0x200800
P00.09	电机最高转速	5800	5800	6000	0	50000	rpm	停机修改	立即生效	0x200900
P00.0A	电机最大转矩	0.750	0.750	基于硬件	0	100	Nm	停机修改	立即生效	0x200A00
P00.0B	电机最大电流	13.576	13.576	基于硬件	0.1	100	Apeak	停机修改	立即生效	0x200B00
P00.0C	电机转动惯量	0.16000	0.16000	基于硬件	0	2000000	kg*cm^2	停机修改	立即生效	0x200C00
P00.0F	负载转动惯量比	5.000	5.000	1	0	600	倍	任意修改	立即生效	0x200F00
P00.11	电机极对数	10	10	5	2	32		停机修改	配置生效	0x201100
P00.12	电机线电感	0.22000	0.22000	基于硬件	0	1000	mH	停机修改	立即生效	0x201200
P00.13	电机线电阻	0.60000	0.60000		0	75	Ω	停机修改	立即生效	0x201300
P00.14	电机反电势系数	5.70000	5.70000	基于硬件	0	500	Vrms/krpm(线)	停机修改	立即生效	0x201400
P00.15	电机转矩系数	0.05515	0.05515	基于硬件	1	800	Nm/Apeak	停机修改	立即生效	0x201500
P00.17	电机相序	0-正序UVW	0-正序UVW	0	0	1	-	停机修改	配置生效	0x201700
P00.18	电机额定电压	48.000	48.000	220	0	400	Vrms	停机修改	立即生效	0x201800
P00.23	编码器分辨率(L...	131072	131072	32768	1	2147483...	lpr	停机修改	配置生效	0x202300
P00.25	编码器单圈分辨...	524288	524288	131072	0	2147483...	Count	不可修改		0x202500
P00.26	编码器初始角	92	92	0	0	359	°	停机修改	立即生效	0x202600

图 5-3 关节参数

Figure 5-3 Joint Parameters

5.5 零点标定与回零流程

Zero Calibration and Zeroing Process

串口上位机零点标定是确保关节当前位置为零点。请按照指示操作：

Zero calibration ensures the current position of the joint is set to zero. Please follow the instructions:

① 位置控制参数 P02.62（回零模式）默认 35 设置当前位置为零位；

Position control parameter P02.62 (zeroing mode) default setting 35 will set the current position to zero.

② 位置控制参数 P02.63（回零启动），双击即可设置当前位置为零位。

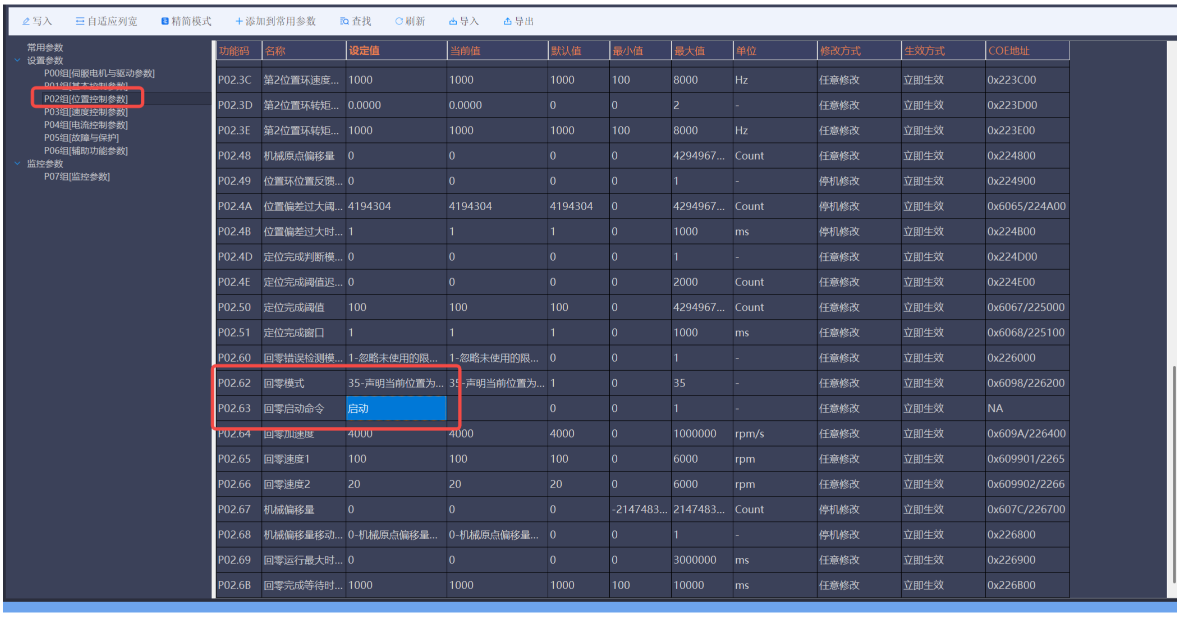
Position control parameter P02.63 (zeroing start) allows you to set the current position to zero with a double-click.

③ 点击保存。

Click Save

SDO 控制：设置模式 0x6060=6，0x6040=0x10，然后当前位置就是 0 点了，保存指令 0x2130=1。

SDO control: set mode 0x6060=6, 0x6040=0x10, then the current position will be zero, and save the command 0x2130=1.



功能码	名称	设定值	当前值	默认值	最小值	最大值	单位	修改方式	生效方式	COE地址
P02.3C	第2位置环速度...	1000	1000	1000	100	8000	Hz	任意修改	立即生效	0x223C00
P02.3D	第2位置环转矩...	0.0000	0.0000	0	0	2	-	任意修改	立即生效	0x223D00
P02.3E	第2位置环转矩...	1000	1000	1000	100	8000	Hz	任意修改	立即生效	0x223E00
P02.48	机械原点偏移量	0	0	0	0	4294967...	Count	任意修改	立即生效	0x224800
P02.49	位置环位置反馈...	0	0	0	0	1	-	停机修改	立即生效	0x224900
P02.4A	位置偏差过大阈...	4194304	4194304	4194304	0	4294967...	Count	停机修改	立即生效	0x6065/224A00
P02.4B	位置偏差过大时...	1	1	1	0	1000	ms	停机修改	立即生效	0x224B00
P02.4D	定位完成判断模...	0	0	0	0	1	-	任意修改	立即生效	0x224D00
P02.4E	定位完成阈值迟...	0	0	0	0	2000	Count	任意修改	立即生效	0x224E00
P02.50	定位完成阈值	100	100	100	0	4294967...	Count	任意修改	立即生效	0x6067/225000
P02.51	定位完成窗口	1	1	1	0	1000	ms	任意修改	立即生效	0x6068/225100
P02.60	回零错误检测模...	1-忽略未使用的限...	1-忽略未使用的限...	0	0	1	-	任意修改	立即生效	0x226000
P02.62	回零模式	35-声明当前位置为...	35-声明当前位置为...	1	0	35	-	任意修改	立即生效	0x6098/226200
P02.63	回零启动命令	启动		0	0	1	-	任意修改	立即生效	NA
P02.64	回零加速度	4000	4000	4000	0	1000000	rpm/s	任意修改	立即生效	0x609A/226400
P02.65	回零速度1	100	100	100	0	6000	rpm	任意修改	立即生效	0x609901/2265
P02.66	回零速度2	20	20	20	0	6000	rpm	任意修改	立即生效	0x609902/2266
P02.67	机械偏移量	0	0	0	-2147483...	2147483...	Count	停机修改	立即生效	0x607C/226700
P02.68	机械偏移量移动...	0-机械原点偏移量...	0-机械原点偏移量...	0	0	1	-	停机修改	立即生效	0x226800
P02.69	回零运行最大时...	0	0	0	0	3000000	ms	任意修改	立即生效	0x226900
P02.6B	回零完成等待时...	1000	1000	1000	100	10000	ms	任意修改	立即生效	0x226B00

图 5-4 设置零位参数

Figure 5-4 Set Zero Position Parameter

5.6 制动器控制与使能逻辑

Brake Control and Enabling Logic

关节模组中的制动器是断电抱闸、通电松闸，在上位机参数控制里可选择制动器的打开是手动还是自动。

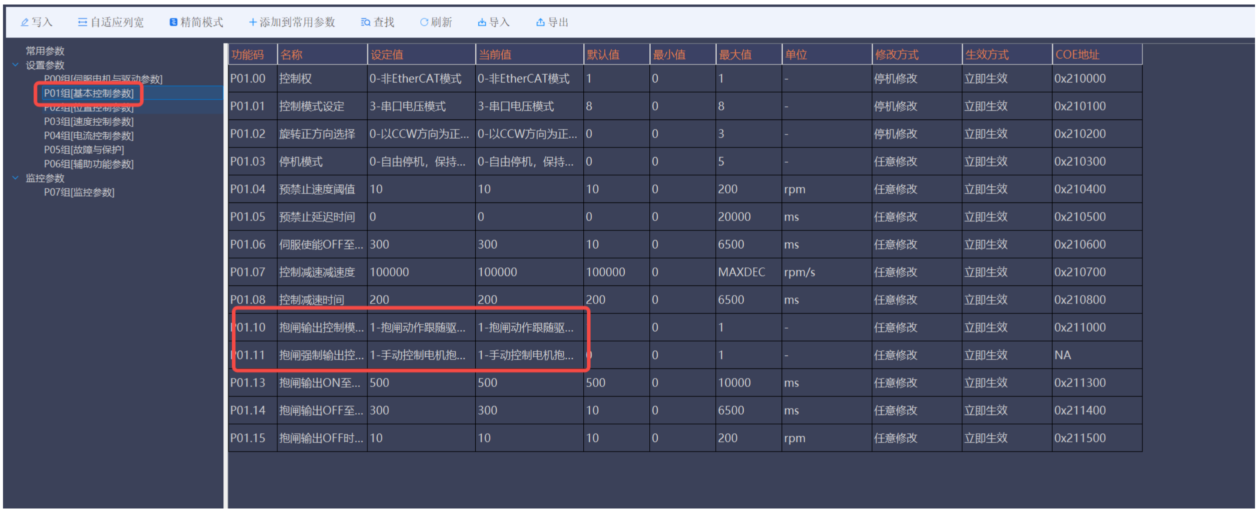
The brake in the joint module is a power-off brake, which releases when powered on. In the upper computer parameter control, you can select whether the brake should be manually or automatically released.

① 手动开制动器：与驱动器使能无关，只需给控制指令，即可打开关闭制动器。

Manual Brake Release: It is independent of the driver enable signal; the brake can be released or applied simply by sending a control command.

② 自动开制动器：跟随驱动器使能命令，在伺服驱动器已使能且稳定（已经建立足够扭矩）后，才能松开抱闸，在制动器完全抱闸后，才能断开伺服驱动器使能。

Automatic Brake Release: Follows the driver enable command. The brake will only release once the servo driver is enabled and stable (enough torque has been established). The servo driver enable will only be disabled after the brake is fully engaged.



功能码	名称	设定值	当前值	默认值	最小值	最大值	单位	修改方式	生效方式	COE地址
P01.00	控制权	0-非EtherCAT模式	0-非EtherCAT模式	1	0	1	-	停机修改	立即生效	0x210000
P01.01	控制模式设定	3-串口电压模式	3-串口电压模式	8	0	8	-	停机修改	立即生效	0x210100
P01.02	旋转正方向选择	0-以CCW方向为正...	0-以CCW方向为正...	0	0	3	-	停机修改	立即生效	0x210200
P01.03	停机模式	0-自由停机，保持...	0-自由停机，保持...	0	0	5	-	任意修改	立即生效	0x210300
P01.04	预禁止速度阈值	10	10	10	0	200	rpm	任意修改	立即生效	0x210400
P01.05	预禁止延迟时间	0	0	0	0	20000	ms	任意修改	立即生效	0x210500
P01.06	伺服使能OFF至...	300	300	10	0	6500	ms	任意修改	立即生效	0x210600
P01.07	控制减速减速度	100000	100000	100000	0	MAXDEC	rpm/s	任意修改	立即生效	0x210700
P01.08	控制减速时间	200	200	200	0	6500	ms	任意修改	立即生效	0x210800
P01.10	抱闸输出控制模...	1-抱闸动作跟随驱...	1-抱闸动作跟随驱...		0	1	-	任意修改	立即生效	0x211000
P01.11	抱闸强制输出控...	1-手动控制电机抱...	1-手动控制电机抱...	0	0	1	-	任意修改	立即生效	NA
P01.13	抱闸输出ON至...	500	500	500	0	10000	ms	任意修改	立即生效	0x211300
P01.14	抱闸输出OFF至...	300	300	10	0	6500	ms	任意修改	立即生效	0x211400
P01.15	抱闸输出OFF时...	10	10	10	0	200	rpm	任意修改	立即生效	0x211500

图 5-5 制动器控制参数

Figure 5-5 Brake Control Parameters

5.7 STO（安全转矩关闭）功能

STO (Safe Torque Off) Function

表 5-1 STO 配置参数

Table 5-1 STO Configuration Parameters

参数名称 Parameter Name	串口地址 Serial Port Address	范围 Range	默认值 Default Value	功能说明 Function Description
STO 故障	P053B	0~3	0	定义 STO 安全输入被触发（断开）时，驱动

模式 STO Fault Mode	（0x253B）			器的响应行为： Defines the drive's response behavior when the STO safety input is triggered (open): **0：关闭** -STO 功能失效，不产生任何提示。（仅用于调试，禁止用于运行） Disabled - The STO function is deactivated, and no prompt is generated. (For debugging only, prohibited during operation) **1：智能响应**-STO 触发时，若驱动器未使能则产生报警；若已使能则报故障并紧急停机。 Smart Response - When STO is triggered, an alarm is generated if the drive is not enabled; if enabled, a fault is reported, and an emergency stop is initiated. **2：仅报警**-STO 触发时，产生报警，恢复 STO 信号即正常。 Alarm Only - When STO is triggered, an alarm is generated; normal operation resumes once the STO signal is restored. **3：仅故障**-STO 触发时，立即报故障并紧急停机。 Fault Only - When STO is triggered, a fault is reported immediately, and an emergency stop is initiated.
STO 故障恢复模式 STO Fault	P053C （0x253C）	0~1	0	定义 STO 安全输入恢复（闭合）后，驱动器的恢复方式： Defines the drive's recovery method after the

Recovery Mode				STO safety input is restored (closed): **0：手动恢复**- STO 故障/报警需手动清除（复位）后才能继续运行。 Manual Recovery - The STO fault/alarm must be cleared manually (reset) before operation can continue. **1：自动恢复**- STO 信号恢复后，故障/报警自动清除，驱动器可根据使能信号继续或恢复运行。 Auto Recovery - After the STO signal is restored, the fault/alarm is cleared automatically, and the drive can continue or resume operation based on the enable signal.

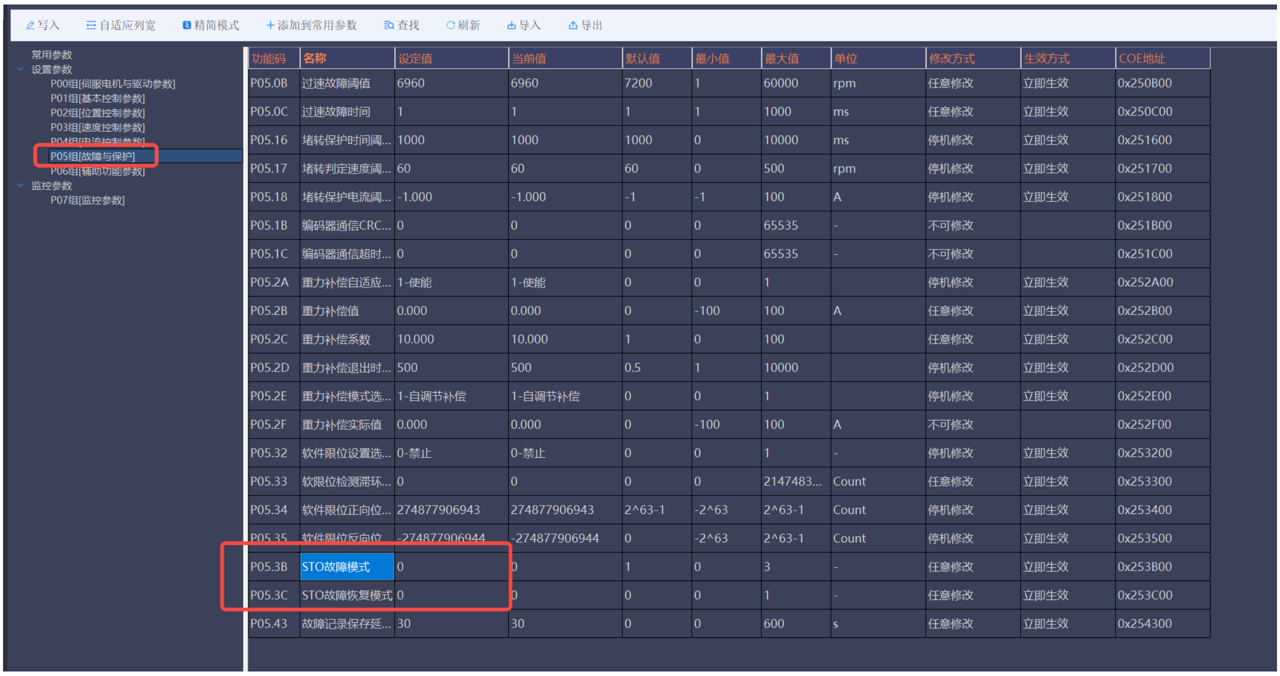


图 5-6 STO 功能地址

Figure 5-6 STO Functional Address

配置建议说明：

Configuration Recommendations:

安全应用：在需要功能安全的场合，建议将 STO 故障模式设置为 1 或 3，以确保运行中断动能立即停机；STO 故障恢复模式 应设置为 0（手动恢复），强制人工介入确认安全。

Safety Applications: In applications requiring functional safety, it is recommended to set the STO Fault Mode to 1 or 3 to ensure immediate stoppage upon activation. The STO Fault Recovery Mode should be set to 0 (Manual Recovery) to enforce manual intervention for confirmation of safety.

调试应用：在调试阶段，可临时设置为 2-仅报警和 1-自动恢复，以便于排查线路，但必须注意安全风险。

Debugging Applications: During the debugging phase, it can be temporarily set to 2 (Alarm Only) and 1 (Auto Recovery) to facilitate troubleshooting wiring, but safety risks must be noted.

警告：禁止在正常运行中将 STO 故障模式设为 0（关闭）。

Warning: It is prohibited to set the STO Fault Mode to 0 (Disabled) during normal operation.

5.8 固件升级

Firmware Upgrade

重要警告： 固件升级存在风险，操作不当可能导致设备损坏。请确保在升级前已备份重要参数。

Important Warning: Firmware upgrades carry risks, and improper operation may cause device damage. Please make sure to back up important parameters before upgrading.

1) 串口升级方法

Serial Port Upgrade Method

① 选择固件升级，选择.yam 文件；确定使用串口，点击升级，**部分版本有警告提醒可保存重启即可清除**。

Select firmware upgrade. Select the .yam firmware file.Confirm using the serial port.

Click Upgrade, Some versions have warnings that can be cleared by saving and

restarting.

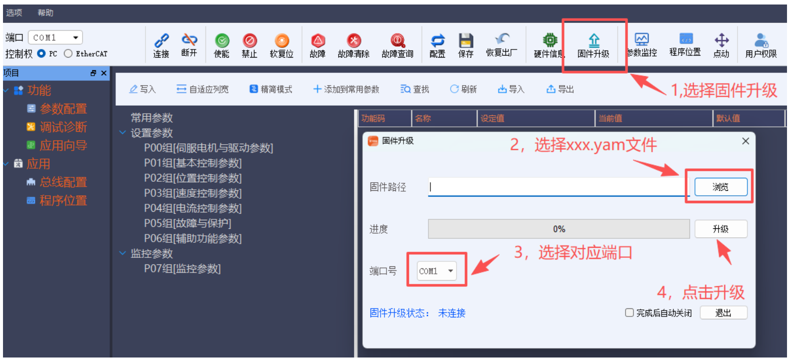


图 5-7 串口升级方法

Figure 5-7 Serial Port Upgrade Method

2) FOE 升级方法

FOE Upgrade Method

Twincat 主站升级：online 界面选择升级，切换至 boot 状态，选择升级固件，输入密码：0x87654321。等升级完成，切换至 init 状态，电机即可重启。

Twincat master station upgrade: In the online interface, select upgrade, switch to boot mode, choose to upgrade firmware, and enter the password: 0x87654321. Wait for the upgrade to complete, switch to init mode, and the motor can restart.

5.9 SDK 二次开发

SDK Secondary Development

PHU 系列 SDK 是一套完整的软件开发工具包，它提供了丰富的应用程序编程接口(API)。通过调用这些 API，高级用户或开发者可以在 PC、嵌入式主板（如树莓派、Jetson 系列）或自定义控制系统中，直接、灵活地控制关节。

获取 SDK：请访问我们的官方网站或联系技术支持获取最新版本的 SDK 及完整开发文档。

The PHU series SDK is a complete software development toolkit that provides a rich set of application programming interfaces (APIs). By calling these APIs, advanced users or developers can directly and flexibly control the joints in a PC, embedded motherboard (e.g., Raspberry Pi, Jetson series), or custom control system, achieving complex motion trajectories.

Getting the SDK: Please visit our official website or contact technical support to obtain the latest version of the SDK and complete development documentation.

六、操作与运行

Operation and Running

6.1 正常运行模式

Normal Operating Mode

6.1.1 位置模式

Position Mode

① 点击使能；

Click enable;

② 运动控制选项卡：运行模式选择位置模式；

Motion control tab: Select position mode for the operating mode;

③ 运行方式：增量、绝对、往复（增量：相对于当前位置移动的偏移量，绝对：运动到绝对的位置点，往复：当前位置与目标位置之间往返运动）；

Running mode: Incremental, Absolute, or Repetitive (Incremental: Movement offset relative to the current position, Absolute: Move to an absolute position, Repetitive: Back-and-forth motion between current and target positions);

④ 移动距离：脉冲数；

Distance: Pulse count;

⑤ 速度与加速度：给定允许的速度与加速度；

Speed and acceleration: Specify the allowed speed and acceleration;

⑥ 运行与停止：点击运行即可运动，停止则会停止运行。

Run and Stop: Click run to move, click stop to halt movement.

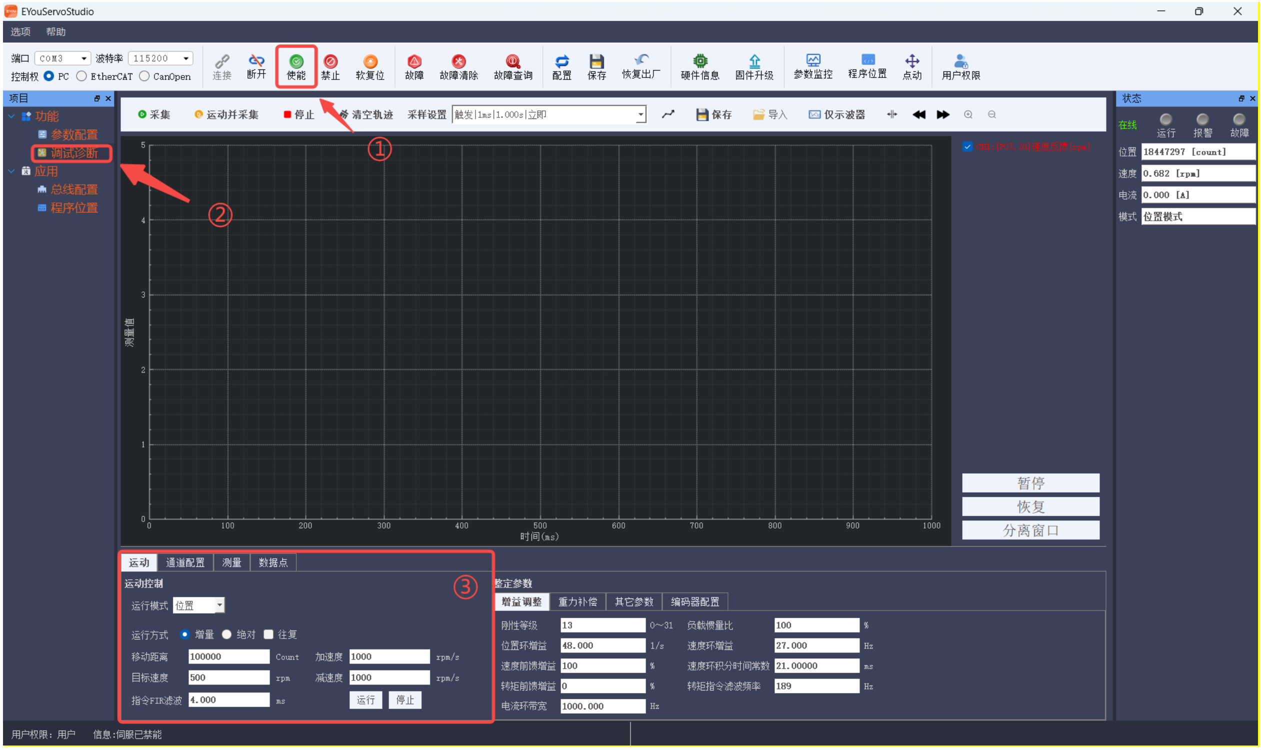


图 6-1 位置模式

Figure 6-1 Position Mode

6.1.2 速度模式

Speed Mode

① 点击使能；

Click enable;

② 运动控制选项卡：运行模式选择速度模式；

Motion control tab: Select speed mode for the operating mode;

③ 运行方式：单次、往复；

Running mode: Single, Repetitive;

④ 目标速度：目标速度 1、持续时间 1（0 为一直运行），目标速度 2、持续时间 2；

Target speed: Target speed 1, Duration 1 (0 for continuous operation), Target speed 2, Duration 2;

⑤ 速度与加速度：给定允许的速度与加速度；

Speed and acceleration: Specify the allowed speed and acceleration;

⑥ 运行与停止：点击运行即可运动，停止则会停止运行。

Run and Stop: Click run to move, click stop to halt movement.

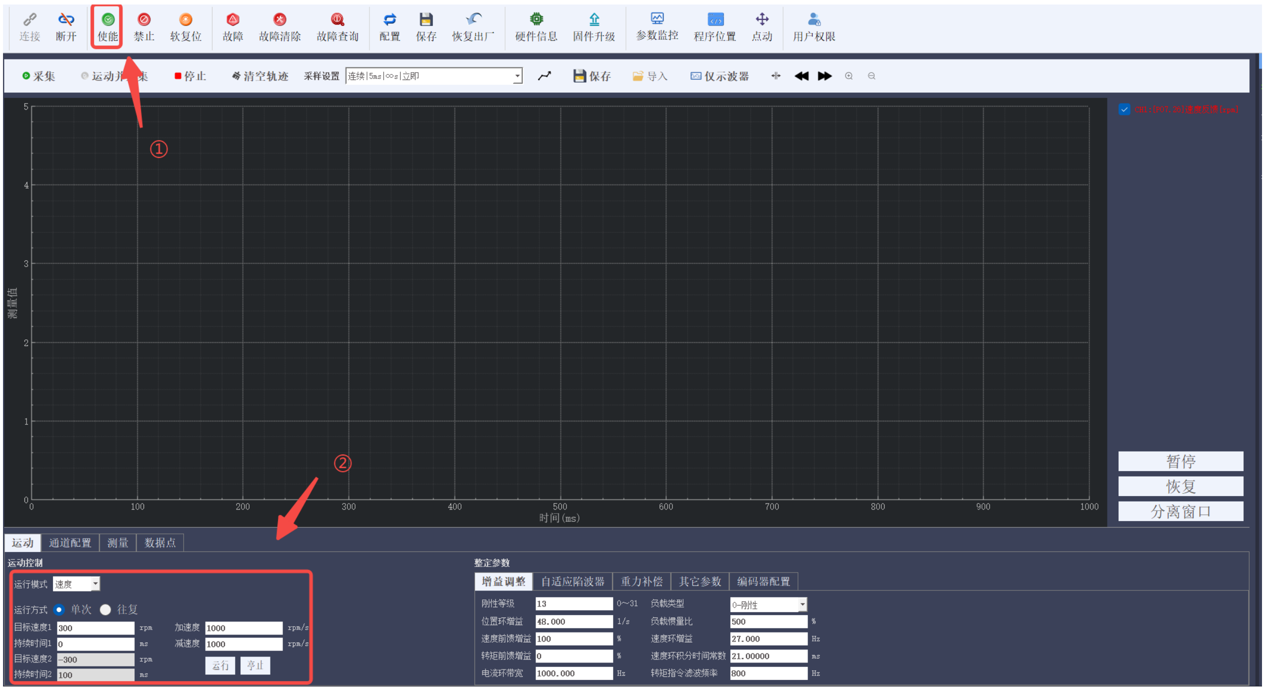


图 6-2 速度模式

Figure 6-2 Speed Mode

6.1.3 电流模式

Current Mode

① 点击使能；

Click enable;

② 运动控制选项卡：运行模式选择电流模式；

Motion control tab: Select current mode for the operating mode;

③ 运行方式：阶跃、脉冲；

Running mode: Step, Pulse;

④ 电流指令：给定 Iq 电流；

Current command: Specify the Iq current;

⑤ 持续时间：脉冲模式生效；

Duration: Pulse mode active;

⑥ 运行与停止：点击运行即可运动，停止则会停止运行。

Run and Stop: Click run to move, click stop to halt movement.

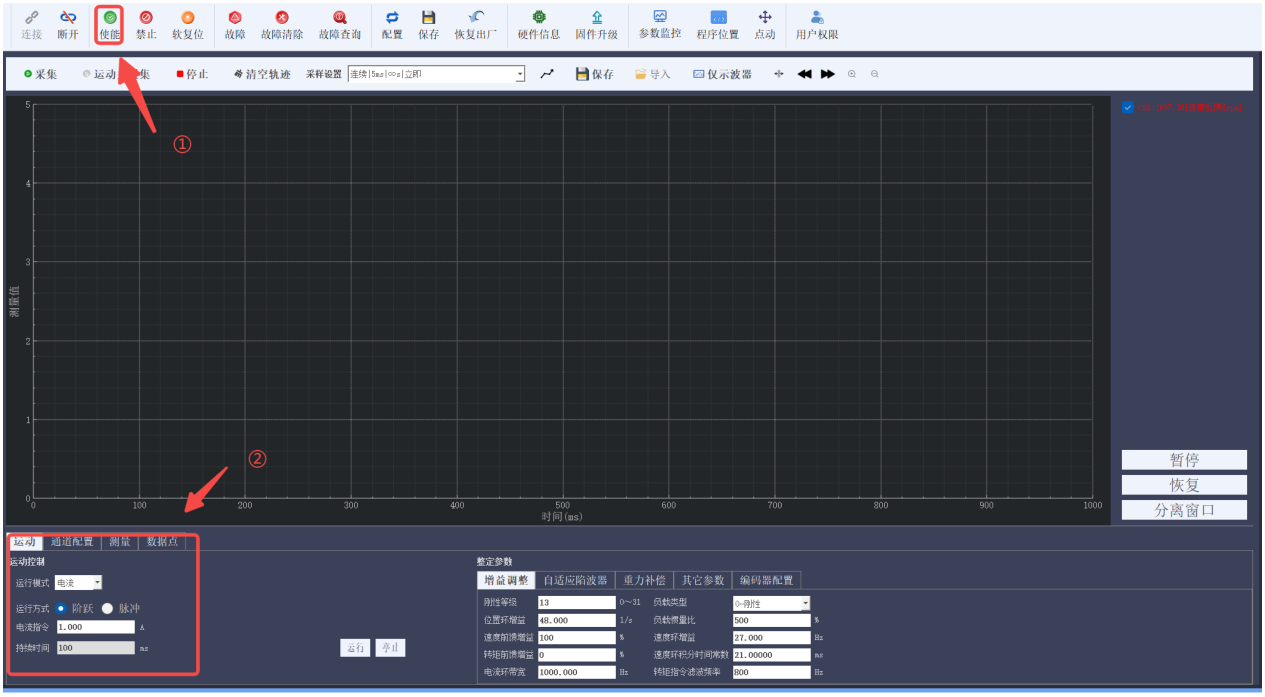


图 6-3 电流模式

Figure 6-3 Current Mode

6.1.4 力矩模式

Torque Mode

①写入扭矩偏移量：空载状态将扭矩反馈（P07BD/0x27BD）写入扭矩偏移量（P0452/0x2452）；

Write torque offset: In no-load condition, write the torque feedback (P07BD/0x27BD) into the torque offset (P0452/0x2452);

②点击使能；

Click to enable;

③关节在外力为 0，扭矩反馈为 0 状态不会运动，当扭矩反馈与给定不一致时关节会运动；

The joint will not move when external force is zero and torque feedback is zero. The joint will move when the torque feedback does not match the set value;

④力矩控制环参数：额定扭矩（P0087/0x2087）、扭矩环比例增益（P0450/0x2450）、扭矩环积分增益（P0451/0x2451）、扭矩反馈低通滤波器频率（P0453/0x2453）；

Torque control loop parameters: rated torque (P0087/0x2087), torque loop proportional gain (P0450/0x2450), torque loop integral gain (P0451/0x2451), torque feedback low-pass filter frequency (P0453/0x2453);

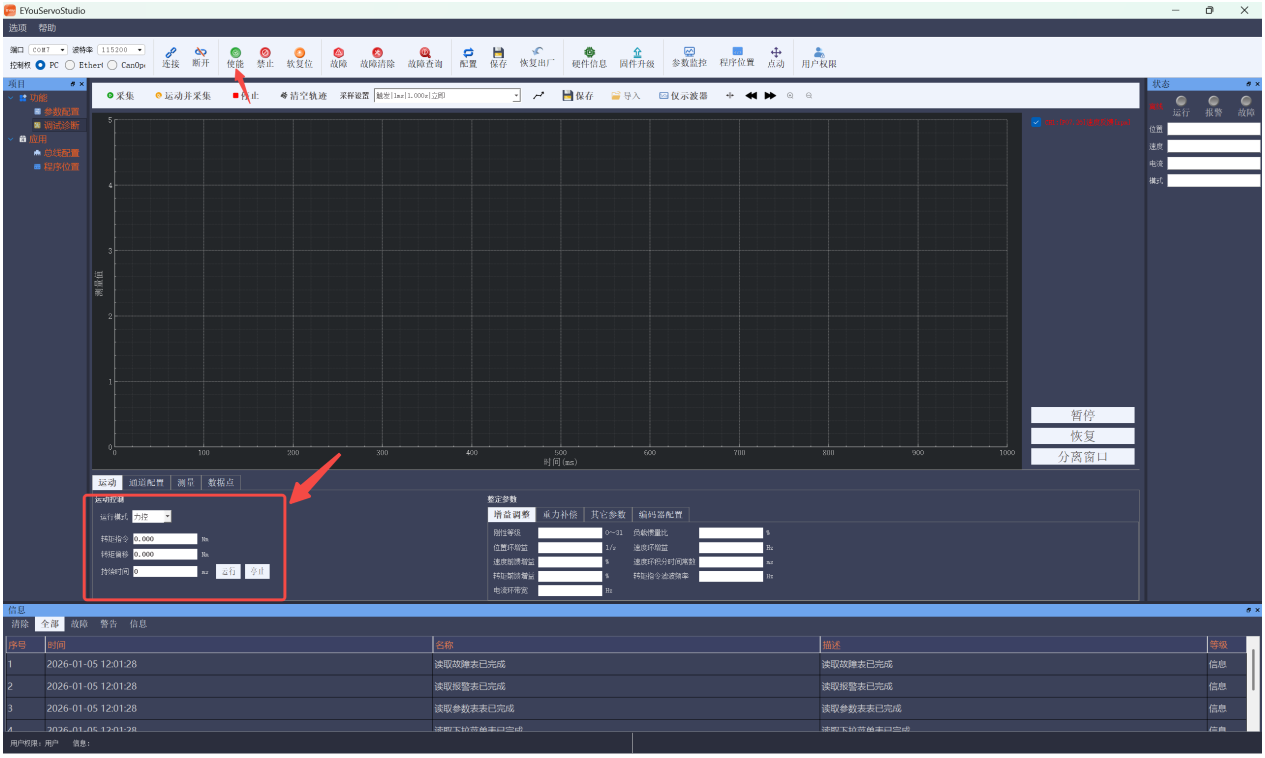


图 6-4 力控模式

Figure 6-4 Current Mode

6.2 状态监控与数据读取

Status Monitoring and Data Collection

6.2.1 状态监控

Status Monitoring

选项窗口有个状态显示，可以显示关节的报警、位置、速度、电流、模式。

The options window has a status display that shows joint alarms, position, speed, current, and mode.

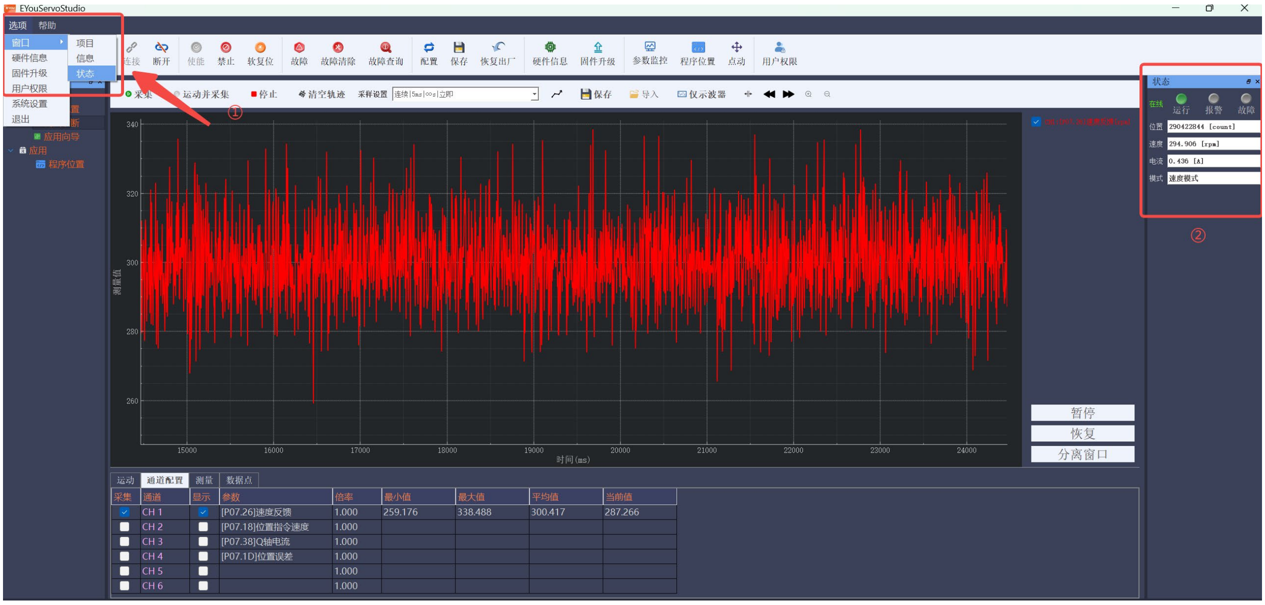


图 6-5 状态显示

Figure 6-5 Status Display

6.2.2 数据采集

Data Collection

① 通道配置：配置需要采集的数据；

Channel configuration: Configure the data to be collected;

② 选择数据采集模式：触发、连续；

Select the data collection mode: Triggered, Continuous;

③ 采集频率：最快 200HZ；

Collection frequency: Up to 200Hz;

④ 点击采集：即可显示波形。

Click collect: The waveform will be displayed.

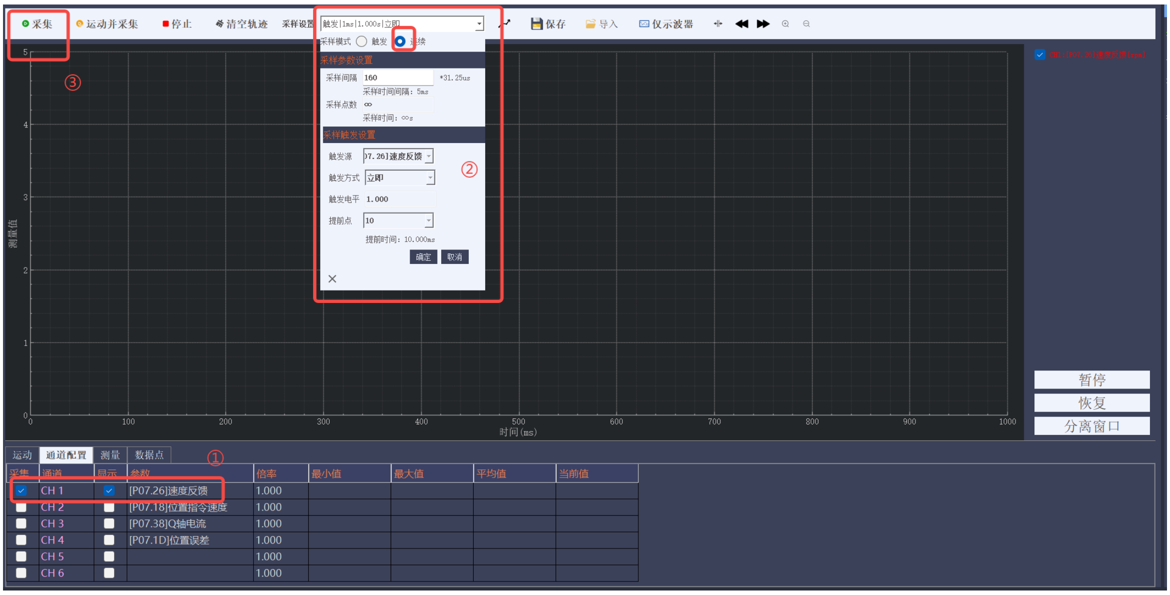


图 6-5 数据采集

Figure 6-5 Data Collection

6.2.3 数据保存

Data Saving

① 停止采集：点击停止采集数据；

Stop collection: Click to stop data collection;

② 保存数据：保存为图片或者文件 CSV。

Save data: Save as an image or CSV file.

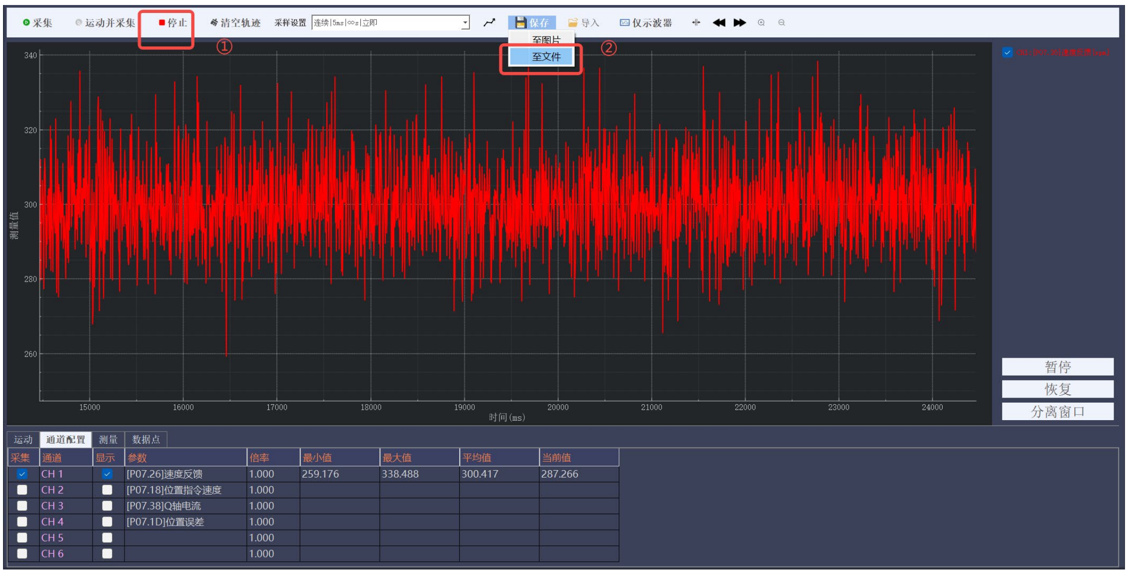


图 6-6 数据保存

Figure 6-6 Data Saving

6.3 异常状态处理与复位

Abnormal Status Handling and Reset

若关节出现报警，可以尝试清除故障，在故障与报警栏可以显示故障原因。

If an alarm occurs in the joint, you can attempt to clear the fault. The fault cause will be displayed in the fault and alarm section.

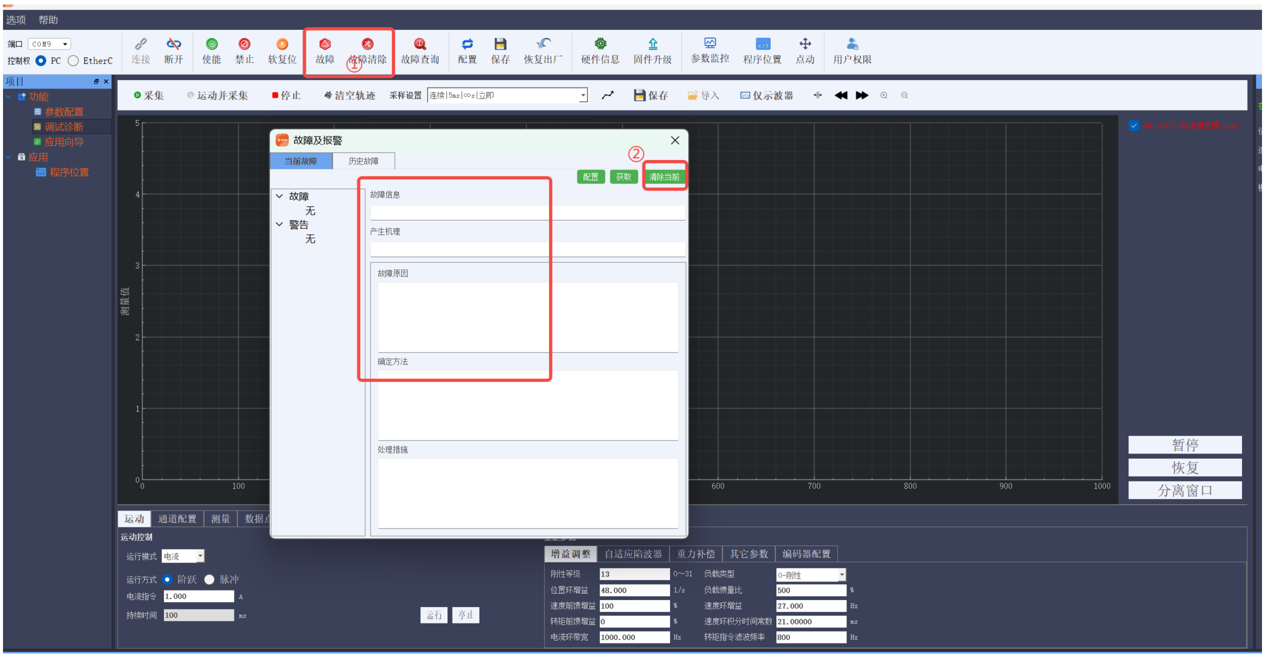


图 6-7 异常报警

Figure 6-7 Abnormal Alarm

6.4 常用功能

Common Parameter Functions

6.4.1 重力补偿

Gravity Compensation

表 6-1 重力补偿参数

Table 6-1 Gravity Compensation Parameters

参数串口地址/总线地址 Parameter Serial Address / Bus Address	参数名称 Parameter Name	功能 Function
P05.2A/0x252A	重力补偿自适应使能 Gravity Compensation	0- 禁止；1-使能; 0 - Disable; 1 - Enable

	Adaptive Enable	在“自调节补偿”模式下，系统会自动学习和更新重力补偿所需的电流值。在“固定值补偿”模式下，此功能通常不激活。 In the "Self-tuning Compensation" mode, the system will automatically learn and update the current value required for gravity compensation. In the "Fixed Value Compensation" mode, this function is typically not activated.
P05.2B/0x252B	重力补偿值 Gravity Compensation Value	这是一个基准值。在“固定值补偿”模式下，驱动器会持续输出这个电流。 This is a base value. In the "Fixed Value Compensation" mode, the drive will continuously output this current.
P05.2C/0x252C	重力补偿系数 Gravity Compensation Coefficient	一个比例系数，用于微调“重力补偿值”的实际输出效果。 A proportional coefficient used to fine-tune the actual output effect of the "Gravity Compensation Value".
P05.2D/0x252D	重力补偿退出时间 Gravity Compensation Ramp-down Time	当重力补偿需要停止时（例如切换到非重力补偿模式，或收到停止命令），补偿电流逐渐减小到零所用的时间。 The time it takes for the compensation current to gradually decrease to zero when gravity compensation needs to stop (e.g., when switching to a non-gravity

		compensation mode, or upon receiving a stop command).
P05.2E/0x252E	重力补偿模式选择 Gravity Compensation Mode Selection	用法（各模式详解）： Usage (Explanation of Each Mode): 0 - 固定值补偿：最简单。驱动器持续输出设定的“重力补偿值”。适用于负载恒定、重力力矩不变的场景。 0 - Fixed Value Compensation: The simplest. The drive continuously outputs the set "Gravity Compensation Value". Suitable for scenarios with constant load and unchanging gravitational torque. 1 - 自调节补偿：系统（通常在使能时）自动运行一小段距离，通过检测维持速度/位置所需的电流，自动计算并更新“重力补偿值”。适用于负载会变化，或每次安装负载不同的场合。 1 - Self-tuning Compensation: The system (usually upon enabling) automatically moves a small distance, detects the current required to maintain speed/position, and automatically calculates and updates the "Gravity Compensation Value". Suitable for situations where the load may change, or the installed load differs each time.
P05.2F/0x252F	重力补偿实际值	这是一个只读的监视参数，显示当前重

	Gravity Compensation Actual Value	力补偿实际输出的电流值。 This is a read-only monitoring parameter, displaying the current actual output current value of the gravity compensation.

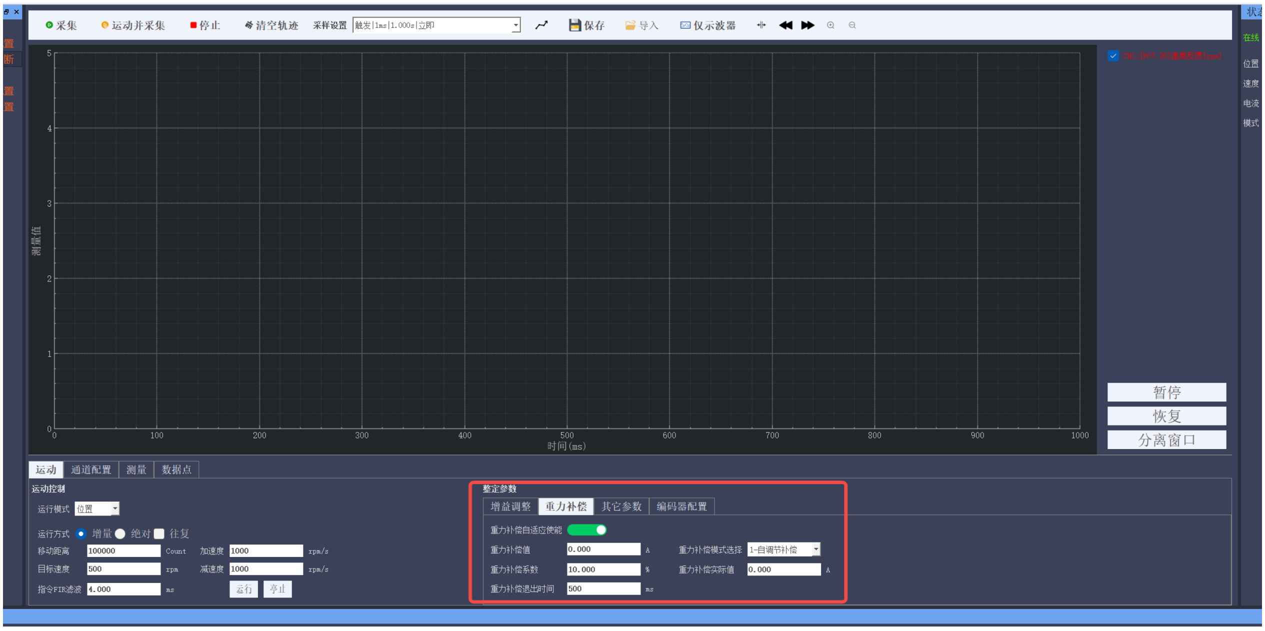


图 6-8 重力补偿

Figure 6-8 Gravity Compensation

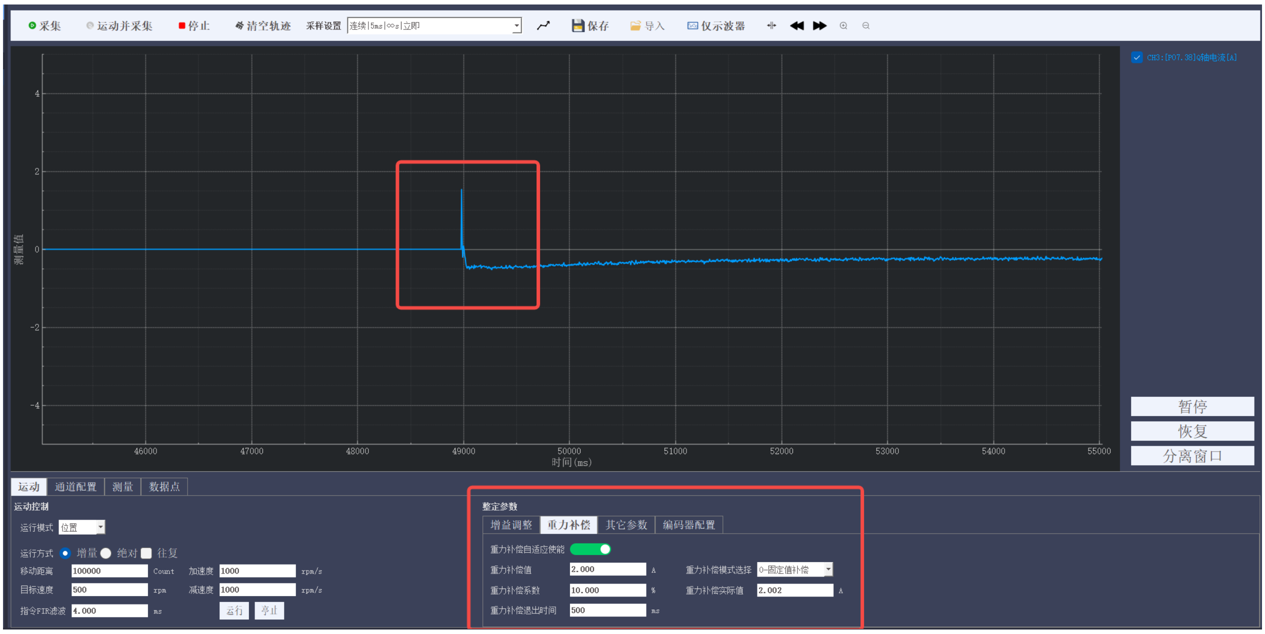


图 6-9 重力补偿效果

Figure 6-9 Gravity Compensation Effect

七、维护与保养

Maintenance and Care

7.1 日常检查项目

Daily Inspection Items

日常检查旨在通过感官和简单工具发现潜在问题，防患于未然。

Daily inspections aim to detect potential problems through sensory checks and simple tools to prevent issues before they arise.

表 7-1 日常检查项目

Table 7-1 Routine Inspection Items

检查项目 **Inspection Item**	方法 **Method**	正常 **Normal Condition**	异常 **Abnormal Condition**
异响与振动检查 Abnormal Noise & Vibration Check	在空载和负载低速运行状态下，仔细听运行声音 Listen carefully during low-speed operation under both no-load and loaded conditions.	平稳、均匀的齿轮啮合声 Smooth, uniform gear meshing sound.	尖锐的嘎吱声、周期性的撞击声、不规则的咔哒声或异常振动。这通常是润滑不良、齿轮磨损或轴承损坏的早期迹象 Sharp screeching, periodic knocking, irregular clicking, or abnormal vibration. These are often early signs of poor lubrication, gear wear, or bearing damage.
温升检查 Temperature Rise Check	运行停止后（或在安全条件下通过红外测温枪），触摸或测量模组外壳温度 Touch or measure the	温升在合理范围内（通常外壳温度不超过环境温度+60°C） Temperature rise	局部或整体过热、烫手。表明可能存在过载、润滑不足或装配不当等问题 Localized or overall overheating, too hot to touch.

	module housing temperature after operation stops (or safely with an infrared thermometer).	is within a reasonable range (typically, housing temperature does not exceed ambient temperature +60°C).	Indicates potential issues like overloading, insufficient lubrication, or improper assembly.
泄漏检查 Leakage Check	目视检查减速器密封处（输出壳体外圈）是否有油渍或油脂渗出 Visually inspect the reducer seals (around the output shaft housing) for any oil stains or grease seepage.	完全干燥，无任何油渍 Completely dry, no signs of oil or grease.	有明显的油脂泄漏。表明密封圈可能老化或损坏，需要关注 Visible grease leakage. Indicates that seals may be aged or damaged and requires attention.
外观与清洁检查 Appearance & Cleanliness Check	检查模组表面是否有磕碰、划伤、锈蚀，并清除表面的灰尘、油污、切削液等污染物 Check the module surface for bumps, scratches, corrosion, and remove contaminants like dust,	外观干净 Clean appearance.	外观有异常锈蚀或污染物 Presence of abnormal corrosion or contaminants.

	oil, or cutting fluid.		
运行性能 Operational Performance	观察运行是否平稳，有无卡顿、抖动或定位精度下降现象 Observe operation for smoothness, checking for jitter, stuttering, or degradation in positioning accuracy.	运行正常 Normal operation.	任何性能下降都可能是内部故障的前兆 Any performance degradation could be a precursor to internal failure.

7.2 定期维护

Periodic Maintenance

定期维护是计划性的、更深入的保养工作，需根据运行时间或使用周期严格执行。

Periodic maintenance is planned and more in-depth maintenance that must be performed strictly according to operating time or usage cycles.

表 7-2 日常检查项目

Table 7-2 Routine Maintenance Items

维护项目 **Maintenance Item**	周期 **Frequency**	内容与步骤 **Procedure and Details**
紧固件检查 Fastener Inspection	每月或每运行 500 小时 Monthly or every 500 operating hours	检查并重新紧固模组与机器人结构之间的安装螺栓、电机连接螺栓等所有紧固件。 Inspect and re-tighten all fasteners, including the mounting bolts between the module and the robot structure, as well as the motor connection bolts.

密封件更换 Seal Replacement	每次更换润滑脂时或发现泄漏时 When replacing grease or if leakage is detected	检查输入轴和输出端的密封圈（油封），如有老化、硬化、裂纹或磨损，必须更换。 Inspect the sealing rings (oil seals) on the input shaft and output side. They must be replaced if aging, hardening, cracking, or wear is found.
润滑脂更换 Grease Replacement	/	/

7.3 存储条件与期限

Storage Conditions and Period

1） 存储环境：

Storage Environment:

① 温度与湿度：应存储在干燥、无冷凝的环境中。理想温度为 5°C ~ 35°C，相对湿度应低于 70%。

Temperature and Humidity: Store in a dry, non-condensing environment. The ideal temperature is 5°C ~ 35°C, and relative humidity should be below 70%.

② 环境：远离腐蚀性化学品、酸、碱及盐分。防止灰尘和金属粉末侵入。

Environment: Keep away from corrosive chemicals, acids, bases, and salts. Prevent dust and metal powders from entering.

③ 振动：存放地点应无强烈振动或冲击。

Vibration: The storage location should be free from strong vibrations or shocks.

2） 存储姿态：以正常工作姿态放置于货架或垫板上，避免不必要的受力。

Storage Orientation: Store in the normal operating orientation on shelves or pads to avoid unnecessary stress.

3） 防护措施：

Protective Measures:

① 所有外露加工面（如法兰面、轴端）应涂抹防锈油（如薄层凡士林或专用防锈剂）。

Apply anti-rust oil (such as a thin layer of petroleum jelly or special anti-rust agent) to all exposed processed surfaces (e.g., flange faces, shaft ends).

② 用防锈纸、塑料罩或防护帽覆盖接口和开口，防止灰尘和湿气进入。

Cover interfaces and openings with anti-rust paper, plastic covers, or protective caps to prevent dust and moisture from entering.

③ 对于全新未使用模组：内部已注有适量润滑脂，无需特殊处理。但长期存储需定期运行（建议每 3-6 个月空载运行），使润滑脂均匀分布，防止轴承和齿轮工作表面锈蚀。

For new, unused modules: An appropriate amount of grease is pre-lubricated internally, requiring no special initial handling. However, for long-term storage, it is recommended to perform periodic no-load operation (suggested every 3-6 months) to redistribute the grease evenly, preventing corrosion on the bearing and gear working surfaces.

4）存储期限：通常不超过 1 年，如果超过 1 年，在使用前，必须进行全面的检查，特别是密封件是否老化。

Storage Period: Typically, do not exceed 1 year. If stored for longer than 1 year, a comprehensive inspection must be carried out before use, especially for sealing components that may have aged.

7.4 维修与更换建议

Repair and Replacement Recommendations

维修注意事项：

Repair Notes:

① 不可维修性：谐波减速器（柔轮、刚轮、波发生器）是高度精密的部件。严禁用户或现场工程师自行拆解维修。任何非授权的拆装都可能导致永久性损坏和精度完全

丧失；

Irreparability: The harmonic reducer (flexible wheel, rigid wheel, and wave generator) is a highly precise component. Users or on-site engineers must not attempt disassembly or repair. Unauthorized disassembly may result in permanent damage and loss of precision.

② 故障判断：如果日常和定期维护中发现严重异常（如持续异响、严重过热、巨大背隙、卡死），应立即停机；

Fault Diagnosis: If serious abnormalities (e.g., continuous abnormal sounds, severe overheating, large backlash, or jamming) are found during daily or periodic maintenance, stop the machine immediately.

③ 专业返厂：唯一的维修途径是将其返回给原制造商或授权维修中心。

Professional Return-to-Factory Service: The only authorized repair method is to return the module to the original manufacturer or an authorized service center.

寿命与更换建议：

Service Life and Replacement Recommendations:

谐波减速器的寿命通常以额定负载和转速下的运行小时数定义。即使保养得当，它也是一个磨损件。当出现以下情况时，应计划更换：

The service life of a harmonic reducer is typically defined by its operating hours under rated load and speed. Even with proper maintenance, it is a wear part. Replacement should be planned when any of the following conditions occur:

① 精度下降，无法通过控制系统补偿；

Positioning accuracy degrades beyond the compensation capability of the control system.

② 背隙超过允许值，影响机器人绝对精度；

Backlash exceeds the allowable tolerance, adversely affecting the robot's absolute accuracy.

③ 运行噪音和振动显著增大，经检查确认为内部损坏。

Operational noise and vibration increase significantly, and internal damage is confirmed through inspection.

八、故障诊断与排除

Fault Diagnosis and Troubleshooting

8.1 故障代码

Fault Codes

故障及警告：关节的故障和警告按严重程度分级，可分为三级，第一类故障（I）、第二类故障（II）、警告（III），严重程度：第一类故障 > 第二类故障 > 警告。

Faults and Warnings: The faults and warnings of the joint are classified into three levels based on severity: Class I faults (I), Class II faults (II), and warnings (III), with severity ranked as: Class I faults > Class II faults > warnings.

故障记录：EYouDrive 可记录最近 40 次的故障码以及故障发生时的时间等信息，可以通过故障记录回溯功能查询过往故障记录清单。

Fault Records: EYouDrive can record the last 40 fault codes along with the time of occurrence. The fault record trace-back function allows you to query past fault records.

故障清除：在关节使能命令撤销，同时故障条件不再存在时，通过外部指令清除故障，使得驱动器恢复到正常工作状态。若同一故障反复出现，请立即停止操作，并联系技术支持，以免造成设备进一步损坏。清除故障的方法包括：

Fault Clearing: Faults are cleared through external instructions when the joint enable command is canceled and the fault condition no longer exists, allowing the driver to return to normal operation. If the same fault repeatedly occurs, stop operations immediately and contact technical support to prevent further damage to the equipment. Methods of clearing faults include:

通过 EYouStudio 软件工具栏的"故障清除"按键；

Using the "Fault Clear" button on the EYouStudio software toolbar;

通过 EtherCAT 上位机设置控制字 0x6040 的 bit7 产生上升沿；

Using the EtherCAT upper computer to set the control word 0x6040 bit7 to trigger a rising edge;

对于警告，当警告源消除后，警告自动清除。

Using the EtherCAT upper computer to set the control word 0x6040 bit7 to trigger a rising edge;

表 8-1 故障与警告码

Table 8-1 Fault and Warning Codes

故障码 Code	故障名称 Description	故障类型 Type	故障等级 Level	故障恢复 Auto Reset
E103 （0x5210）	电流传感器异常 Current Sensor Error	驱动器故障 Drive Fault	I	否 No
E105 （0x5211）	使能时电流传感器异常 Current Sensor Error at Enable	驱动器故障 Drive Fault	I	否 No
E107 （0x6321）	写 EEPROM 失败 EEPROM Write Failure	驱动器故障 Drive Fault	II	是 YES
E120 （0x2222）	硬件过流 Hardware Over-Current	驱动器故障 Drive Fault	I	否 No
E121 （0x2221）	软件过流 Software Over-Current	驱动器故障 Drive Fault	II	是 YES
E123 （0x4210）	电机过温 Motor Over-Temperature	驱动器故障 Drive Fault	II	是 YES
E124 （0x4310）	驱动 MOS 过温 DriveMOSFETOver-Temperature	驱动器故障 Drive Fault	II	是 YES
E125 （0x2350）	驱动过载 Drive Overload	驱动器故障 Drive Fault	II	是 YES
E126 （0x7121）	堵转故障 Motor Stall	驱动器故障 Drive Fault	II	是 YES
E127 （0x2351）	电机过载 Motor Overload	驱动器故障 Drive Fault	II	是 YES
E129	电机缺相	驱动器故障	II	是

（0x7122）	Motor Phase Loss	Drive Fault		YES
E130 （0x3210）	直流母线过压 DC Bus Over-Voltage	驱动器故障 Drive Fault	II	是 YES
E131 （0x3220）	直流母线欠压 DC Bus Under-Voltage	驱动器故障 Drive Fault	II	是 YES
E139 （0xFF04）	内部错误 Internal Error	驱动器故障 Drive Fault	I	否 No
E140 （0xFF10）	驱动器参数初始化出错 Drive Parameter Initialization Error	驱动器故障 Drive Fault	I	否 No
E141 （0x6322）	MCU 参数初始化校验错误 MCU Parameter Verification Error	驱动器故障 Drive Fault	I	否 No
E157 （0x6323）	EEPROM 版本信息错误 EEPROM Version Error	驱动器故障 Drive Fault	I	否 No
E158 （0x6320）	参数 EEPROM 错误 Parameter EEPROM Error	驱动器故障 Drive Fault	I	否 No
E160 （0xFF24）	请执行保存和断电重启动作 Please Save Parameters and Power Cycle	驱动器故障 Drive Fault	I	否 No
E161 （0xFF03）	请执行配置动作 Please Perform Configuration	驱动器故障 Drive Fault	II	是 YES
E203 （0x7510）	第一编码器通讯断线 Primary Encoder Communication Loss	电机编码器故障 Motor/Encoder Fault	I	是 YES
E204 （0x7511）	第二编码器通讯断线 Secondary Encoder Communication Loss	电机编码器故障 Motor/Encoder Fault	I	是 YES

E206 （0x7521）	编码器型号初始化失败 Encoder Model Initialization Failure	电机编码器故障 Motor/Encoder Fault	I	是 YES
E209 （0x7515）	第一编码器 CRC 异常 Primary Encoder CRC Error	电机编码器故障 Motor/Encoder Fault	I	是 YES
E210 （0x7513）	第二编码器 CRC 异常 Secondary Encoder CRC Error	电机编码器故障 Motor/Encoder Fault	I	是 YES
E212 （0x7514）	双编码器位置偏差过大 Dual Encoder Position Mismatch	电机编码器故障 Motor/Encoder Fault	I	是 YES
E230 （0xFF11）	加载电机参数出错 Motor Parameter Loading Error	电机编码器故障 Motor/Encoder Fault	I	是 YES
E301 （0x5454）	驱动禁止输入保护（STO） Drive Safe Torque Off (STO)	应用故障 Application Fault	I	否 No
E302 （0x8611）	位置偏差过大 Position Error Excess	应用故障 Application Fault	II	是 YES
E303 （0x8400）	速度偏差过大 Velocity Error Excess	应用故障 Application Fault	II	是 YES
E304 （0x8401）	过速故障 Over-Speed Fault	应用故障 Application Fault	II	是 YES
E308 （0xFF35）	电子齿轮比计算溢出 Electronic Gear Ratio Calculation Overflow	应用故障 Application Fault	II	是 YES
E309 （0xFF36）	电子齿轮比超范围 Electronic Gear Ratio Out of Range	应用故障 Application Fault	II	是 YES
E326 （0x8613）	回零失败 Homing Failure	应用故障 Application Fault	II	是 YES

E401 （0xFF31）	EtherCAT 丢包 EtherCAT Packet Loss	现场总线故障 Fieldbus Fault	II	是 YES
E402 （0xFF32）	EtherCAT 通讯状态未进 OP EtherCAT State Not Operational (OP)	现场总线故障 Fieldbus Fault	I	是 YES
E403 （0xFF30）	EtherCAT 网线未接 EtherCAT Cable Disconnected	现场总线故障 Fieldbus Fault	I	是 YES
E404 （0xFF34）	指令插补周期不支持 Command Interpolation Cycle Not Supported	现场总线故障 Fieldbus Fault	I	是 YES
E405 （0xFF33）	PDO 配置超出数量限制 PDO Configuration Count Exceeded	现场总线故障 Fieldbus Fault	I	是 YES
警告码 Warning Codes	**警告名称 Warning Name**	**警告类型 Warning Type**		
A102	电机过载警告 Motor Overload Warning	驱动器警告 Drive Warning		
A104	母线欠压警告 Bus Under-Voltage Warning	驱动器警告 Drive Warning		
A105	驱动过载警告 Drive Overload Warning	驱动器警告 Drive Warning		
A126	软正限位开关有效 Soft Positive Limit Switch Active	驱动器警告 Drive Warning		
A127	软负限位开关有效 Soft Negative Limit Switch Active	驱动器警告 Drive Warning		

A128	软正负限位开关有效 Soft Positive/Negative Limit Switch Active	驱动器警告 Drive Warning		
A130	恢复出厂设置失败 Restore Factory Defaults Failed	驱动器警告 Drive Warning		
A131	清除故障历史失败 Clear Fault History Failed	驱动器警告 Drive Warning		
A132	参数超范围 Parameter Out of Range	驱动器警告 Drive Warning		
A401	SYNC0 同步与插补时间不一致 SYNC0 Mismatch	现场总线警告 Drive Warning		

8.2 异常现象排查

Troubleshooting for Abnormal Phenomena

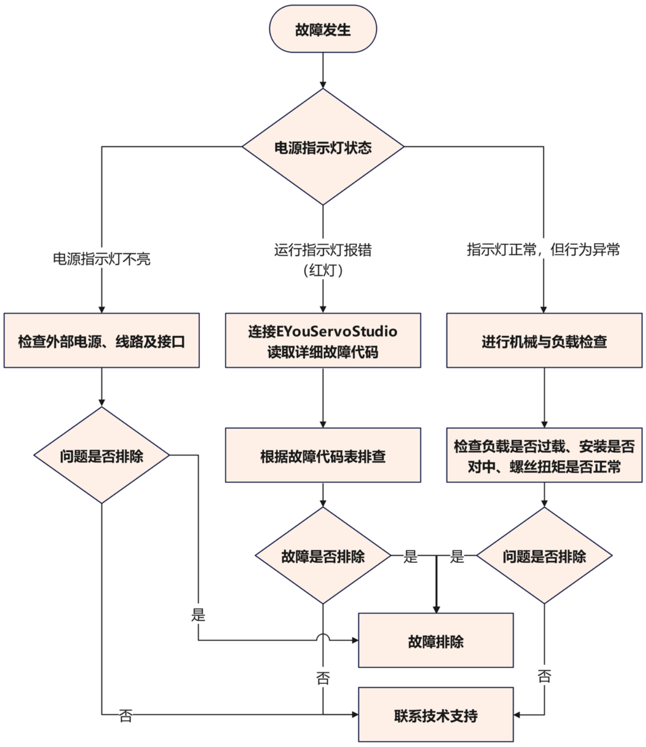


图 8-1 问题排查流程

Figure 8-1 Troubleshooting Flowchart

1) 基础检查

Basic Checks

① 现象：完全无反应，指示灯不亮。

Phenomenon: No response, indicator light is off.

② 可能原因：

Possible Causes:

◆ 外部电源未供电或电源损坏；

External power is not supplied or the power supply is damaged;

◆ 电源线（XT30）松动、接触不良或损坏；

Power cable (XT30) is loose, has poor contact, or is damaged;

◆ 关节模组内部严重短路（此情况较少见）。

Serious internal short circuit in the joint module (rare case).

③ 解决措施：

Solution:

◆ 使用万用表检测电源空载输出电压，确认是否为 48V±10%；

Use a multimeter to check the unloaded output voltage of the power supply to confirm whether it is 48V±10%;

◆ 检查电源线两端（电源端和关节端）是否接插牢靠；

Check if the power cable is securely connected at both ends (power side and joint side);

◆ 尝试更换完好确认无异常的电源线；

Try replacing the power cable with a confirmed working one.

◆ 检查电源接口定义，确保正负极没有接反。

Check the power interface definition to ensure that the positive and negative terminals are not reversed.

2) 机械类异常

Mechanical Abnormalities

① 现象：运行中有异响、抖动、卡顿，或者无法达到预期扭矩、速度。

Phenomenon: No response, indicator light is off.

② 可能原因：

Possible Causes:

- 负载过载：超过关节额定或者峰值扭矩；

  Overload: Exceeding the joint's rated or peak torque.

- 安装不当：连接法兰的止口配合过紧、不同心、端面不平，导致输出轴承受额外弯矩；

  Improper Installation: Excessive bending moment on the output shaft due to an overly tight spigot fit, misalignment, or non-flat end surface of the connecting flange.

- 紧固不当：安装螺丝未按要求的扭矩和对角顺序拧紧，或者未加螺纹紧固剂；

  Incorrect Fastening: Mounting screws not tightened according to the specified torque and diagonal sequence, or lack of thread-locking adhesive.

- 内部损坏：谐波减速器因冲击负载等原因导致柔轮或钢轮损坏。

  Internal damage: The harmonic reducer experiences damage to the flexible wheel or steel wheel due to impact loads and other reasons.

③ 解决措施：

Solutions:

- 空载测试：将关节与负载分离，进行空载运行，如果空载正常，问题可能出在负载端或者安装上；

  No-load Test: Disconnect the joint from the load and run it under no-load conditions. If it operates normally, the issue may lie with the load or the installation.

- 检查负载：重新核算负载的惯量和扭矩需求，确保在模组能力范围内；

  Check the Load: Recalculate the load's inertia and torque requirements to ensure they are within the module's capabilities.

- 重新安装：参照手册 3.3 机械安装步骤，确保连接法兰的加工精度（止口圆度、平面度）、同心度符合要求，并使用扭矩扳手严格按照标准重新安装；

  Reinstall: Refer to section 3.3 of the manual for mechanical installation steps. Ensure the machining accuracy of the connecting flange (spigot roundness, flatness) and alignment meet the requirements, and reinstall strictly using a torque wrench according to the standard.

- ◆ 联系技术支持：如果空载时仍有异响或者卡滞，可能为内部损伤，需返厂检测。
  Contact Technical Support: If abnormal noise or sticking persists during no-load operation, there may be internal damage, and the unit should be returned to the factory for inspection.

3) 通信类异常

Communication Abnormalities

① 现象：软件无法连接、时报时通、或者控制器报警通信中断。

Phenomenon: Software fails to connect, intermittent communication, or controller alarms indicating communication interruption.

② 可能原因：

Possible Causes:

- ◆ 接线错误：CAN 总线未使用双绞线或者线序错误（CAN_H/L 接反），EtherCAT 总线线序异常；
  Wiring errors: CAN bus not using twisted pair or wire order is wrong (CAN_H/L reversed), EtherCAT bus wire order incorrect ; Power cable (XT30) is loose, has poor contact, or is damaged;
- ◆ 终端电阻缺失：CAN 总线网络两端未安装 120Ω 终端电阻；
  Missing terminal resistors: 120Ω terminal resistors not installed at both ends of the CAN bus network;
- ◆ 节点 ID 冲突：网络中多个设备设置相同的节点 ID；
  Node ID conflict: Multiple devices in the network are set to the same node ID;
- ◆ 波特率不匹配：主站与关节的 CAN 通信波特率设置不一致;
  Baud rate mismatch: The CAN communication baud rate between the master station and the joint is mismatched;

- 电磁干扰：通信线缆未采用屏蔽线，或屏蔽层未接地。

  Electromagnetic interference: Communication cables are not shielded, or the shielding layer is not grounded.

③ 解决措施：

Solution:

- 核对接线，确保 CAN 或者 EtherCAT 线序没有接反；

  Check wiring and ensure CAN or EtherCAT wire order is correct; Check if the power cable is securely connected at both ends (power side and joint side);

- 检查并确定 CAN 总线两端有 120Ω 终端电阻；

  Check and confirm that 120Ω terminal resistors are installed at both ends of the CAN bus;

- 通过上位机为每个关节节点设置唯一节点 ID；

  Set a unique node ID for each joint node via the upper computer;

- 确认主站与关节通信波特率一致（通常为 1Mbps）。

  Ensure the baud rate of communication between the master station and joint is consistent (typically 1Mbps).

8.3 详细故障代码及原因

Detailed Fault Codes and Causes

详见《CANopen 与 EtherCAT 通信手册》。

For details, please refer to the 《*CANopen and EtherCAT Communication Manual*》.

九、附录

Appendix

9.1 外形尺寸图

Outline Dimensions Diagram

详见相关 2D/3D 总成图。

Refer to the relevant 2D/3D assembly drawing(s).

9.2 保修政策与期限

Warranty Policy and Duration

1） 总则

General Provisions

感谢您选用本公司生产的谐波关节模组（以下简称“产品”）。本公司承诺，产品在材料和制造工艺上均符合严格的质量标准。本章节详细说明了产品享有的保修服务、期限、范围以及相关责任限制，请您在使用前仔细阅读。

Thank you for choosing the harmonic joint module (hereinafter referred to as "the product") produced by our company. We guarantee that the product meets strict quality standards in materials and manufacturing processes. This section provides detailed information about the warranty service, duration, scope, and related liability limitations. Please read this section carefully before use.

2） 保修期限

Warranty Duration

本产品享有自客户收货之日起 12 个月的有限保修服务。

This product comes with a limited warranty service of 12 months from the date the customer receives it.

3） 保修范围

Warranty Scope

在保修期内，在正常安装、使用和维护的前提下，产品如发生由于材料缺陷或制造工艺问题导致的故障，本公司将提供免费的维修或更换服务。包括但不限于：

Within the warranty period, under the conditions of normal installation, use, and maintenance, if the product experiences failure due to material defects or manufacturing issues, our company will provide free repair or replacement services. This includes, but is not limited to:

① 谐波减速器（柔轮、刚轮、波发生器）的断裂。

Cracks in the harmonic reducer (flexible wheel, rigid wheel, and wave generator).

② 集成电机绕组短路、磁钢脱落。

Short circuit in the integrated motor windings, magnetic steel detachment.

③ 绝对值编码器、制动器等功能性失效。

Functional failure of the absolute encoder, brake, etc.

④ 轴承的非正常磨损或损坏。

Abnormal wear or damage of the bearings.

4） 非保修范围

Non-Warranty Scope

下述情况导致的故障或损坏不在免费保修范围之内，本公司可提供有偿维修服务：

The following situations that cause faults or damage are not covered by the free warranty. Our company can provide paid repair services:

① 超限使用：超出产品手册中规定的额定扭矩、转速、惯量匹配或过载能力运行。

Overuse: Operating beyond the rated torque, speed, inertia matching, or overload capacity specified in the product manual.

② 错误安装与操作：未遵循手册要求进行安装（如不对中、过紧的配合）、接线（如电源反接、电压不符）、参数配置或操作。

Incorrect Installation and Operation: Installation (e.g., misalignment, overly tight mating), wiring (e.g., reversed polarity, incorrect voltage), parameter configuration, or operation that does not follow the manual's instructions.

③ 外部环境与灾害：在不符合要求的环境（如多粉尘、潮湿、腐蚀性气体、高温、低

温）中使用，或因火灾、水浸、地震、雷击、电压突变等不可抗力造成的损坏。

External Environment and Disasters: Damage caused by use in environments that do not meet the requirements (e.g., excessive dust, humidity, corrosive gases, high or low temperatures) or due to disasters such as fire, flooding, earthquakes, lightning, or voltage surges.

④ 外部因素：因运输磕碰、跌落、外部冲击或其他人为意外造成的物理损伤。

External Factors: Physical damage caused by shipping bumps, drops, external impacts, or other human accidents.

⑤ 未经授权的维修：用户或非本公司授权技术人员进行的拆卸、维修、改装。

Unauthorized Repairs: Disassembly, repair, or modification performed by the user or technicians not authorized by the company.

⑥ 正常磨损与消耗：属于正常的性能降低或磨损（如润滑脂随时间的自然损耗）。

Normal Wear and Tear: Normal performance degradation or wear (e.g., natural loss of lubricant over time).

⑦ 随附配件：随产品提供的标准电源线、信号线等标准件。

Accessories: Standard components such as power cords, signal cables, and other accessories provided with the product.

5） 保修服务流程

Warranty Service Process

① 如发现产品故障，请首先联系我公司技术支持部门进行远程诊断。

If a fault is detected in the product, please first contact our technical support department for remote diagnosis.

② 如确需返厂检测或维修，请您提前与销售人员确认。未确认寄回的产品我司有权拒收。

If factory inspection or repair is required, please confirm with the sales staff in advance. Products returned without confirmation may be rejected by our company.

③ 请将产品妥善包装（建议使用原包装），并附上详细的故障描述、联系方式，寄往

我公司指定的地址。运输途中产生的损坏由寄方承担。

Please package the product properly (preferably using the original packaging), and include a detailed fault description and contact information. Send it to the address designated by our company. Any damage during shipping will be the responsibility of the sender.

④ 我司收到产品后，将进行检测并判断故障原因是否属于保修范围。若属于保修范围：我司将免费维修或更换，并承担修好产品寄回的费用。若不属保修范围：我司将出具检测报告并提供付费维修报价，待您确认并付款后执行维修。

After receiving the product, our company will inspect it and determine whether the fault is covered under warranty. If it is within the warranty scope, we will repair or replace it free of charge and cover the cost of returning the repaired product. If it is outside the warranty scope, we will provide an inspection report and a paid repair quote. The repair will proceed after your confirmation and payment.

6） 责任限制

Liability Limitations

① 本保修政策取代并排除所有其他明示或暗示的保修条款。

This warranty policy replaces and excludes all other express or implied warranty terms.

② 本保修政策是您享有的唯一且排他的救济方式。

This warranty policy is your sole and exclusive remedy.

③ 在任何情况下，我司均不对因产品使用或无法使用所引发的间接性、偶然性、特殊性或后果性损失（包括但不限于数据丢失、生产停工、利润损失）承担赔偿责任。

Under no circumstances shall our company be liable for any indirect, incidental, special, or consequential damages arising from the use or inability to use the product (including, but not limited to, data loss, production downtime, or loss of profits).

7） 其他

Others

本保修政策仅适用于从本公司官方渠道或授权经销商处购买的产品。

This warranty policy only applies to products purchased through official channels or authorized distributors of our company.

本公司保留对本保修政策的最终解释权，以及随时修订此政策的权利，恕不另行通知。最新版本以公司官网公布为准。

Our company reserves the right to make the final interpretation of this warranty policy and to revise this policy at any time without prior notice. The latest version will be published on the company's official website.

温馨提示：正确安装、规范操作和定期维护是延长产品寿命的关键。请务必仔细阅读本手册的其他章节，确保产品在最佳状态下运行。请妥善保管您的购买凭证。

Friendly Reminder: Proper installation, standardized operation, and regular maintenance are key to extending the product's lifespan. Please be sure to read other sections of this manual carefully to ensure the product operates in optimal condition. Please keep your proof of purchase safe.

9.3 联系方式与技术支持

Contact Information and Technical Support

1） 重要提示

Important Notes

在联系我们之前，建议您：

Before contacting us, we recommend that you:

① 查阅本手册：您遇到的问题，很可能已在手册的【操作、安装、故障排除】等相关章节中有详细说明。

Review this manual: The problem you are facing may have detailed instructions in the manual's sections on [Operation, Installation, Fault Troubleshooting], etc.

② 准备产品信息：请将产品型号和机身序列号（S/N） 准备好，我们的技术支持人员需要这些信息来为您服务。

Prepare product information: Please have the product model and serial number (S/N) ready, as our technical support staff will need this information to assist you.

③ 描述故障现象：请详细描述问题现象、出现频率以及您已尝试过的解决方法，这将极大提高支持效率。

Describe the fault phenomenon: Please provide a detailed description of the issue, its frequency, and any solutions you have already tried, which will greatly improve the efficiency of our support.

2） 技术支持与服务

Technical Support and Services

我们为您提供多种技术支持渠道：

We offer multiple channels for technical support:

① 技术支持联系方式：

Technical Support Contact Information:

表 9-1 技术支持联系方式

Table 9-1 Technical Support Contact Information

工程师 **Engineer**	电话 **Phone**	邮箱 **Email**
沈工（华东地区） Eng. Shen (East China Region)	15251673298	jun.shen@eyoubot.com
陆工（华东地区） Eng. Lu (East China Region)	17851788985	congsheng.lu@eyoubot.com
陈工（华南地区） Eng. Chen (South China Region)	18961828318	yin.chen@eyoubot.com
庹工（华南地区） Eng. Tuo (South China Region)	18033095498	wenyu.tuo@eyoubot.com
李工（华北地区） Eng. Li (North China Region)	15833086282	jinzhao.li@eyoubot.com
李工（淘宝） Eng. Li (Taobao)	18362391134	xiaoen.li@eyoubot.com

② 服务时间：周一至周五，上午 9:00 - 12:00，下午 13:30 - 18:00 (北京时间，公共节

假日除外)；

Service Hours: Monday to Friday, 9:00 AM - 12:00 PM, 1:30 PM - 6:00 PM (Beijing Time, excluding public holidays).

③ 技术支持邮箱：见表 9-1。发送邮件时，请在标题中注明产品型号和问题概要（例如：“关于上电初始化报警的咨询”）。我们将在 1-3 个工作日内回复；

Technical support email: see Table 9-1. When sending an email, please indicate the product model and a brief description of the issue in the subject line (for example: 'Inquiry about power-on initialization alarm'). We will reply within 1-3 working days.

④ 在线支持与文档中心：请访问我公司官方网站 www.eyoubot.com 并导航至【服务与支持】板块，您可以找到：产品技术文档、手册及应用笔记。

Online Support and Documentation Center: Please visit our official website www.eyoubot.com and navigate to the 'Service & Support' section, where you can find product technical documents, manuals, and application notes.

3） 售后服务与维修

After-Sales Service and Repairs

如果您需要返厂维修或检测，请务必先联系销售人员，获取寄送地址：

If you need to return the product for repairs or inspection, please contact the sales staff first to obtain the shipping address:

江苏意优机器人科技有限公司

EYou Robot Technology Co., Ltd.

收件部门：售后服务部

Department: After-Sales Service

地址：与销售人员获取

Address: To be provided by the sales staff

联系电话：与销售人员获取

Contact Number: To be provided by the sales staff

4） 商务联系

Business Contact

销售咨询：

Sales Inquiry:

地区 **Region**	销售 **Sales Manager**	电话 **Phone**	邮箱 **Email**
华东 East China	陈经理 Manager Chen	17508913141	csl@eyoubot.com
华北 North China	王经理 Manager Wang	18611151809	wx@eyoubot.com
华南 South China	彭经理 Manager Peng	17620316328	pj@eyoubot.com
西南 Southwest China	黄经理 Manager Huang	19822950310	hwn@eyoubot.com

5） 官方信息渠道

Official Information Channels

为确保您获得准确的产品信息和安全通知，请仅通过我公司官网和本手册提供的联系方式进行咨询。本公司不对通过未经授权的第三方渠道提供的信息和服务承担任何责任。

To ensure you receive accurate product information and safety notifications, please contact us only through our official website and the contact information provided in this manual. Our company will not be responsible for information or services provided through unauthorized third-party channels.

9.4 术语表

Glossary

名词 **Term**	定义 **Definition**
关节模组	一种高度集成的机电一体化单元，将驱动器、电机、减速器、编

Joint Module	码器、制动器等关键部件融合在一个紧凑的壳体内，直接输出旋转运动，作为机器人的核心关节。 A highly integrated mechatronic unit that combines key components such as the drive, motor, reducer, encoder, and brake within a compact housing. It directly outputs rotational motion and serves as the core joint of a robot.
谐波减速器 Harmonic Drive	一种利用柔性元件（柔轮）的弹性变形来传递运动和动力的精密减速装置。由波发生器、柔轮和刚轮三部分组成。 A precision reduction device that transmits motion and power via the elastic deformation of a flexible component (the Flexspline). It consists of three parts: the wave generator, the flexspline, and the circular spline.
无框力矩电机 Frameless Torque Motor	一种高性能直驱电机，通常由独立的转子和定子组成，无需额外的轴承或外壳，可被直接集成到客户的机械结构中。 A precision speed reduction device that transmits motion and power through the elastic deformation of a flexible element (flexspline). It consists of three parts: a wave generator, a flexspline, and a circular spline.
伺服驱动器 Servo Drive	关节模组的控制核心，接收上位机指令，并驱动电机运行。它集成了控制算法、功率放大电路和通信接口。 The control core of the joint module. It receives commands from the upper-level controller and drives the motor. It integrates control algorithms, power amplification circuits, and communication interfaces.
绝对值编码器 Absolute Encoder	一种能够记录并反馈当前位置信息（绝对坐标）的传感器，即使断电后位置信息也不会丢失，无需回零即可重新获得准确位置。 A sensor capable of recording and feeding back the current position

	information (absolute coordinates). The position data is retained even after a power loss, allowing the accurate position to be known immediately upon restart without a homing procedure.
制动器 Brake	一种安全装置，通常为断电抱闸（常闭型）。在伺服下使能、断电或故障时自动闭合，锁住输出轴，防止负载下滑。 A safety device, typically a fail-safe (normally closed) brake. It engages automatically when the servo is disabled, during a power failure, or upon a fault, locking the output shaft to prevent load drop.
额定转矩 Rated Torque	在额定输入转速（2000r/min）下，减速器输出端允许连续施加的负载转矩。 The continuous load torque permitted at the output side of the reducer when operating at the rated input speed (2000 rpm).
峰值转矩 Peak Torque	关节在启动、停止或短时过载时能够承受的最大转矩。超过此值可能导致损坏。 The maximum torque the joint can withstand during startup, stoppage, or short-term overload. Exceeding this value may cause damage.
瞬间容许最大转矩 Maximum Instantaneous Allowable Torque	减速器能够承受的、来自外部的不可预期的短暂冲击转矩的最大值。 The maximum value of short-duration, unexpected impact torque from external sources that the reducer can endure.
回零/零点标定 Homing / Zero Point Calibration	确定关节机械零点的过程。通过此操作，将编码器的计数值与机械位置对应起来，是保证绝对定位精度的基础。 The process of determining the mechanical zero point of the joint. This operation aligns the encoder count value with the mechanical position and is fundamental for ensuring absolute positioning accuracy.

位置模式 Position Mode	控制器向关节发送目标位置指令，关节内部闭环控制以实现精确位置跟踪。 The controller sends target position commands to the joint, which uses internal closed-loop control to achieve precise position tracking.
速度模式 Velocity Mode	控制器向关节发送目标速度指令，关节维持给定速度运行。 The controller sends target velocity commands to the joint, which maintains the given speed.
电流/扭矩模式 Current / Torque Mode	控制器向关节发送目标电流（Iq）指令，直接控制电机的输出扭矩。 The controller sends target current (Iq) commands to the joint to directly control the motor's output torque.
使能 Enable	一种控制信号，用于激活伺服驱动器。上使能后，驱动器准备接收指令并驱动电机；下使能后，驱动器停止输出，制动器通常会抱闸。 A control signal used to activate the servo drive. When enabled, the drive is ready to receive commands and power the motor. When disabled, the drive ceases output, and the brake is typically engaged.
EtherCAT	一种基于以太网的高性能实时工业通信协议。以其高带宽、低延迟和精确的同步能力而著称，是工业自动化领域的先进总线标准。 A high-performance real-time industrial communication protocol based on Ethernet. Renowned for its high bandwidth, low latency, and precise synchronization capabilities, it is an advanced bus standard in industrial automation.
CANopen	建立在 CAN 总线基础上的高层通信协议，定义了标准的通信对象和设备配置文件，广泛应用于嵌入式控制网络。 A higher-layer communication protocol built upon the CAN bus. It defines standard communication objects and device profiles and is

	widely used in embedded control networks.
UART	一种异步串行通信协议，常用于设备间的点对点通信，在 PHU 系列中主要用于本地调试和固件升级。 An asynchronous serial communication protocol often used for point-to-point communication between devices. In the PHU series, it is primarily used for local debugging and firmware updates.
过压保护 Overvoltage Protection	当检测到母线电压超过安全阈值（如 60V）时，驱动器自动触发的保护功能，会报错停机以防止设备损坏。 A protection function automatically triggered by the drive when the DC bus voltage exceeds a safe threshold (e.g., 60V). It reports an error and shuts down to prevent equipment damage.
欠压保护 Undervoltage Protection	当检测到母线电压低于最低工作电压（如 24V）时，驱动器自动触发的保护功能，可能导致制动器无法正常打开。 A protection function automatically triggered by the drive when the DC bus voltage falls below the minimum operating voltage (e.g., 24V). This may prevent the brake from releasing properly.
螺纹紧固剂 Thread locker	一种涂在螺纹上，防止螺丝因振动而松动的化学制剂。在安装关节法兰螺丝时必须使用，以确保连接可靠性。 A chemical adhesive applied to threads to prevent screws from loosening due to vibration. It must be used when installing joint flange screws to ensure connection reliability.

*注：随着产品迭代，部分术语定义或功能可能更新，请以最新版官方文档为准。

Note: Definitions and functionalities are subject to change with product iterations. Please refer to the latest official documentation for the most current information.